

Hybrid Inverter

KR-L8-15K-SPUS Series



KR-L8K-SPUS

KR-L10K-SPUS

KR-L12K-SPUS

KR-L15K-SPUS



User Manual

All-in-One Hybrid Inverter — Quick Start

KR-L8K-SPUS · KR-L10K-SPUS · **KR-L12K-SPUS** · KR-L15K-SPUS | 120/240 V split phase, 60 Hz

KOOR

Keep this sheet with the inverter. It covers the steps you use on the job. Full detail, settings and fault codes are in the **User Manual** in this same book. **Qualified electrical personnel only.**

1 WHAT IS ON THE UNIT

Front panel — indicator LEDs

- 1**  **Green** — ON: inverter is running. Flashing: standby.
- 2**  **Blue** — ON: communication with the battery BMS is normal.
- 3**  **Yellow** — ON: inverter is in EPS (backup) mode.
- 4**  **Red** — inverter is in fault mode.

Buttons

- 5** **UP** — go to previous selection
- 6** **DOWN** — go to next selection
- 7** **ENTER** — confirm the selection
- 8** **ESC** — exit setting mode

Outside of the unit

- 9** PV DC disconnect
- 10** LCD screen
- 11** BAT breaker
- 12** Parallel RJ45 ports
- 13** Input pinouts for sensors & accessories
- 14** **BMS RJ45 ports** (RS485 / CAN)
- 15** Load breaker
- 16** Wi-Fi dongle port
- 17** ON / OFF button

Inside the wiring compartment

- 18** Battery terminals
- 19** Loads
- 20** 3 × PV inputs
- 21** Generator terminals
- 22** Load terminals
- 23** Ground busbar
- 24** Grid terminals

Parallel systems

Up to **6 units** in parallel — up to **90 kW**. All units must be on the **same software version** (check MCU numbers under System Setup). A parallel system **requires one joint battery bank**, and all inputs and outputs must be shared across every inverter.

2 WHERE IT CAN HANG

Site limits

Max ambient temp	+60 °C
Min ambient temp	–25 °C
Relative humidity	up to 100% (no condensation)
Altitude	up to 4000 m
Enclosure	High sealed design, NEMA 4X
Clearance	500 mm free on every side

Yes

- Solid vertical wall, unit straight up and down
- Out of direct sunlight, rain and snow
- Out of reach of children
- Dry, ventilated, 500 mm clear all around

No

- No tilting forward, backward or laying it down
- No water pipes or cables buried in the wall behind it
- No flammable material, no corrosive or explosive area
- No direct sun, rain or snow on the unit

3 MOUNTING — 5 STEPS

- 1** Mark the bracket
- 2** Drill **10 mm** holes
- 3** Fasten the bracket to the wall
- 4** Lift the inverter onto the bracket
- 5** Lock it down with the M8 anti-collision bolts

4 WHAT IS IN THE BOX

Item	Qty	Description
A	1	Hybrid inverter
B	10	Tubular terminal (20 AWG)
C	4	Stainless steel anti-collision bolt M8
D	1	Communication cable
E	2	Copper terminal (1/0 AWG)
F	12	Tubular terminal (10 AWG)

Item	Qty	Description
G	1	User manual
H	1	Wi-Fi plug (dongle)
I	2	Sensor clamp (CT)
J	10	Tubular terminal (6 AWG)
K	1	Mounting bracket

Check the box and the unit for damage before you sign for it. Note any dents, holes or cracks on the freight paperwork.

Wiring, Start-Up and Checks

Koor All-in-One KR-L8K / L10K / **L12K** / L15K-SPUS | Quick Start page 2

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5 CABLES

#	Cable	Type	Size
1	Grounding	Single multi-core yellow/green copper	10 AWG or bigger
2	AC output	Two or three different color multi-core copper	2/0 AWG or bigger
3	PV input	PV cable (such as PV1-F)	12 AWG – 10 AWG
4	Battery	Red and black multi-core copper	2/0 – 4/0 AWG
5	Communication	CAT5E	—

Strip lengths

Generator conductors (6 – 4 AWG)	16.5 mm
Grid, load and battery conductors	16.5 mm
PV conductors (12 – 10 AWG)	16 mm

Before you touch a terminal

- **Every switch OFF** — PV, battery, grid, load, generator. Never work on it with power on.
- The lithium battery has its **own** power line and comm line — use the ones that came with it.
- The inverter ships with the basic comm line — use it directly.
- **Do not reverse battery polarity.** It damages the system and voids the warranty.
- Never land the grid on the **LOAD** output terminal.

PV check before you land it

Meter every string open-circuit before connecting: **≤**

525 V

PV+ = RED · PV– = BLACK. Crimp the pin, push the cable in firmly and press the catch until it clicks.

6 COMMUNICATION PORTS

	Port	What it is for
A	NTC	Temperature sensor for a lead-acid battery
B	CT interface	Current sensor clamps
C	GEN (NO)	Generator auto-start dry contact. When the GEN signal is active the contact closes — no voltage output .
D	±12 V RSD	Power for the RSD PLC transmitter (500 mA max)
E	EPO	Reserved for an external emergency stop
F	Meter RS485	Energy meter communication
G	ATS	240 V output port, live when the inverter is on
H	Battery comm	CAN & RS485 to the battery BMS
I	Parallel comm	CAN & RS485 between inverters
J	Parallel CAN resistor	DIP switch — set it when inverters run in parallel

BMS RJ45 pinout (battery port)

Pin	Signal
1	BAT RS485 A
2	BAT RS485 B
3	NC
4	BAT CAN H
5	BAT CAN L
6	NC
7	BAT RS485 B
8	BAT RS485 A

Only the **master** battery gets a comm cable to the inverter. Set battery type and brand on the screen — comms come up in 1–2 minutes. Wrong protocol is not covered by warranty.

7 TURN IT ON — IN THIS ORDER

- 1 Press and hold the **ON/OFF** button to turn the unit on.
- 2 Turn on the **PV** switch.
- 3 Turn on the switch between the **grid** and the inverter.
- 4 Turn on the switch between the **battery** and the inverter, and **wake the battery up**.
- 5 Set the inverter up — see the User Manual for every setting.

8 SHUT IT DOWN

- 1 Battery switch off.
- 2 Grid switch off.
- 3 PV switch off.
- 4 Hold the **ON/OFF** button to turn the unit off.

Shutdown is start-up backwards. Before power on, make sure every voltage and current is inside the inverter spec — outside it, the inverter gets damaged.

9 POST-INSTALL CHECK

☐ Inverter mounted correctly and firmly	☐ All switches off
☐ Wi-Fi / BLE stick installed correctly and firmly	☐ Ground wire connection confirmed and reliable
☐ Cable runs reasonable, no broken insulation	☐ All wires correct and securely connected
☐ Cable ties trimmed, no sharp corners	☐ Exposed terminals protected, no vacant ports
☐ All packing and scrap material cleaned up and hauled out	

10 APP AND MONITORING

Web	home.solarmanpv.com
iPhone	App Store — search “ Solarman Smart ”
Android	Google Play — search “ Solarman Smart ”

Wi-Fi stick lights. COM off + READY slow flash = not talking to the inverter; reseal it, hold Reset 5 s. NET off + COM on = never joined the router. Fast flashing is normal. Full table: **User Manual 8.5**.

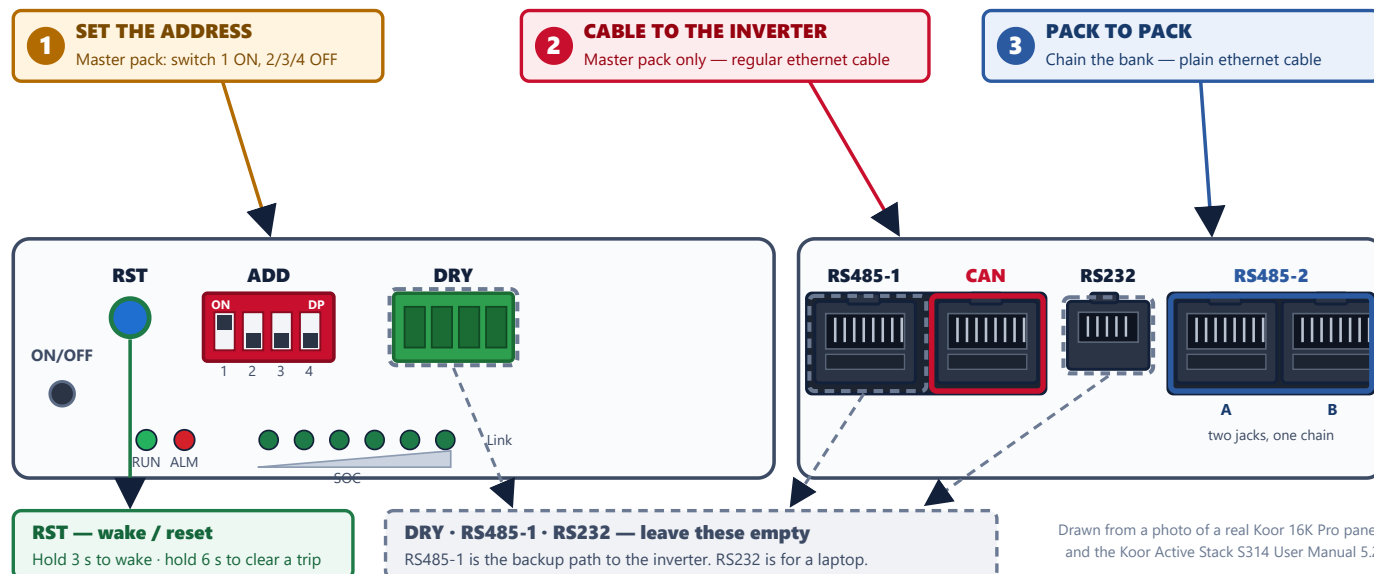
Where Every Comm Cable Plugs In

Koor battery comm panel — 16K Pro and Active Stack S314 · feeding a Koor All-in-One KR-L8K / L10K / **L12K** / L15K-SPUS

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Four of these jacks look the same and three of them say RS485. Only two jacks get a cable on a normal job. Go by the words stamped on the panel.

THE COMM PANEL — AS YOU FACE THE BATTERY



WHAT GOES IN EACH ONE

	Stamped on the panel	What plugs in	Which packs	Cable
1	ADD (red dial switch)	Nothing — it is a switch	Every pack, a different number each	Master = address 1: switch 1 ON, 2/3/4 OFF. Next packs 2, 3, 4... Set them before power-on.
2	CAN — the right-hand jack of the RS485-1/CAN pair	Goes to the inverter BMS RJ45 port	Master pack only	A regular straight ethernet cable. Both ends wired the same. Nothing crossed, nothing special built — this is how Fantastech has done it many times.
3	RS485-2 — the pair on the far right	Pack to pack, down the line	Every pack in the bank	Plain straight ethernet, all 8 pins fine. Out of B on one pack, into A on the next. Keep it the same all the way down.
—	DRY, RS485-1, RS232	Nothing on a normal job	—	RS485-1 is the backup path to the inverter if CAN will not come up. RS232 is the laptop/BMS-software port. DRY is a dry contact.

Two jacks stay empty on purpose. On the **master** pack, the **A** jack of RS485-2 has nothing coming into it. On the **last** pack in the line, the **B** jack has nothing leaving it. That is how you know the chain is a line and not a ring.

16K Pro and Active Stack wire the same. Regular ethernet on both, no crossing, no custom cable. This panel is drawn from a **16K Pro**; the Active Stack has the same layout, but its manual shows the last pair stamped **RS485A / RS485B** instead of RS485-2 — go by what is on the panel in front of you.

Koor Batteries → Koor All-in-One

KOORAppendix · Koor Active Stack S314 (51.2 V, 314 Ah, 16 kWh) and Koor 16K Pro tower → KR-L8K / L10K / **L12K** / L15K-SPUS

Read this first. Two cables and a dial switch is the whole job. One **regular ethernet cable** from the master battery jack marked **CAN** to the inverter, one from each pack to the next on **RS485-2**. Nothing crossed, nothing custom. Set every address before a single pack is powered up.

1 WHICH JACK GOES WHERE

On the inverter		The cable — a regular ethernet cable	
Port	BMS RJ45 port — marked H on the comm board, item 14 on the outside of the unit	Type	Straight CAT5E or CAT6, both ends wired the same
Not this one	The Parallel RJ45 ports (item 12, port I) are inverter-to-inverter only	Battery end	The jack stamped CAN — the right one of the RS485-1/CAN pair
Signal	pin 4 = CAN-H · pin 5 = CAN-L · pins 1, 2, 7, 8 = RS485	Inverter end	The BMS RJ45 port
On the battery		Signals	CAN-H on pin 4, CAN-L on pin 5, straight through
Active Stack S314	The jack stamped CAN — the right one of the RS485-1/CAN pair. Pin 4 = CAN-H, pin 5 = CAN-L. RS485-2 is the pack-to-pack pair. RS485-1 is the backup path to the inverter.	Both Koor batteries wire the same. Active Stack or 16K Pro, it is a regular ethernet cable off the CAN jack. No crossed cable, no cable to build. Fantastech has put this combination in many times.	
16K Pro tower	Same — the jack stamped CAN . Wires the same way as the Active Stack.		

2 HOW THE PACKS CHAIN — ONE PACK TALKS, THE REST REPORT THROUGH IT



- **Only the master** gets a cable to the inverter. Nothing else does.
- The rest chain **in a line** on the **RS485-2** pair: out of **B** on one pack, into **A** on the next, all the way down.
- **Never** run a cable from the last pack back to the master. It is a line, not a ring.
- The master RS485-2 **A** jack stays empty. The last pack RS485-2 **B** jack stays empty.
- **No two packs share an address.**
- Adding a pack later? Keep it **within 2 V** of the others or it trips charge overcurrent and will not take a charge.

Set every DIP switch before anything is powered up. The BMS reads its address at power-on. Change a switch after the fact and you have to cycle that pack.

Master address — ADDRESS 1: switch 1 ON, switches 2, 3 and 4 OFF. The battery manual calls the master the pack on the 1-0-0-0 switch pattern. If the inverter never sees the bank, try all four switches OFF and power-cycle that pack — some BMS firmware numbers packs starting at 0.

3 DIP SWITCH ADDRESSES — 4 SWITCHES, UP TO 15 PACKS

Addr	1	2	3	4	Addr	1	2	3	4	Addr	1	2	3	4	Addr	1	2	3	4
1	ON	off	off	off	5	ON	off	ON	off	9	ON	off	off	ON	13	ON	off	ON	ON
2	off	ON	off	off	6	off	ON	ON	off	10	off	ON	off	ON	14	off	ON	ON	ON
3	ON	ON	off	off	7	ON	ON	ON	off	11	ON	ON	off	ON	15	ON	ON	ON	ON
4	off	off	ON	off	8	off	off	off	ON	12	off	off	ON	ON					

1 is the master — the one pack cabled to the inverter. Give every other pack its own number going down the line. Switch numbers read left to right. Table is straight out of the Koor battery manual, section 4.3.5.

Settings, Power-Up and Troubleshooting

Koor batteries on a Koor All-in-One | Appendix page 2

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4 INVERTER SETTINGS — BATTERY MENU

Setting	Set it to	Why
Batt1 Type	Lithium	Other choices are Lead-acid, DC Source, No Battery
Com Type	CAN	Both Koor batteries talk on CAN. RS485 is the fallback, not the first pick.
Company	Pylon	Pick the Pylon entry on the list. Depending on the firmware in front of you it reads Pylon LV2 or Pylontech V1 — either one is the Pylon protocol. The inverter then matches the protocol on its own.
Activate Batt	use it once	Wakes a sleeping pack if the inverter sees nothing
Grid Charge	per job	On for grid or generator charging, off for PV only
Parallel Bat & Bat2	see note	Only if one bank feeds both battery ports. Then the comm cable goes to the BMS1 port.
Max A Charge / Disch	lower of the two	Whichever is smaller, the bank limit or the inverter limit. With CAN comms up, the BMS reports its own limits.

Why Pylon. The factory that builds these inverters gave the answer in writing — “*pylontech protocol*” — and the Koor batteries run a JK BMS whose CAN list carries the Pylon low-voltage protocol. A Koor All-in-One in the field came up right on the Pylon setting. Koor batteries on Koor off-grid inverters use the same Pylon-over-CAN pick, and Fantastech has run this combination many times.

5 POWER-UP ORDER

- 1 Everything off.** Set every DIP address before a single pack is powered.
- 2 Cable it.** Chain the packs on RS485-2, then the master CAN jack to the inverter BMS port with a regular ethernet cable.
- 3 Recheck the DIPs.** Master on address 1 (sw1 ON), no two packs alike.
- 4 Batteries on.** DC breaker on every pack, then each power switch. Wake the master if it sits dark.
- 5 Inverter on.** Hold ON/OFF, then the battery switch.
- 6 Set the menu.** Lithium → CAN → Pylon.

How you know it worked — three things, all of them:

- The **blue LED** on the inverter is **ON** (BMS communication normal)
 - A **live SOC %** shows on the screen
 - Charge and discharge limits** show, and they change
- Give it **1 to 2 minutes** after you pick the brand.

6 IT WILL NOT TALK — WORK DOWN THIS LIST

What you see	What to check
Blue LED off, no SOC	Cable in the wrong jack, or a bad cable. It goes in the jack stamped CAN , not RS485-1. Swap in a known-good ethernet cable.
Blue LED off, cable is right	Com Type is on RS485 instead of CAN, or Company is not on a Pylon entry. Reselect and wait 2 minutes.
Only some packs show	Two packs share an address, or the chain is broken. Count them on the battery screen PACKS tab.
Master dark	Pack asleep. Wake it at the battery, then use Activate Batt on the inverter.
Nothing at all	Check the master DC breaker and power switch before anything else.

Do not leave it on Lead-acid and walk away. The manual lets you pick Lead-acid when lithium will not communicate. That runs the bank **open loop** — no SOC, no BMS limits, no protection handoff. Use it to finish a day, never to finish a job. Call us and we will get the comms right.

Damage from the wrong communication protocol is not covered by warranty. Neither is reversed battery polarity. Meter before you land anything.

Where this came from. Koor All-in-One User Manual §6.6 and the Battery Settings screen; Koor Active Stack S314 User Manual §4.3; the Koor off-grid 12K manuals, Appendix A; the factory own written answer on the protocol; and a Koor All-in-One brought up in the field on the Pylon setting. **Fantastech has installed this combination many times** — if it will not talk, the cause is almost always the wrong jack, a bad cable, or a duplicate address.

Limited Warranty — Terms

Koor All-in-One Hybrid Inverter · sold and warranted by Fantastech Energy Solution LLC

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What is covered and for how long

Koor All-in-One hybrid inverter — 10 years Koor batteries — 10 years

Koor off-grid inverters carry a 5-year warranty. The term runs from the date of installation, and in no case more than six months past that term from the date the unit left the factory.

The warranty covers **defects of design, components and manufacturing**. It applies to devices bought from Fantastech and installed at the address on the front of this card. It is non-transferable unless Fantastech agrees in writing.

What is not covered

- Incorrect installation or commissioning.
- Failure to follow the User Manual, the installation guide or the maintenance instructions.
- Unauthorized modifications, changes or attempted repairs.
- Incorrect use or inappropriate operation, including reversed battery polarity and use of the wrong battery communication protocol.
- Insufficient ventilation of the device.
- Failure to observe applicable safety regulations and electrical code.
- Breaking the product seal — opening the casing — without written approval.
- Force majeure: lightning, over-voltage, storm, flood, fire.
- Transport damage, and superficial or cosmetic defects — dents, marks, scratches — that do not affect how the inverter works.

How a claim works

- **Report it within 14 days** of the first sign of a problem. Call 574-218-1973 or email derek@fantastech.org.
- Send this completed warranty card, the purchase invoice, the installation or commissioning report, and any extension warranty card if you have one.
- Send pictures and the fault code or fault screen. It is the fastest way to get a unit moving.
- Fantastech decides whether the unit is repaired on site, repaired at the shop, or exchanged for a replacement device of equivalent value for model and age.
- On an exchange, the remainder of the original warranty carries over to the replacement device. A replacement does not start a new warranty term and does not come with a new certificate.
- Because of ongoing product improvement, a replacement part or unit may not match your system monitoring or other components, and the cost of changing those out is not part of this warranty.

Limits

- This warranty covers the equipment. It does not cover labor, travel, lift or crane rental, lost production, lost profits, or other direct or indirect costs, unless Fantastech agrees to them in writing beforehand.
- Claims beyond the rights stated here — including costs of removal and reinstallation — are not covered.
- Your rights under statutory liability and product liability law are unaffected by this warranty.

Keep this card and your invoice together. Both are needed to support a claim. If the card was never sent in, send it with the claim.

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READ THE INSTRUCTIONS COMPLETELY BEFORE OPERATING THE EQUIPMENT

Check the utility voltage before turning ON the unit.

Verify the inviter's programmed grid type before connecting to the utility.

The unit will be programmed in 120/240V Split-Phase at 60Hz by default.

Disregarding these instructions could result in permanent damages to the unit



DISCLAIMER

UNLESS SPECIFICALLY AGREED TO IN WRITING:

KOOR assumes no responsibility or liability for any damages, property loss, personal injury, or any adverse consequences resulting from improper use and installation of the product or the failure to adhere to the guidelines provided in this document. Users are expressly advised to follow the instructions and guidelines outlined in the documentation accompanying the product. KOOR shall not be liable for any damages or losses incurred due to deviations from recommended usage, installation, or maintenance procedures. By using the product, users acknowledge their understanding of these disclaimers and agree to use the product at their own risk. KOOR reserves the right to update or modify product information, specifications, and guidelines without

KOOR retains the right to final interpretation of this document and all related materials pertaining to this product. This document is subject to modifications, updates, revisions, or termination without prior notice.



Fantastech Energy Solution LLC

Email: derek@fantastech.org | Phone: 574-218-1973 | Goshen, IN 46528, US

1.About This Manual

This manual describes the product information, installation, electrical connection, commissioning, troubleshooting, and maintenance. Read through this manual before installing and operating the product. All the installers and users have to be familiar with the product features, functions, and safety precautions. This manual is subject to update without notice.

For more product details and latest documents, visit [_____](#)

1.1 Applicable Model

Model	Nominal Output Power	Nominal Output Voltage
KR-L8K-SPUS	8kW	120Vac / 240Vac (Split phase) ; 208Vac (2/3 phase) 2L/N/PE
KR-L10K-SPUS	10kW	
KR-L12K-SPUS	12kW	
KR-L15K-SPUS	15kW	

1.2 Target Group

This manual is intended for qualified and knowledgeable electrical technical personnel who are responsible for hybrid inverter installation and commissioning in the energy storage system and electric system.

1.3 Symbol Definition

The following types of safety instructions and general information appear in this document as described below:

 DANGER	 WARNING	 CAUTION	 NOTICE
“Danger” indicates a hazardous situation with a high level of risk that, if not avoided, will result in death or serious injury.	“Warning” indicates a hazardous situation with a medium level of risk that, if not avoided, could result in death or serious injury.	“Caution” indicates a hazardous situation with a low level of risk that, if not avoided, could result in minor or moderate injury.	“Notice” provides some tips and methods to solve product-related problems to save time.

1.4 Updates

The latest document contains all the updates made in earlier issues.

V1.0 2024-11-11

- First Issue

2. Safety Precaution

Please strictly follow these safety instructions in the user manual during the operation.

2.1 General Safety

NOTICE

- The information in this user manual is subject to change due to product updates or other reasons. This guide cannot replace the product labels or the safety precautions in the user manual unless otherwise specified. All descriptions here are for guidance only.
- Before installations, read through the quick installation guide. For additional information, please see the user manual.
- All installations should be performed by trained and knowledgeable technicians who are familiar with local standards and safety regulations.
- Use insulating tools and wear personal protective equipment when operating the equipment to ensure personal safety. Wear anti-static gloves, cloths, and wrist strips when touching electron devices to protect the inverter from damage.
- Strictly follow the installation, operation, and configuration instructions in this manual. The manufacturer shall not be liable for equipment damage or personal injury if you do not follow the instructions. For more warranty details, please visit [_____](#).

2.2 PV String Safety

DANGER

Connect the DC cables using the delivered PV connectors. The manufacturer shall not be liable for the equipment damage if other connectors or terminals are used.

WARNING

- Ensure the component frames and the bracket system are securely grounded.
- Ensure the DC cables are connected tightly, securely, and correctly.
- Measure the DC cables with a multimeter to avoid reverse polarity connection. Also, the voltage should be under the permissible range.
- Do not connect one PV string to more than one inverter at the same time. Otherwise, it may cause damage to the inverter.
- The PV modules used with the inverter must have an IEC61730 class A rating.
- When the photovoltaic array is exposed to light, it supplies a d.c. voltage to the inverter.

2.3 Inverter Safety

WARNING

- The voltage and frequency at the connecting point should meet the on-grid requirements.
- Additional protective devices like circuit breakers or fuses are recommended on the AC side. Specification of the protective device should be at least 1.25 times the AC rated output current.
- Make sure that all the groundings are tightly connected. When there are multiple inverters, make sure that all the grounding points on the enclosures are equip Potential bonding.
- Off-grid function is not recommended if the PV system is not configured with batteries. Otherwise, the risk in system power usage is beyond the equipment manufacturer's warranty scope.
- It should be taken into account the characteristics of photovoltaic power instability, if the battery is not connected, there is no EPS function.
- DO NOT connect the grid to the "LOAD" output terminal.

DANGER

- All labels and warning marks should be visible after the installation. Do not cover, scrawl, or damage any label on the equipment.
- Warning labels on the inverter are as follows:

	DANGER High voltage hazard. Disconnect all incoming power and turn off the product before working on it.		Delayed discharge. Wait 5 minutes after power off until the components are completely discharged.
	Read through the user manual before working on this device.		Potential risks exist. Wear proper PPE before any operations.
	High-temperature hazard. Do not touch the product under operation to avoid being burnt.		Grounding point.
	With CE mark & the inverter fulfills the basic requirements of the guideline governing Low-Voltage and electron-magnetic compatibility.		Do not dispose of the inverter as household waste. Discard the product in compliance with local laws and regulations, or send it back to the manufacturer.
	RCM marking		

2.4 Battery Safety

WARNING

- The battery used with the inverter shall be approved by the inverter manufacturer. The approved battery list can be obtained through the official website.
- Before installations, read through the corresponding battery's User Manual to learn about the product and the precautions. Strictly follow its requirements.
- If the battery discharged completely, please charge it in strict accordance with the corresponding model's User Manual.
- Factors such as: temperature, humidity, weather conditions, etc. may limit the battery's current and affect its load.

- Contact Fantastech immediately if the battery is not able to be started. Otherwise, the battery might be damaged permanently.
- Use the multimeter to measure the DC cable to avoid reverse polarity connection. Also, the voltage should be under the permissible range.
- Do not connect one battery group to several inverters at the same time. Otherwise, it may cause damage to the inverter.
- Lead-acid battery types—including but not limited to Flooded Lead Acid (FLA), Gel, and Absorbent Glass Mat (AGM) batteries—typically require specific charging methods and may perform optimally only under certain operating conditions. The KOOR inverter may not fully support these requirements by default, and adjustments to its charging and discharging parameters may be necessary.
- If you intend to use any of these battery types, it is strongly recommended to consult the battery manufacturer for guidance on best practices, compatibility, and overall system sizing specific to your application.
- Lead Acid battery types, including but not limited to Flooded Lead Acid [FLA], Gel, and AGM battery types, often require specific charging methods or work best in certain conditions. The inverter may not meet such requirements, or it may require modification to the charging and discharging parameters.
- If you plan to use one of these battery types, please contact the battery manufacturer for suggestions on best practices and overall system sizing requirements for your particular application.

2.5 Personal Requirements



NOTICE

- Personnel who install or maintain the equipment must be strictly trained, learn about safety precautions and correct operations.
- Only qualified professionals or trained personnel are allowed to install, operate, maintain, and replace the equipment or parts.

3. Product Introduction

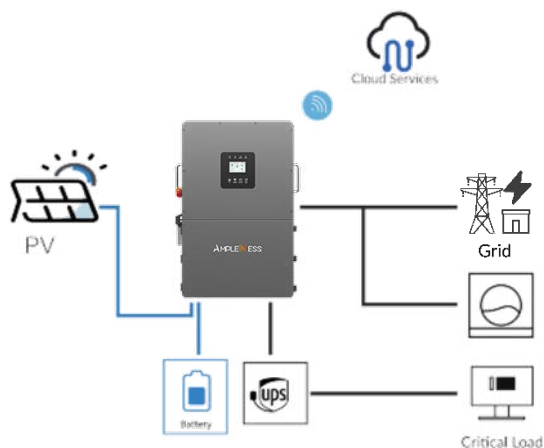
3.1 Product Features

Intended usage

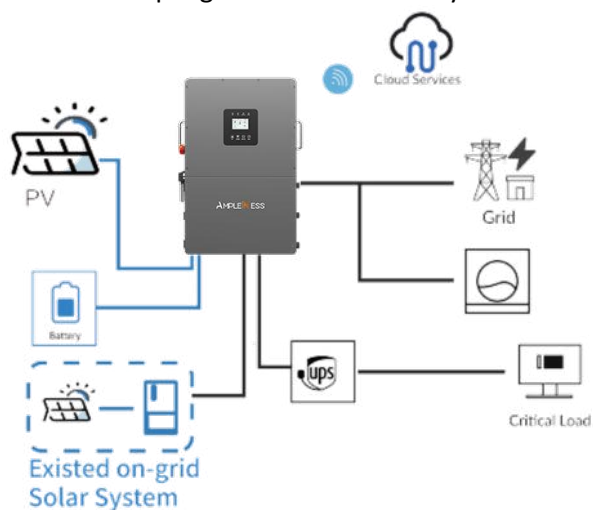
This is a multifunctional inverter, combining functions of inverter, solar charger and battery charger to offer uninterruptible power support with portable size. The inverter is equipped with a powerful APP. Users can view the operating status of the inverter in real time through the APP, including but not limited to battery information, photovoltaic power generation information, alarm information, etc. The brief operation interface makes it easier to set the appropriate working mode in different application scenarios to meet the diversified needs of users.

This unit and its associated system are suitable for the following applications:

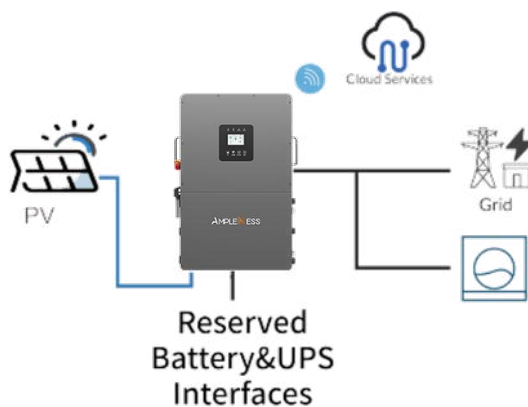
A: Solar and battery storage system



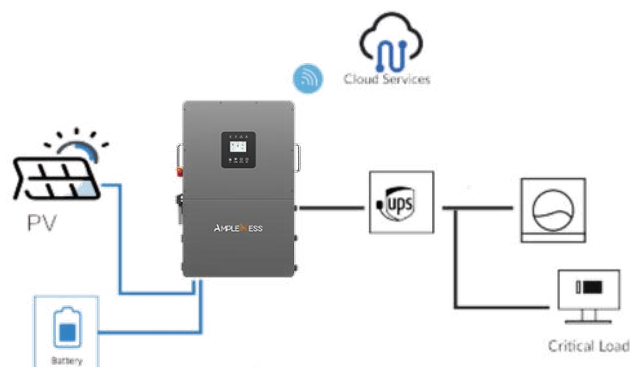
B: AC Coupling with existed solar system



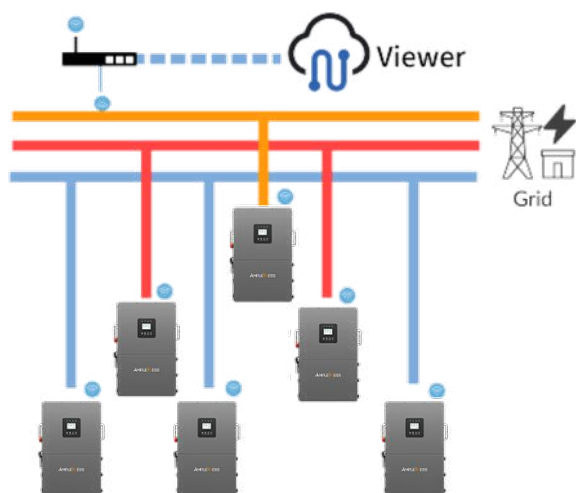
C: On-grid solar system without battery
(Support EPS even without battery)



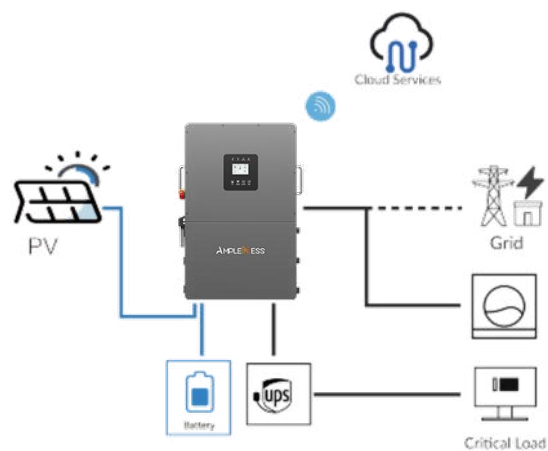
D: off-grid and back-up applications



E: Single and three phase paralleling system



F: Energy storage system with peak shaving Function



3.2 Working Mode

The KR-L8-15K-SPUS Series hybrid inverter has the following work modes based on your configuration and layout conditions.

Work modes	Description
Self Use (with PV Power)	Priority: load>battery>grid The energy produced by the PV system is used to optimize self-consumption. The excess energy is used to charge the batteries, then exported to grid.
Self Use (without PV Power)	Priority: load>battery When no PV supplied, battery will discharge for local loads firstly, and grid will supply power when the battery capacity is not enough.
TOU ¹ Balance	In this mode, users can set the time period, photovoltaic sufficient and time-of-use price area, in the case of large power consumption of users, to achieve the balance of photovoltaic utilization rate and economic benefits. Peak price: Run spontaneous self-use mode. Flat price: The photovoltaic gives priority to the load power supply, and when the photovoltaic power is insufficient, the battery is restricted to discharge to ensure the continuity of energy. Valley price: Charge the battery at full power priority until it is full.
TOU ¹ Eco	This mode can be used to meet users' demand for peak cutting and valley filling and achieve maximum economic benefit in areas with large difference of peak and valley electricity price. Peak price: The battery is discharged at full power to sell electricity to the grid at a high price Flat price: Run spontaneous self-use mode. Valley price: Buy electricity from the grid at a low price to charge the battery at full power
Back up only	Priority: load>battery When entering this mode, the system will start to charge the battery until it is fully charged and remains fully charged, waiting for the power grid to fail. When the grid is off, system will supply emergency power from PV or battery to supply the home loads .
Grid Priority	Priority: load>grid When the system works in this mode, the electricity generated by the photovoltaic will be preferentially connected to the grid. Users can send requests to the grid at peak times, and in this mode, users can set the end of the battery SOC point.

1.TOU means time of use.

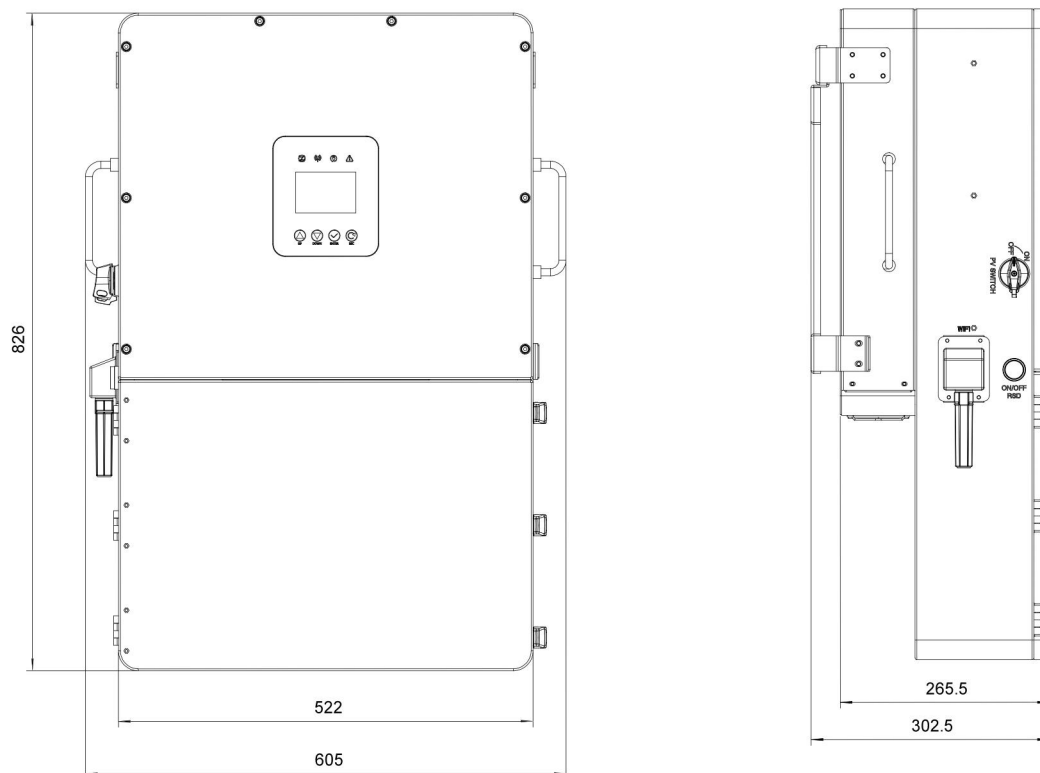


Make sure the load powering rating is within the EPS's output rating. Or the inverter will shut down with an 'over load' warning. When an "over load" is appeared, adjust the load power make sure it is within the range of the EPS output, and turn the inverter on. For the nonlinear load, please pay attention to the inrush power make sure it is within the range of the EPS output.

3.3 Appearance

3.3.1 Dimensions

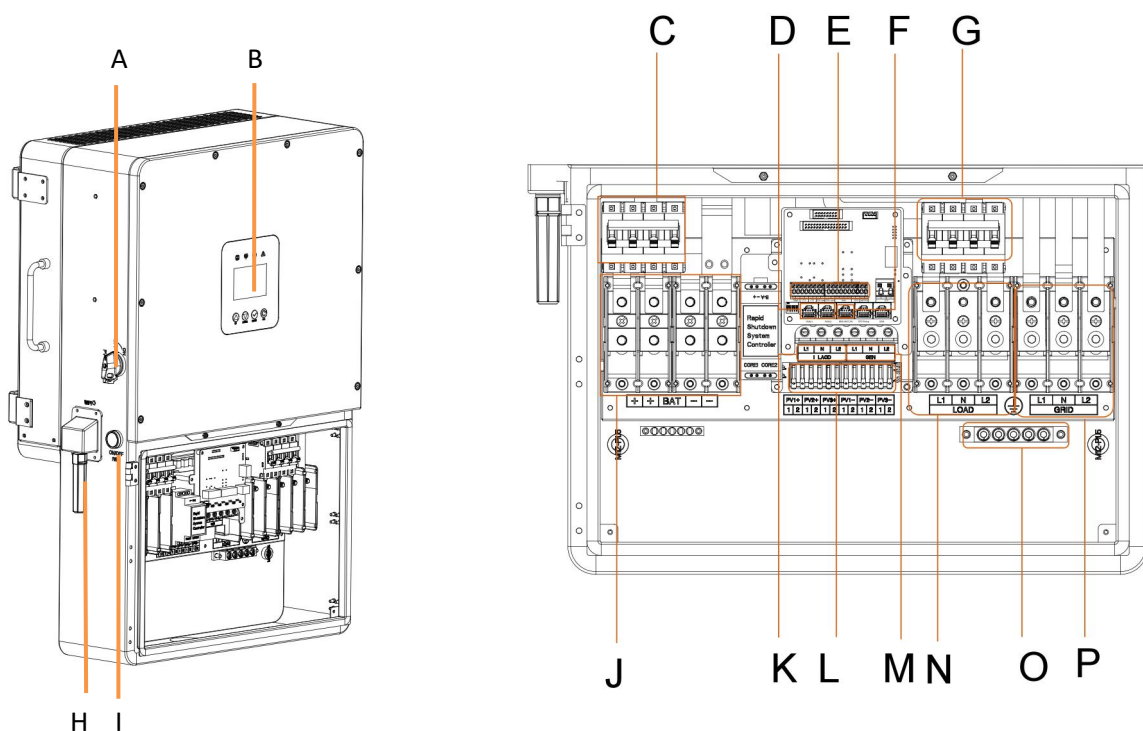
Considering the dimensions of the inverter, find a suitable location for the system. The upper, lower and left and right sides of the inverter must leave a suitable distance to ensure heat dissipation.



PROTECT THE LCD SCREEN from direct exposure to UV light.

The KR-L8-15K-SPUS is a INEMA 4X enclosure that is rated for outdoor installation but can also be installed indoors.

3.3.2 General Description



Component	Name	Component	Name
A	PV DC disconnect	J	Battery terminals
B	LCD Screen	K	I Loads
C	BAT Breaker	L	3*PV Inputs
D	Parallel RJ45 ports	M	Generator Terminals
E	Input pinouts for sensors & accessories	N	Load Terminals
F	BMS RJ45 ports (RS485/CAN)	O	Ground busbar
G	Load Breaker	P	Grid Terminals
H	Wi-Fi		
I	ON/OFF Button		

3.3.3 Nameplate

The nameplate is for reference only.

Hybrid Inverter	
Model	AP-L15K-SPUS
Max PV input voltage	550V d.c
PV input operating voltage range	120-520V d.c
Max operating PV input current	26A d.c*3
Isc PV	39A d.c*3
Grid rated voltage	120V/240V,208V a.c, 2L+N+PE
Grid rated frequency	60Hz
Grid rated output current	62.5A a.c
Max passthrough current	200A
Power Factor	0.8Leading-0.8Lagging
EPS rated output voltage	120V/240V,208V a.c, 2L+N+PE
EPS rated output frequency	60Hz
EPS rated output current	62.5A a.c
EPS rated output apparent power	15000VA
Battery operation voltage range	40-60V d.c
Max.charge and discharge current	275A d.c
Battery type	Li-ion/Lead-acid
Ingress protection degree	NEMA 4X
Inverter topology	Non-isolated
Operating temperature range	-25℃~+60℃(>45℃derating)

SN:









Made in China

3.3.4 Specification

Model	KR-L8K-SPUS	KR-L10K-SPUS	KR-L12K-SPUS	KR-L15K-SPUS
Product Type	Hybrid Inverter			
Battery				
Battery type	Li-ion/Lead-acid			
Battery voltage range	40-60 Vd.c			
Rated battery voltage	51.2 Vd.c			
Max.charge/discharge Power	8 kW	10 kW	12 kW	15 kW
Rated charge/discharge current	167 A d.c.	210 A d.c.	250 A d.c.	275 A d.c.
Max. charge/discharge current	167 A d.c.	210 A d.c.	250 A d.c.	275 A d.c.
Reverse Connect Protection	Yes			
Communication interface	CAN/RS485			
Input (PV)				
Recommended Max. PV array power for each input	12 kW	15 kW	18 kW	22.5 kW
Max. operating PV input current (PV 1 /PV 2 / PV3)	26A d.c+26A d.c+26A d.c			
Max. Isc PV (PV 1 /PV 2 / PV3)	26A d.c+26A d.c+26A d.c			
Vmax PV (Max. PV input voltage)	550 V d.c			
MPPT Voltage Range	120-480V dc			
Full power MPPT voltage range	120-480 V d.c	125-480 V d.c	145-480 V d.c	195-480 V d.c
Start-up Voltage	120V dc			
Number of MPP Trackers	3			
Strings per MPP Tracker	2			
Number of PV input	6			
Grid AC input and AC output				
Grid rated voltage	120Vac / 240Vac (Split phase) , 208Vac (2/3phase)			
Grid rated frequency	60Hz			
Grid rated input active power	8 kW	10 kW	12 kW	15 kW
Grid rated input apparent power	8 kVA	10 kVA	12 kVA	15 kVA
Grid max. input active power	48kW	48kW	48kW	48kW
Grid max. input apparent power	48 kVA	48 kVA	48 kVA	48 kVA
Grid rated output active power	8 kW	10 kW	12 kW	15 kW
Grid rated output apparent power	8 kVA	10 kVA	12 kVA	15 kVA
Grid max. output active power	8 kW	10 kW	12 kW	15 kW
Grid max. output apparent power	8 kVA	10 kVA	12 kVA	15 kVA
Grid rated input current	33.4 A a.c	41.7 A a.c	50 A a.c	62.5 A a.c
Grid max. input current	200A a.c	200A a.c	200A a.c	200A a.c
Grid rated output current	33.4 A a.c	41.7 A a.c	50 A a.c	62.5 A a.c
Grid max. output current	33.4A ac	41.7 A a.c	50 A a.c	62.5 A a.c
Grid power factor	0.8 leading to 0.8 laggings			
Grid input and output Inrush current	500A a.c @ 3μs			
Max. Grid output fault current	67A a.c @0.2μs	84A a.c @0.2μs	100A a.c @0.2μs	125 A a.c @0.2μs
Max. Grid output overcurrent protection	240V a.c /200 A a.c			
Grid input Icc (Rated conditional short-circuit current)	500A a.c			
Grid input Icw (Rated short-time withstand current)	500A a.c			
Total Harmonic Distortion(THDi, rated power)	<3%			

EPS output	KR-L8K-SPUS	KR-L10K-SPUS	KR-L12K-SPUS	KR-L15K-SPUS
EPS rated output Voltage	120Vac / 240Vac (Split phase) , 208Vac (2/3phase)			
EPS rated output frequency	60Hz			
EPS rated output active power	8 kW	10 kW	12 kW	15 kW
EPS rated output apparent power	8 kVA	10 kVA	12 kVA	15 kVA
EPS max. output active power	8 kW	10 kW	12 kW	15 kW
EPS Max. output apparent power	8 kVA	10 kVA	12 kVA	15 kVA
EPS rate output current	33.4 A a.c	41.7 A a.c	50 A a.c	62.5 A a.c
EPS output power factor	0.8 leading to 0.8 lagging			
EPS output peak power	200% overload(<10s)			
EPS output Inrush current	200A a.c @ 10ms			
EPS max. output fault current	200A a.c @ 10ms			
Switch Time	<10ms			
Total Harmonic Distortion(THDv, linear Load)	<2%			
Compatible with the Generator	Optional			
Efficiency				
MPPT Efficiency	99.50%			
California-efficiency	96.50%			
Max.efficiency	97.50%			
Standard				
Safety	IEEE1547, CA RULE21, RULE14(HECO Compliant), UI1741, UL1741SA/SB, CSA C22.2, PREPA, UL1699B			
EMC	FCC Part15 Class B, Sunspec			
General Parameter				
Degree of ingress protection	NEMA 4X			
Protection class	I			
Environment category	Indoor&Outdoor			
Wet location classification	Yes			
Pollution degree	PD3			
Operating altitude	4000 m(>2000 Derating)			
Operating ambient temperature	-25 - +60 °C (linely derating to 60% when exceed +45 to +60 °C)			
Operating relative Humidity	0-100% (non-condensing)			
Storage Temperature	-25- +60 °C			
Storage relative Humidity	0-100% (non-condensing)			
Noise Emission(typical)	<45 db			
Over voltage Category	AC: III, PV: II			
Electrical supply system	TN, TT			
Dimension (WxHxD)	496*806*270.5mm			
Net Weight	55kg			
Cooling Mode	Intelligent Air Cooling			
Topology	Non-isolated			
Active anti-islanding method	Active frequency drift			
Communication	RS485/Wifi/Bluetooth/LAN/CAN/DRM/Meter Yes/ Yes/ Opt/Opt/ Yes/ Yes/ Opt			
LED indicator	LCD+LED+APP			
Warranty	5/10 years			

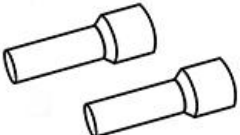
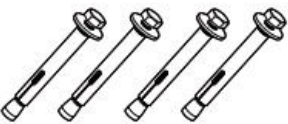
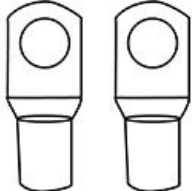
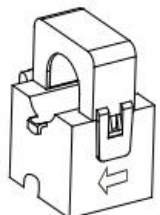
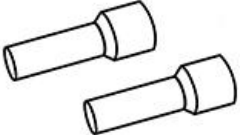
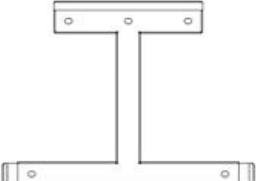
4. Check and Storage

4.1 Check Before Receiving

Check the following items before receiving the product.

1. Check the outer packing box for damage, such as holes, cracks, deformation, and others signs of equipment damage. Do not unpack the package and contact Fantastech as soon as possible if any damage is found.
2. Check the inverter model. If the inverter model is not what you requested, do not unpack the product and contact Fantastech .
3. Check the deliverable for correct model, complete contents, and intact appearance. Contact Fantastech as soon as possible if any damage is found.

4.2 What's in the box?

 Hybrid inverter x1	 Tubular Terminal (20AWG) x10	 Stainless steel anti-collision bolt M8 x5	 communication cable x1
 Copper Terminal (1/0 AWG) x4	 Tubular Terminal (10AWG) x12	 Certification x1 Warranty card x1 Quick installation manual x1	 Wi-Fi-Plug x1
 Sensor Clamp x 2	 Tubular Terminal (6 AWG) x10	 Mounting Bracket x 1	 Desiccant 50g 1x

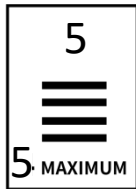
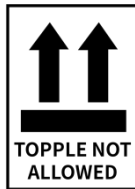
4.3 Storage

If the equipment is not to be installed or used immediately, please ensure that the storage environment meets the following requirements:

1. Do not unpack the outer package or throw the desiccant away.
2. Store the equipment in a clean place. Make sure the temperature and humidity are appropriate and no

condensation.

3. The height and direction of the stacking inverters should follow the instructions on the packing box.
4. The inverters must be stacked with caution to prevent them from falling.
5. If the inverter has been long term stored, it should be checked by professionals before being put into use.
6. The storage temperature range is: -25°C~60°C, and the storage humidity is 0~100%.
7. The box should be suitable for loads more than 75kg.

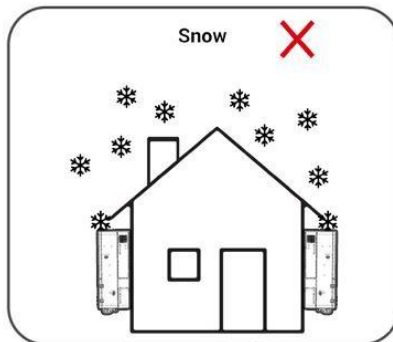
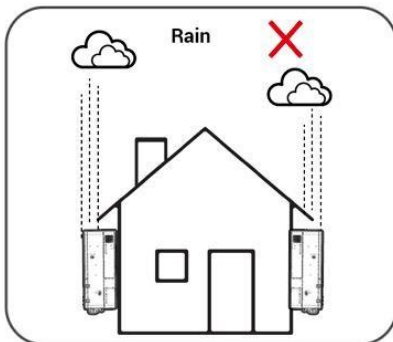
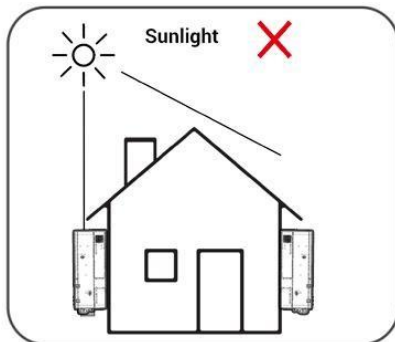
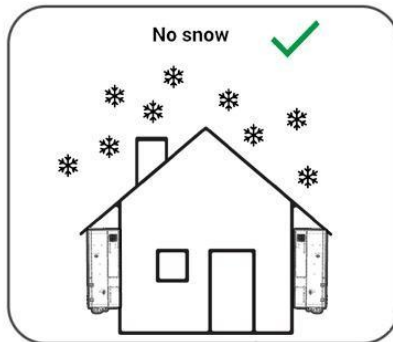
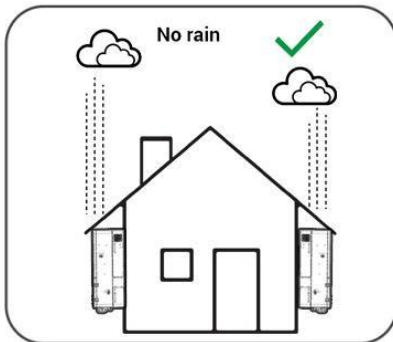
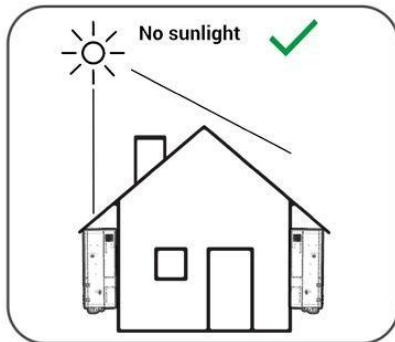
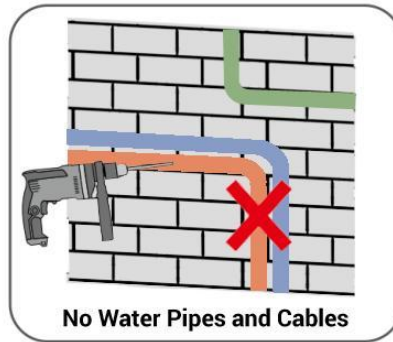
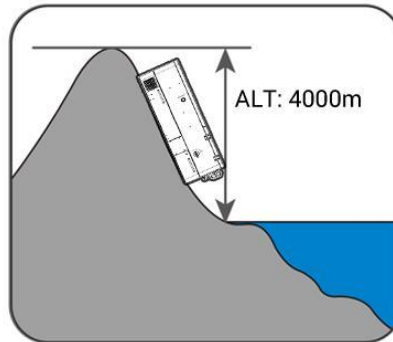
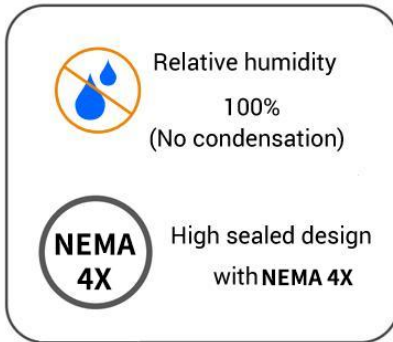
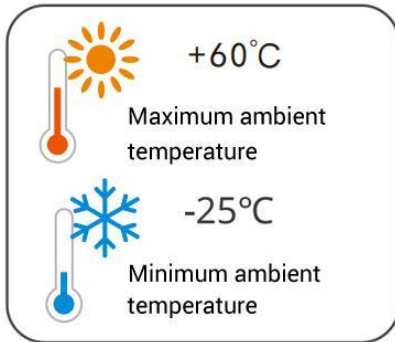
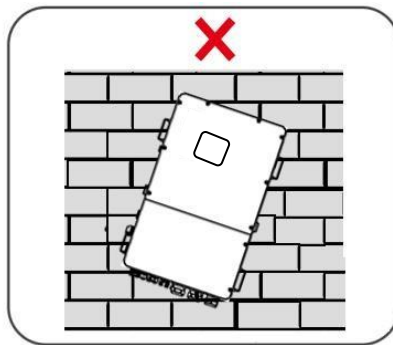
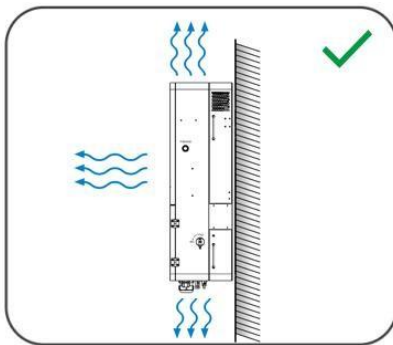
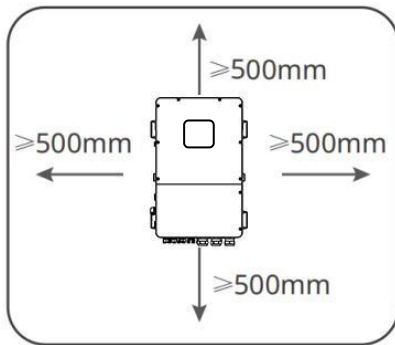


5. Installation

5.1 Installation Requirements

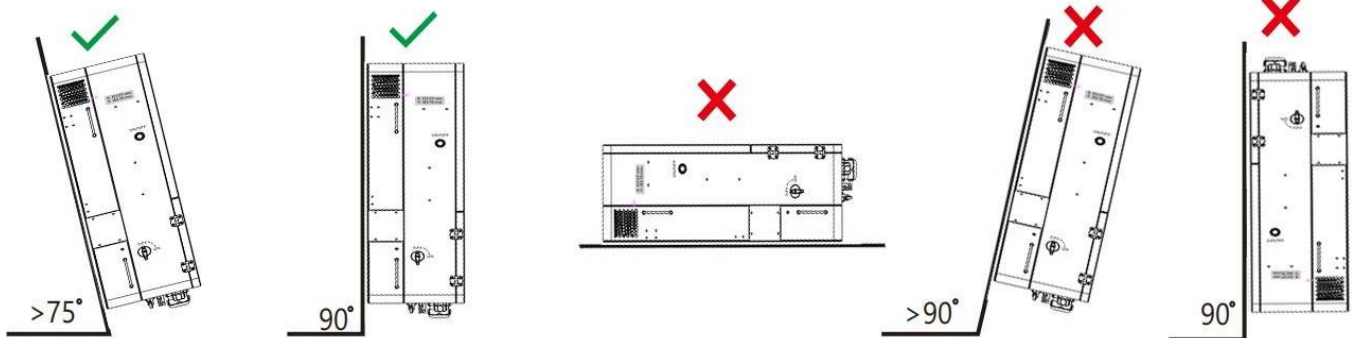
Installation Environment Requirements

1. Do not install the equipment in a place near flammable, explosive, or corrosive materials.
2. Do not install the equipment in a place that is easy to touch, especially within children's reach. High temperature exists when the equipment is working. Do not touch the surface to avoid burning.
3. Avoid the water pipes and cables buried in the wall when drilling holes.
4. Install the equipment in a sheltered place to avoid direct sunlight, rain, and snow. Build a sunshade if it is needed.
5. The place to install the equipment shall be well-ventilated for heat radiation and large enough for operations.
6. The equipment with a high ingress protection rating can be installed indoors or outdoors. The temperature and humidity at the installation site should be within the appropriate range.
7. Install the equipment at a height that is convenient for operation and maintenance, electrical connections, and checking indicators and labels.
8. The altitude to install the inverter shall be lower than the maximum working altitude 4000m.
9. The PV modules used with the inverter must have an IEC61730 class A rating.
10. There should be provided an overcurrent protection (such as a breaker rated 400V a.c /200 A a.c) before AC input and after LOAD output, and make sure that the installation position shall not prevent access to the disconnection means.
11. Please ensure that there is adequate ventilation space for the inverter after installation, refer to the installation diagram below.
12. This inverter does not provide an internal isolated transformer between PV input and Battery / AC output circuits, But a basic insulation is provided between PV input / Battery / AC output circuits and metal enclosure / earth, and reinforced / double insulation between PV input / Battery / AC output and communication circuits (DRM / Meter / WiFi/ RS485)
13. The PV input ratings please refer to the specification table of subclause 3.3.5, and please make sure that PV array should not be grounded.
14. Install the equipment away from electromagnetic interference. If there are radio stations or wireless communication equipment below 30 MHz near the installation location, please install the equipment as follows:
 - Add a multi-turn winding ferrite core at the DC input line or AC output line of the inverter, or add a low-pass EMI filter.
 - The distance between the inverter and the wireless EMI equipment is more than 30m.



Installation Angle Requirements

- Install the inverter vertically or at a maximum back tilt of 15 degrees.
- Do not install the inverter upside down, forward tilt, back forward tilt, or horizontally.



Installation Tool Requirements

The following tools are recommended when installing the equipment. Use other auxiliary tools on site if necessary.

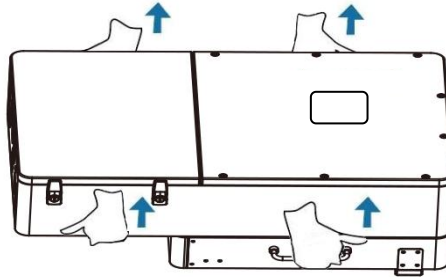


5.2 Inverter Installation

5.2.1 Moving the Inverter



•The unit is heavy. Do not lift it alone During lifting procedures ensure that the unit is firmly secured to avoid the risk of accidental tipping or dropping. Parts serving for support or immobilization of unit shall be designed and manufactured so as to minimize the risk of physical injuries and of accidental loosening of fixing. Ensure that the method of lifting will not allow the unit to slip from chains and slings or turn-over or slide from lifting devices.



transport

- Transportation must be carried by specialized person (truck operators. Hook-up personal), equipped with the necessary protection equipment(overalls, safe shoes, protective gloves, helmets, goggles)
- Do not walk or stand beneath or in the proximity of the load. Avoid sudden movements and jolts when unloading and positioning the unit, Internal handling procedures must be conducted with care.
- Do not exert leverage on the components of the machine. If the unit is not balanced apply ballast, Any protruding parts should not be supported by hand. The inverter should be installed so that the operating panel shall be easily accessible- easy access to the electrical power connection point.
- Accessible for maintenance and repair work. Parts serving for support or immobilization of unit shall be designed and manufactured so as to minimize the risk of physical injuries and accidental loosening of fixings.
- Loading capacity and hardness of the supporting surface, load rating of mounting bracket should be at least four times the weight of the devices according to IEC62109-1. And supporting characteristics will be impaired by wear, corrosion, material fatigue or ageing, This should be calculated by inspection of the design data of supporting material and consulting construction engineer.

5.2.2 Installing Steps



- Use screws or anchors suitable for the support surface and capable of supporting the weight of the inverter.
- Avoid the water pipes and cables buried in the wall when drilling holes.
- Wear goggles and a dust mask to prevent the dust from being inhaled or contacting eyes when drilling holes.
- Make sure the inverter is firmly installed in case of falling down.

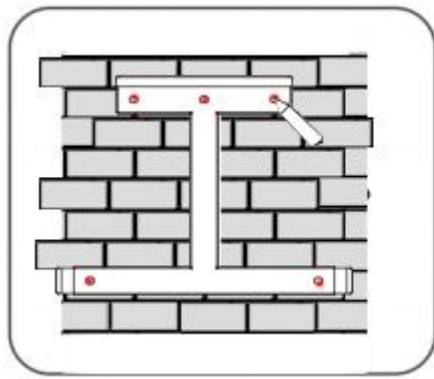
Step 1 : Install the mounting bracket flush against the wall and mark the positions of the holes.

Step 2 : Mark the positions of the holes accordingly. Then drill the holes at the marked positions.

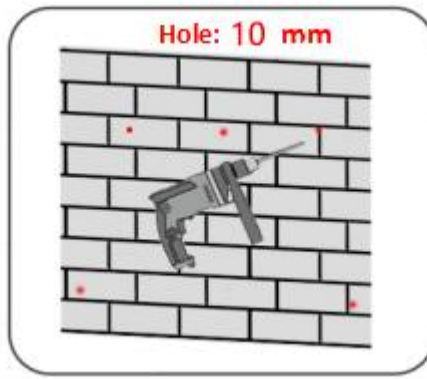
Step 3 : Fix the installation brackets on the wall according to the pre-drilled holes and tighten the screws.

Step 4 : Hang the equipment on the installation bracket.

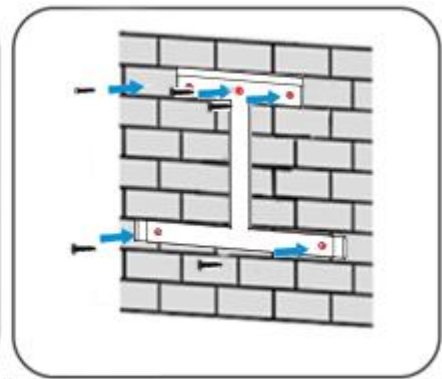
Step 5: After the equipment is hung up, lock the four screws at both ends.



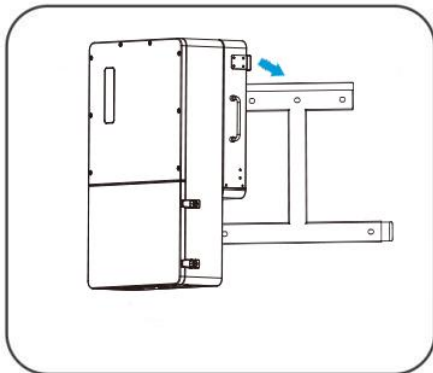
Step 1



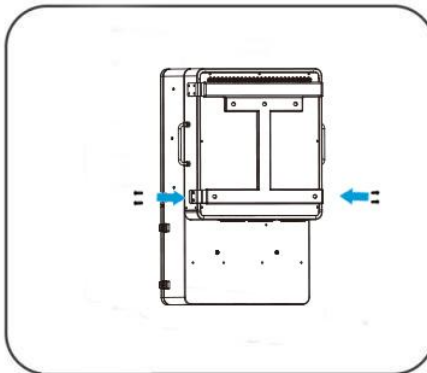
Step 2



Step 3



Step 4



Step 5

6. Electrical Connection

6.1 Safety Precaution



DANGER

- All operations, cables and parts specification during the electrical connection shall be in compliance with local laws and regulations.
- Disconnect the DC switch and the AC output switch of the inverter to power off the inverter at least 5 minutes for the capacitor to be electrically discharged before any electrical connections. Do not work with power on. Otherwise, an electric shock may occur.
- Tie the same type cables together, and place them separately from cables of different types. Do not place the cables entangled or crossed.
- If the cable bears too much tension, the connection may be poor. Reserve a certain length of the cable before connecting it to the inverter cable port.
- When crimping the terminals, ensure that the conductor part of the cable is in full contact with the terminals. Do not crimp the cable jacket with the terminal. Otherwise the inverter may not operate, or its terminal block getting damaged due to heating and other phenomenon because of unreliable connection after operation.



NOTICE

- Wear personal protective equipment like safety shoes, safety gloves, and insulating gloves during electrical connections.
- All electrical connections should be performed by qualified professionals.
- Cable colors in this document are for reference only. The cable specifications shall meet local laws and regulations.

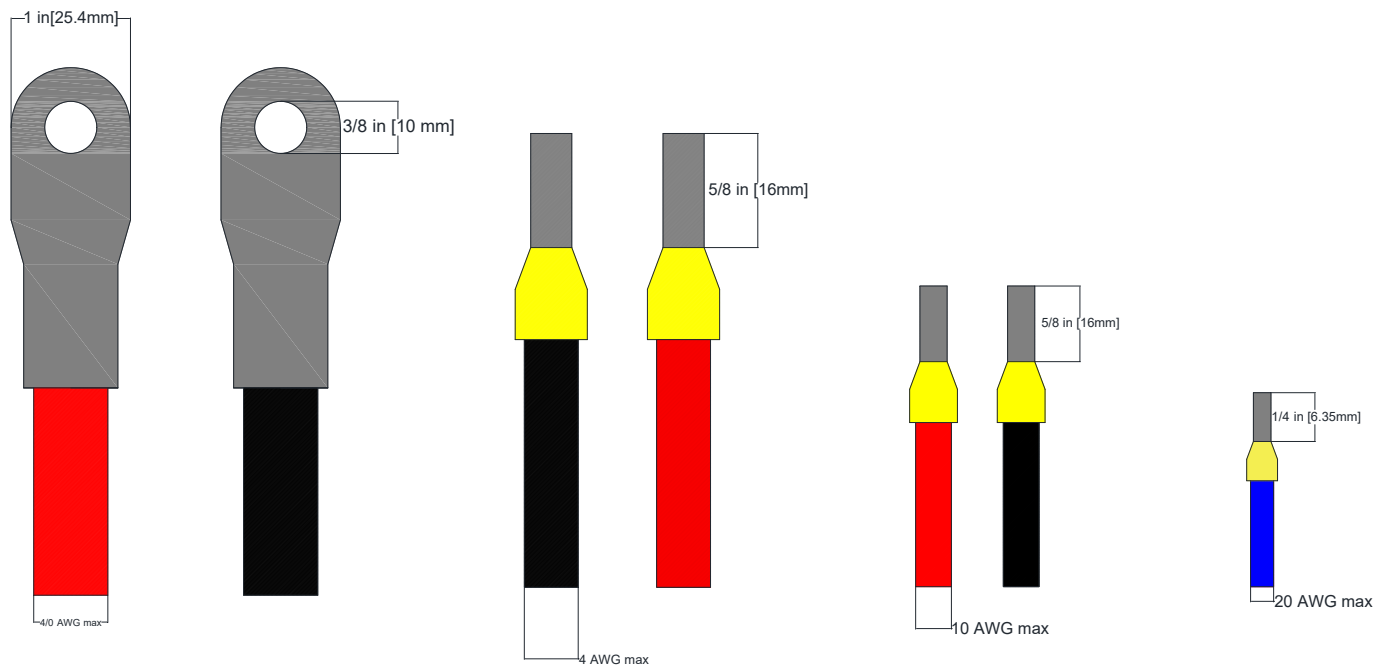
6.2 Connection Requirements

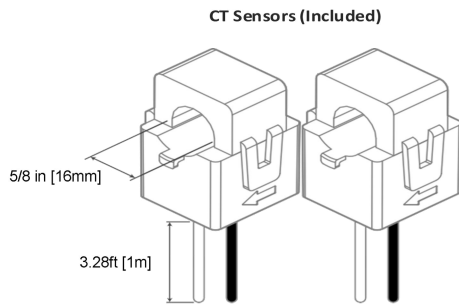
6.2.1 AC / DC Connection Requirements

Port	Terminal / Breaker Rating	Terminal Wire Size Range (min-max)
GRID	200A AC(15kW)	2/0 – 4/0 AWG
LOAD	200A AC(15kW)	2/0 – 4/0 AWG
GEN	90A AC	6 – 4 AWG
MPPT	27A DC	12 – 10 AWG
Battery Port	200A DC×2	2/0 – 4/0 AWG
I Load	90A AC	6 – 4 AWG

6.2.2 Sensors and Communications Requirements

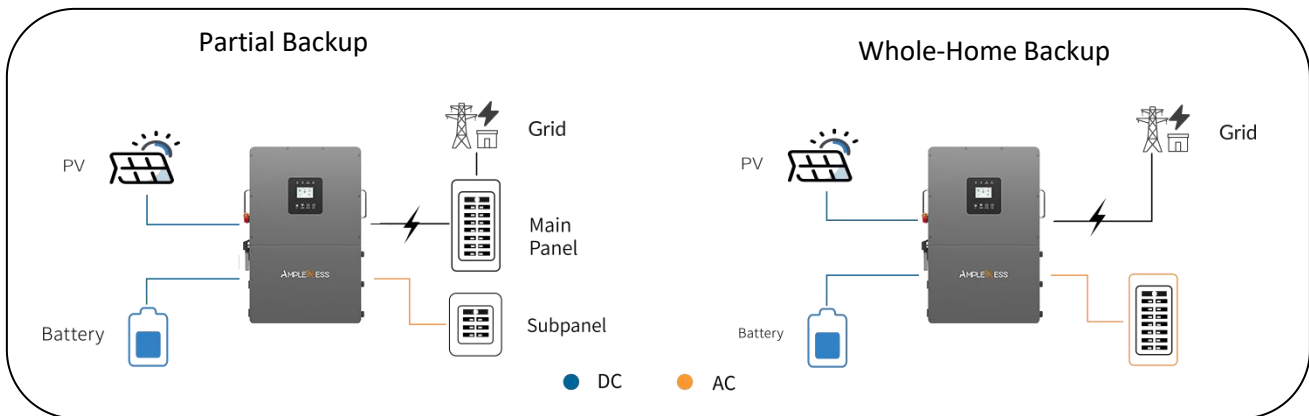
Component	Wire Size Range	Max Distance
CT Sensor	16-20 AWG	0' – 13' [4 m]: 16 AWG included 13' – 20' [6 m]: CAT6 extendable
Communications	24 – 23 AWG	0' – 100' [30 m]: 24 AWG 100' – 400' [120 m]: 23 AWG
RJ45 Parallel Communication	CAT 5E or better	0' – 7' [2.1 m]: Included 7' – 20' [6m]: Extendable





6.3 System Integration

The system connection diagram is as follows (US. version):



Single System Installation Guide

For Partial Backup (Essential Loads Only):

Connect the output of the back-feed breaker or line-side tap (depending on your interconnection method) to the “GRID” terminal of the inverter.

Install an external disconnect switch between the point of interconnection and the inverter. The disconnect must be properly sized according to local electrical code.

Connect the “LOAD” terminal of the inverter to the Essential Loads Panel. Select the appropriate wire gauge in compliance with electrical code.

For Whole-Business Backup (Entire Load):

Connect the utility grid directly to the inverter’s “GRID” terminal.

Install an external disconnect switch between the grid and the inverter. Size the disconnect according to code requirements.

Connect the “LOAD” terminal to the Main Service Panel, ensuring correct wire sizing as per code.

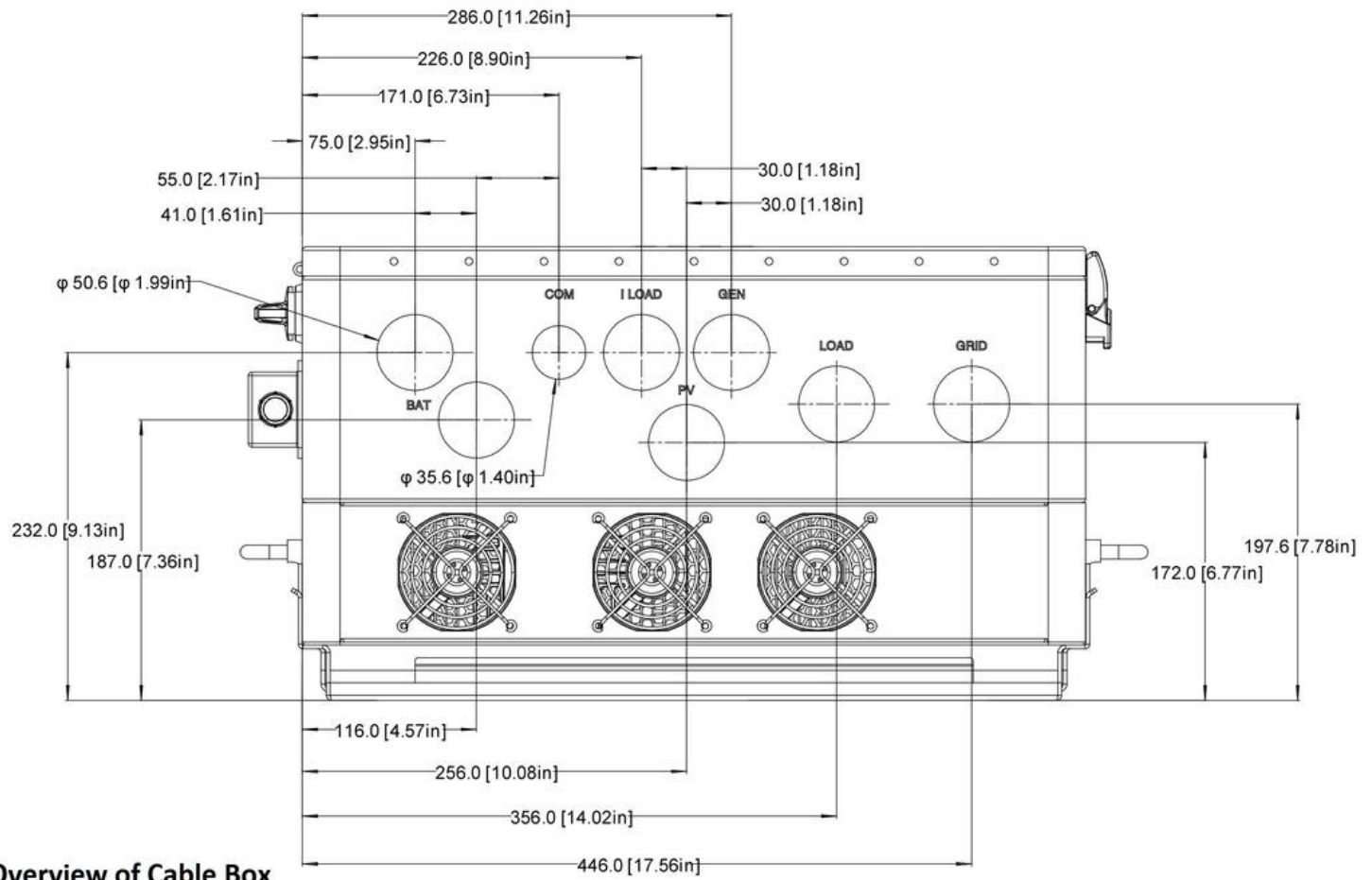
AC Generator or AC-Coupled Source (Optional):

A generator or AC-coupled power source (e.g., string inverter or microinverter) can be connected to the “GEN” terminal.

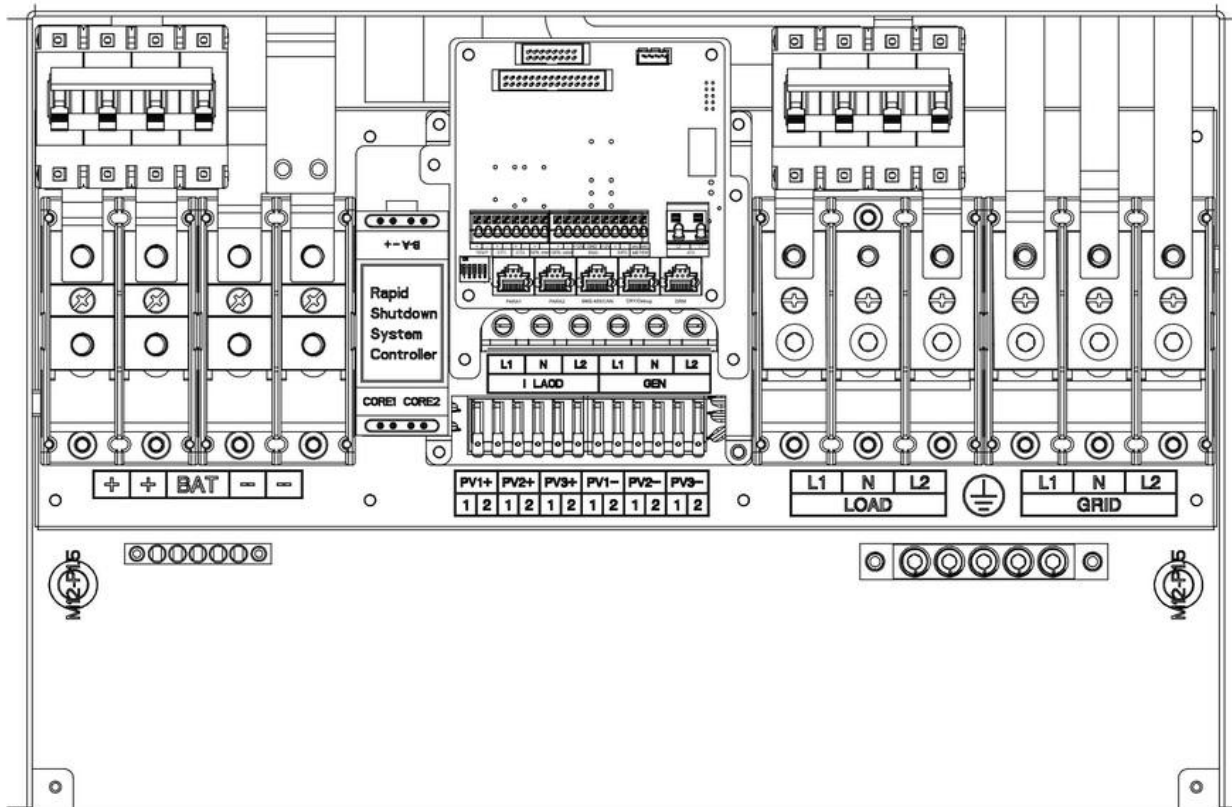
The maximum input for the GEN terminal is 63A (15kW).

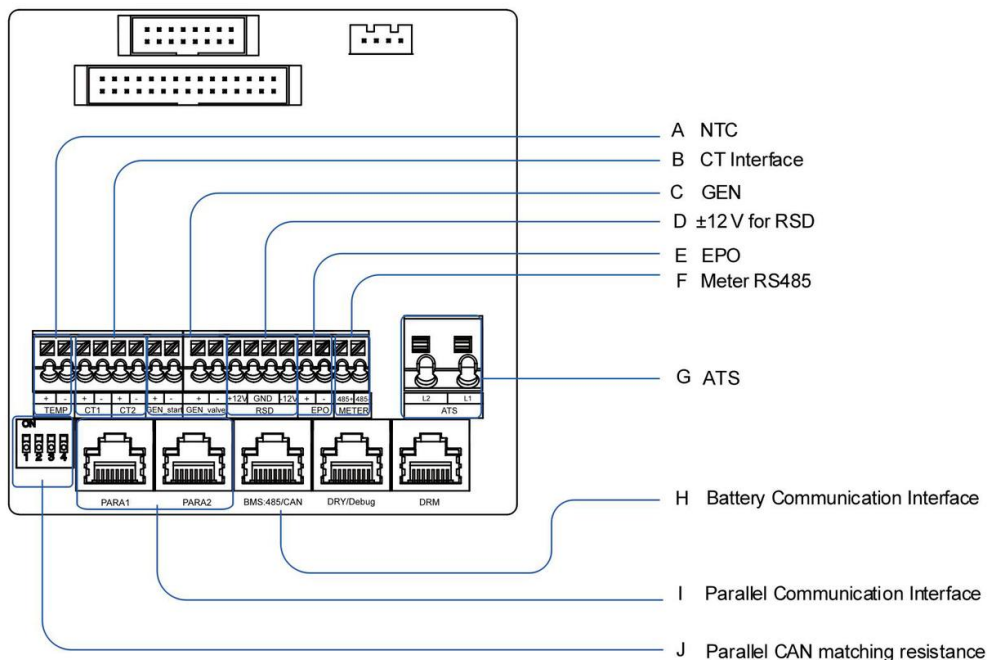
Only one AC source may be connected to the GEN terminal at a time.

Bottom View of Cable Box Knockouts



Overview of Cable Box



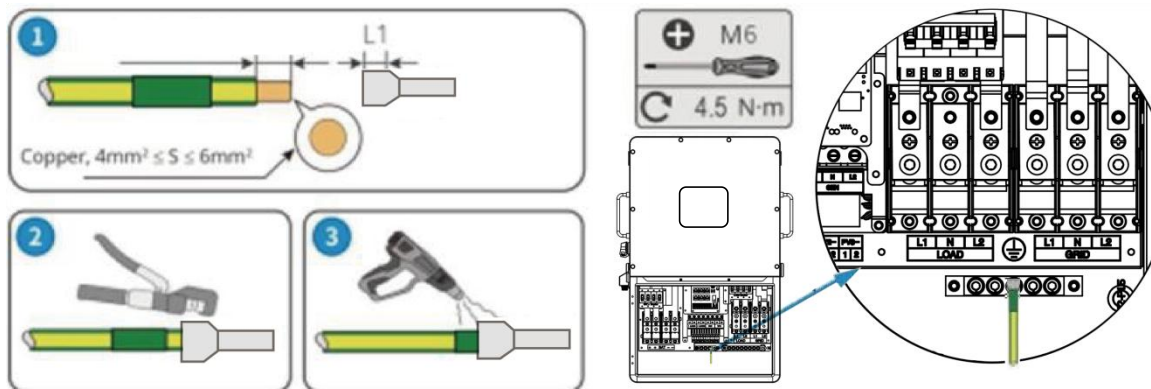


- A. NTC: Connection for temperature sensor of Lead acid battery.
- B. CT Interface: please check Chapter 6.7.4 for Pin definition.
- C. GEN(NO): Connection for generator auto-start function. dry contact signal for startup the diesel generator. When the "GEN signal" is active, the open contact (GS) will switch on (no voltage output).
- D. $\pm 12V$ for RSD : Power supply for RSD PLC transmitter(Max current 500mA).
- E. EPO : Reserved for external EPO.
- F. Meter RS485 :for meter communication.
- G. ATS: 240V output port when inverter is on.
- H. Battery Communication port (CAN&485) : please check Chapter 6.6.2 for Pin definition.
- I. Parallel Communication port (CAN&485) : please check Chapter 6.9.1 for Pin definition.
- J. Parallel CAN matching resistance: Set DIP Switch when use inverters in Parallel.

6.4 PE Cable Connection



- The PE cable connected to the enclosure of the inverter cannot replace the PE cable connected to the AC output port. Both of the two PE cables must be securely connected
- Make sure that all the grounding points on the enclosures are equipotential connected when there are multiple inverters.
- To improve the corrosion resistance of the terminal, it is recommended to apply silica gel or paint on the ground terminal after installing the PE cable.
- Prepare PE cables with the recommended specification:
 - Type: Outdoor single-core copper wire
 - Cross-sectional area: 6mm^2 (10AWG)



6.5 PV Connection

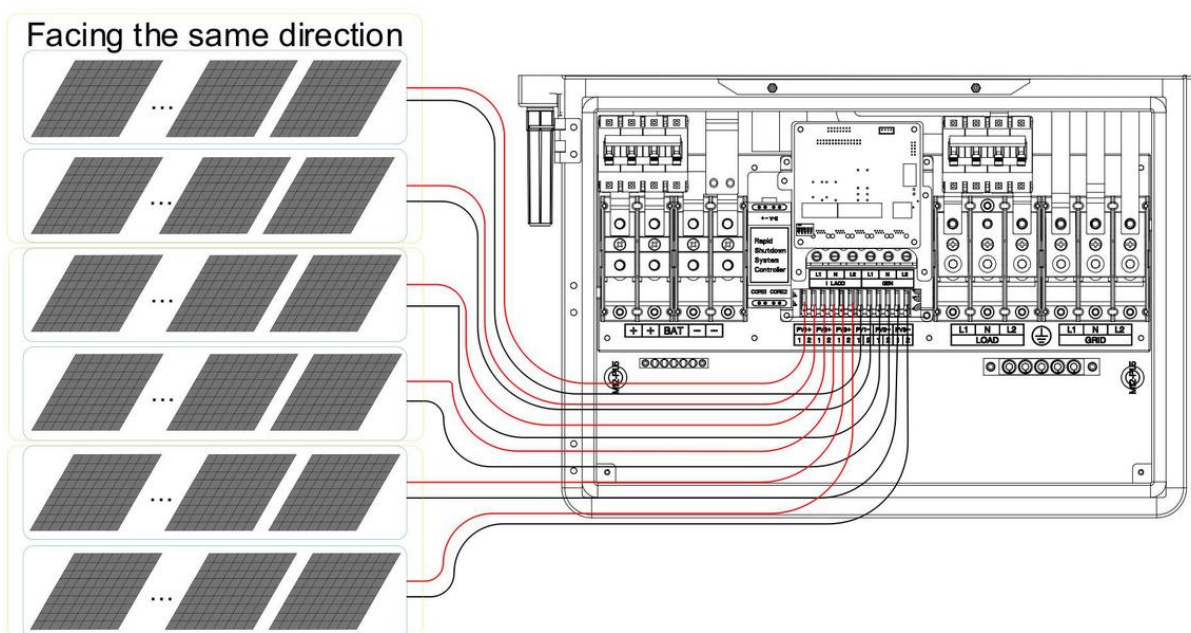
The photovoltaic connection of this hybrid inverter is the same as that of a traditional grid-tied solar inverter (string inverter).

WARNING

Please double-check the minimum ambient temperature at the installation location. The thermal Voc on the solar panel nameplate is measured at 25°C. As the ambient temperature decreases, the solar panel Voc increases. Ensure that the maximum solar string voltage, corrected for the lowest temperature, does not exceed the inverter's maximum input voltage of 525V.

Note

- The inverter has three MPPTs. For each MPPT, the user can connect two strings of solar panels.
- When the user connects two strings to the same MPPT, ensure that the number of solar panels in each string is the same, and that the installation location and orientation are as consistent as possible. The inverter will automatically limit the total input current of each MPPT to 26A.
- The inverter limits the maximum solar input power to 22.5kW.
- Strings in parallel on the same MPPT must have the same designed open-circuit voltage (Voc), otherwise the system will be limited to the lowest string voltage.
- If the solar panels are oriented in different directions and connected in the same MPPT, there will be a loss in PV efficiency.
- For ground mounted arrays, KOOR recommends installing an auxiliary grounding electrode placed near the array to ensure optimal earth-to-ground resistance of the grounding system. This auxiliary electrode would need to follow the Standard specification.
- Connect the solar panel string using the following configuration:



PV Connection Steps

- a. Strip 1/4-5/16 inch (6-8mm) of insulation from the positive and negative conductors of the PV string.
- b. If the PV string wires are stranded, use wire ferrules.
- c. Insert the conduit connector into the PV connection port.
- d. Pass the photovoltaic conductors through the conduit fittings and insert them into the inverter.
- e. Confirm that the cable connections are correct and secure. Then take appropriate measures to ensure that the

conduit and conduit fittings are properly tightened and sealed into the cable hole.

6.6 Battery Connection

Battery Connection Guidelines for KR-L8-15K-SPUS

Important: Ensure the KR-L8-15K-SPUS inverter is turned OFF before connecting any batteries.

Connect your battery bank according to the appropriate configuration based on the battery voltage (see wiring diagrams below if applicable).

Battery breakers must be OFF during wiring.

If your battery bank does not have built-

in breakers, take all necessary safety precautions when handling exposed terminals or cables.

The maximum battery charge/discharge current is 275A when both sets of battery terminals are used.

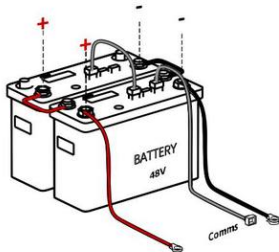
If you are using only one set of battery terminals, the charge/discharge current will be limited accordingly.

This inverter is designed for a 48VDC nominal system.

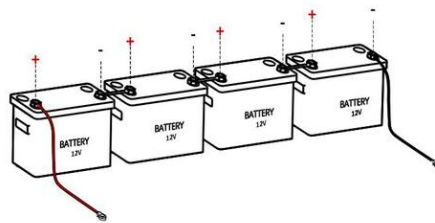
❌ DO NOT connect it to battery configurations outside the 40–60VDC range.

❌ If you are using 12V batteries, you must not exceed 4 batteries in series (i.e., max 48V nominal).

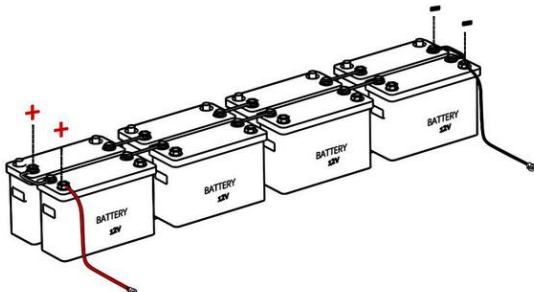
The inverter supports various battery chemistries, provided the voltage remains within 40VDC to 60VDC. Depending on the battery voltage, wire up the battery bank in the possible configurations shown below.



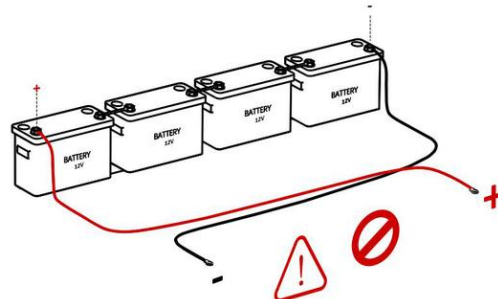
48V batteries in parallel connection



12V batteries in series connection



Series and parallel connections for complete 48V battery bank



DO NOT reverse polarity. The system will be damaged, and warranty voided!

6.6.1 Battery Power Cable Connection

Procedure for connecting battery terminals:

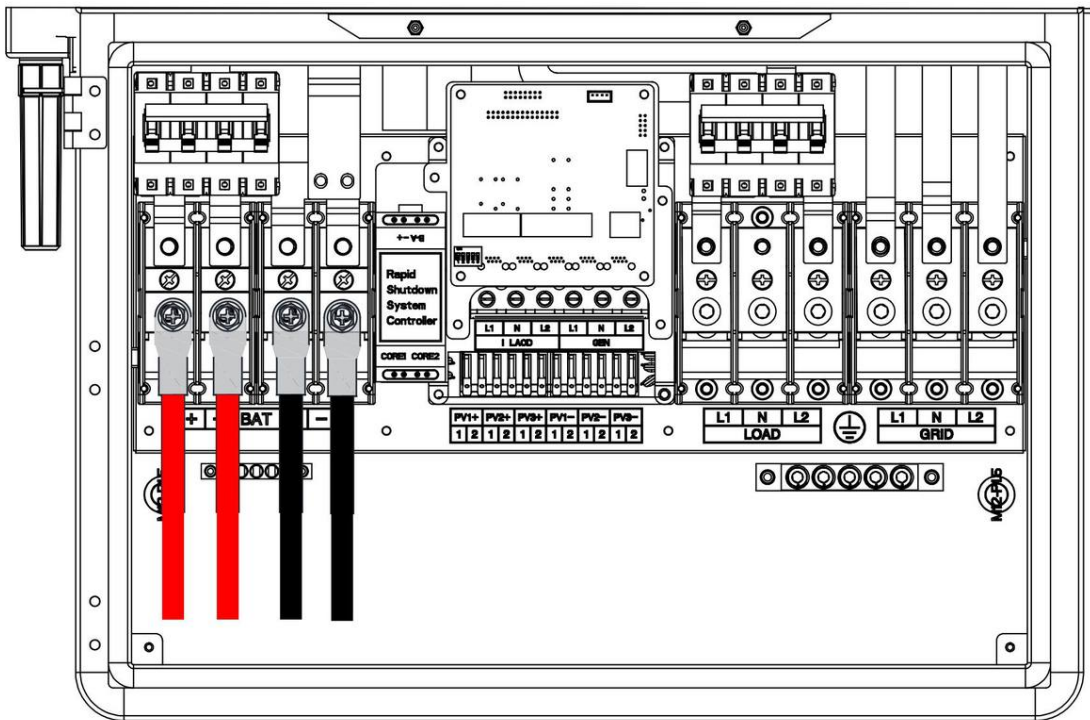
Step 1: Strip 1/4 ~ 5/16 inch (6 ~ 8mm) of insulation from the cable ends and crimp the OT rings.

Step 2: Lay the battery power cables, connecting the positive terminal to BAT+ and the negative terminal to BAT-. Step 3: Use the reverse nut to secure the conduit fittings to the enclosure.

Step 4: Secure the OT rings of the battery positive and negative cables onto the terminal block according to the markings.

Multi-Terminal Installation

The two battery input terminals of the KR-L8-15K-SPUS will parallel batteries internally to ensure a common connection between battery banks and simplify battery installations. If a charge / discharge rate of 275A is needed, the batteries must be connected to both input terminals. If you use 3 or more batteries, use external busbars for (+) and (-) connections.



**WARNING**

Multiple inverters in parallel system

All parallel inverters must connect to a single battery bank. Otherwise, the system will NOT operate properly.

DO NOT use separate battery banks in parallel systems.

Follow all battery manufacturer-specified values to ensure proper charging and discharging.

BATTERY CHARGING SETPOINT EXAMPLES (48V NOMINAL)

Battery Type	Absorption	Float	Equalize (Every 30 days)
AGM	57.6V	53.6V	57.6V
Gel	56.4V	54.0V	
Lithium	54.6V	54.3V	-

CALCULATING BATTERY BANK AMP-HOURS EXAMPLES (AGM)

# of batteries	Voltage per Battery	Ah per battery	Ah @48V(Chain of 4 batteries of 12V in series)	Max Charge / Discharge (A)
4	12V	230Ah	230Ah	100A
8	12V	230Ah	460Ah	200A
12	12V	230Ah	690Ah	275A
16	12V	230Ah	920Ah	275A

Please refer to Chapter **9.7 Battery Settings Menu** for the operation method of the screen.

**WARNING**

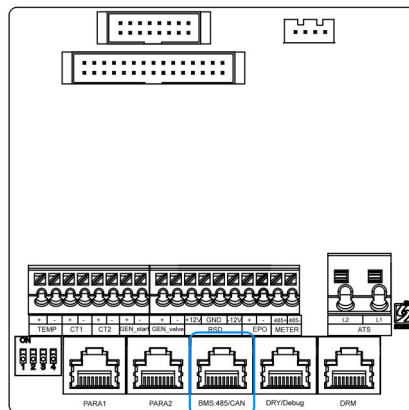
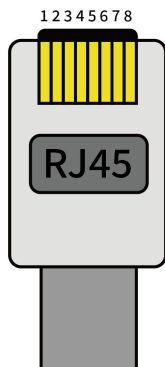
Reverse polarity will damage the inverter!

Before connecting or disconnecting the battery, ensure that the built-in battery circuit breaker is in the "off" position.

6.6.2 Battery Communication Cable Connection

When the user selects the lithium-ion battery type, the correct battery communication cable must be used to connect the battery to the inverter. If the lithium battery cannot communicate with the inverter, please select the "lead-acid" type. The inverter's battery communication port is an RJ45 socket, and the pins of the RJ45 connector are as follows. Create the communication cable according to the correct pins of the inverter and the battery communication port. The inverter supports both CAN and RS485 communication.

Pin	Description
1	BAT RS485 A
2	BAT RS485 B
3	NC
4	BAT CAN H
5	BAT CAN L
6	NC
7	BAT RS485 B
8	BAT RS485 A



After the battery socket and communication socket are connected, you need to enter the advanced settings in the inverter's app to change the battery type and brand. After selecting the appropriate battery protocol, communication will be established within 1-2 minutes.

WARNING

Any damage caused by the improper use of the communication protocols (CANBUS or MODBUS) will not be covered by warranty. Modbus map is available upon request for "READ" operations only. Contact Fantastech to obtain the MODBUS map.

6.7 Grid& EPS Load Connection

6.7.1 Grid Type and Regulation Selection

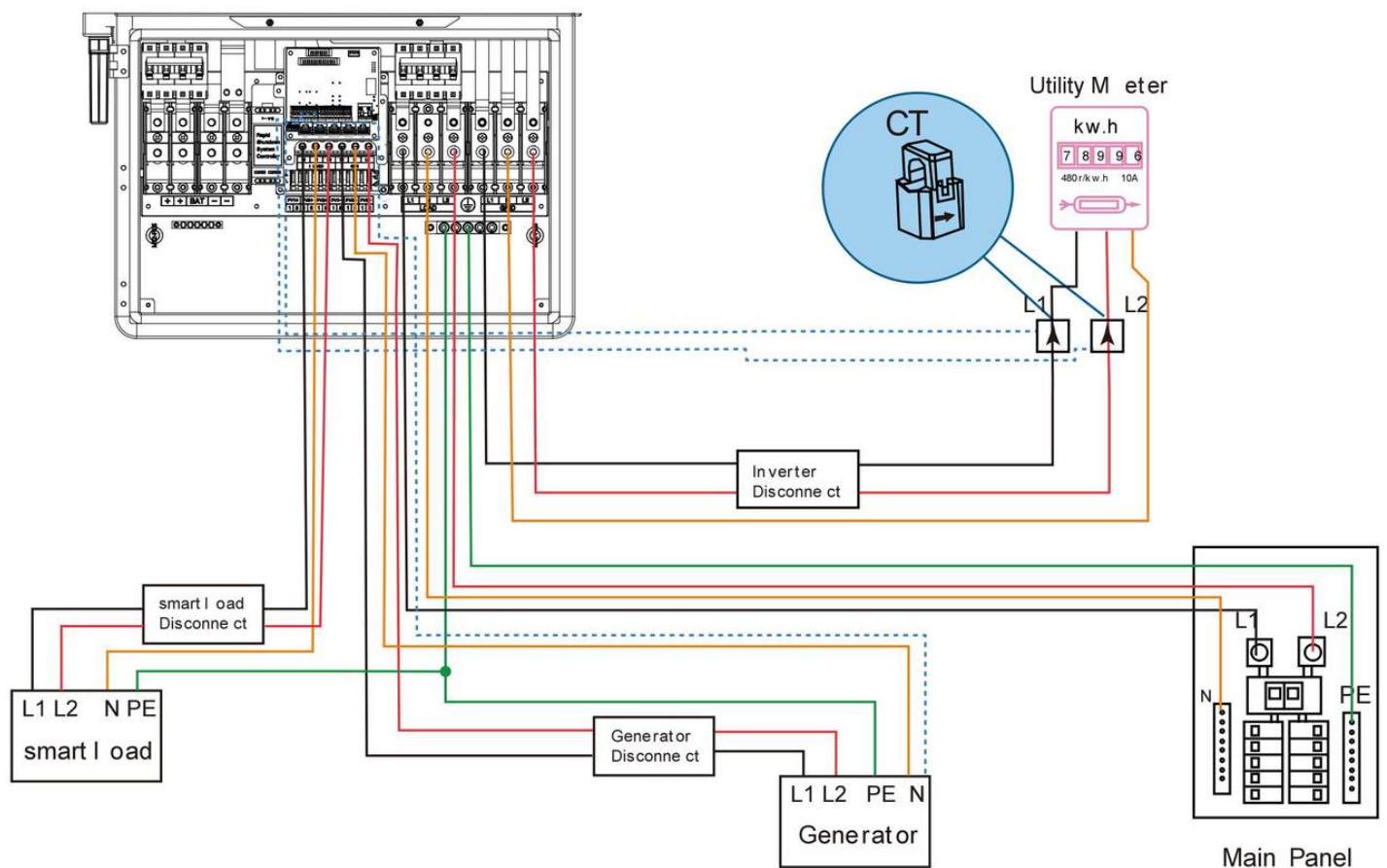
This inverter can be used with 120/240V split-phase or 120/208V split-phase systems. If connecting to the grid, ensure that the grid settings are configured correctly. The user can select different grid types and regulations in the advanced settings of the app, as shown in the following diagram.

6.7.2 Split-phase Grid and EPS Load Connection

The wiring diagram for 120/240V is shown below: The wiring diagram for the 120/208V split-phase service is largely the same, except it does not support a generator.

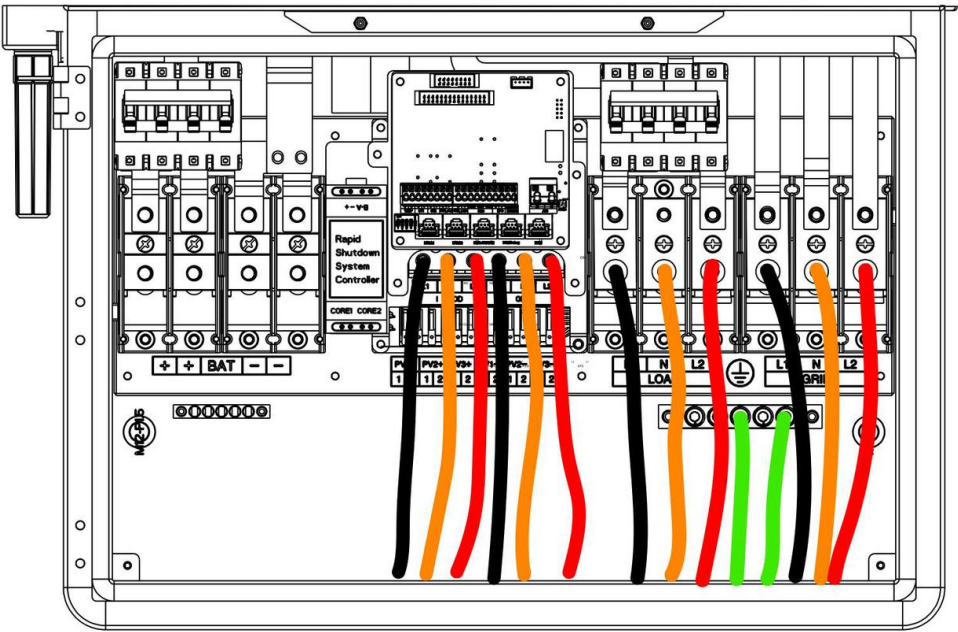
When the main panel bus rating meets the NEC705.12(B)(3) requirements, the inverter can be connected to the load side of the service disconnecting device. Otherwise, a line-side connection can be made to avoid the costly upgrade of the main panel.

Grid and EPS Load Connection



6.7.3 AC Cable Connection

- Strip 5/16-3/8 inch (8-10mm) of insulation from the cable.
- a. If the cable is stranded, use a cable sheath.
 - b. Secure the conduit fitting to the enclosure using the top nut of the fitting.
 - c. Secure the grid line and EPS load line onto the terminal block according to the markings.
 - d. Fix the conduit onto the conduit fitting.
 - e. Check that the cable connections are correct and secure. Take appropriate measures to ensure that the conduit and conduit fittings are properly tightened and sealed into the cable entry hole.



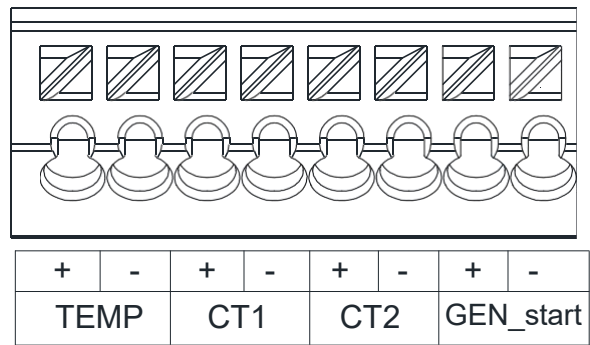
 **WARNING**

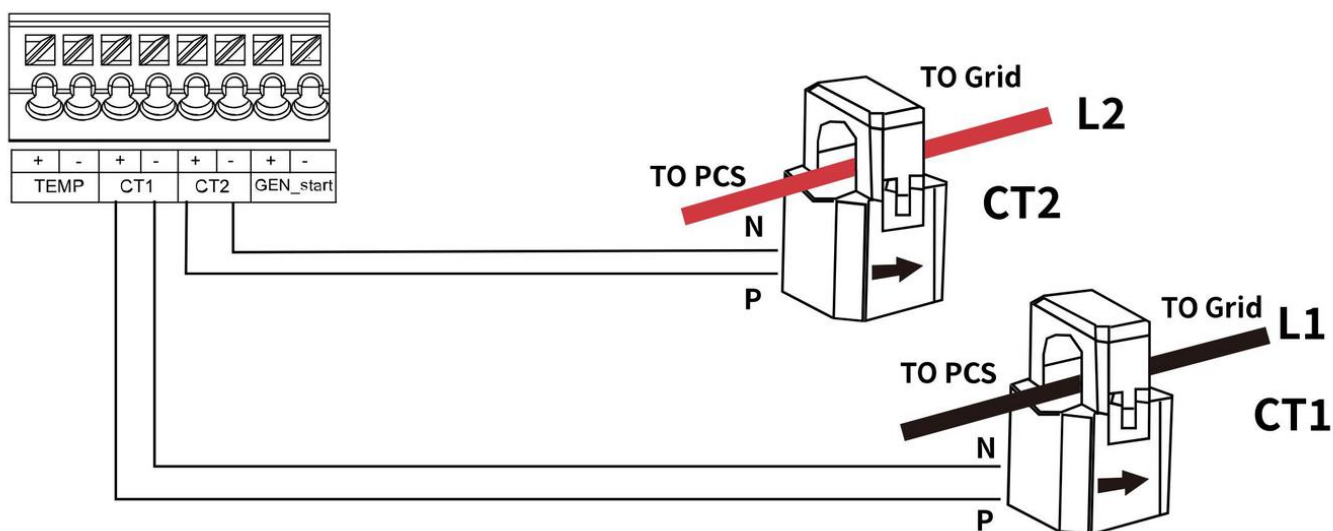
Before supplying power to the EPS load, ensure that the built-in load circuit breaker is in the "ON" position.

6.7.4 CT / Energy Meter Connection

To measure the input and output grid power, a pair of CTs or a three-phase meter must be installed at the service entrance point, either inside or near the main service panel. We standardly supply 2 CTs for one inverter.

CT Interface Pin Definition





Please refer to the wiring diagram to determine the correct placement of the CTs. Attach the two CTs to the L1 and L2 lines at the service entrance point of the main service panel.

- **CT1** (labeled L1) should be placed on **L1**.
- **CT2** (labeled L2) should be placed on **L2**.

⚠ WARNING

- Install sensors on incoming electrical service wires (L1, L2, and L3 if the system is 3Φ).
- Embossed arrows on the sensors must point toward the grid.
- If the system is 3Φ, the arrows must point toward the inverter(s).

(Incorrect installation of the CTs may result in incorrect readings on the display and may prevent the inverter from functioning properly.) If the CT is installed in the wrong direction, there is an option to reverse the direction in the inverter settings: **CT Direction Reversal** (only applicable if the CT1 or CT2 is not installed correctly). This can be done in the **Advanced Settings** tab. However, you should not need to change this if the CTs are installed correctly.

CT Clamping Ratio

CT (Current Transformer) Compatibility and Configuration

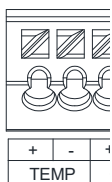
The inverter supports a standard CT ratio of 2000:1.

The CT included in the accessory package has a 2000:1 ratio and is pre-calibrated for use with the system.

If using a third-party CT, ensure that its ratio is either 1000:1, or 2000:1

⚠ Be sure to set the correct CT ratio in the inverter's monitoring interface or LCD display, depending on the CT used. Incorrect CT ratio settings may result in inaccurate power measurements or system misbehavior.

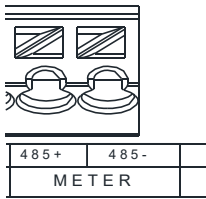
Temperature Sensor



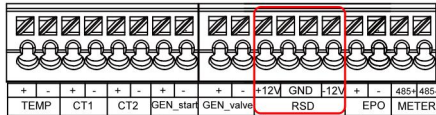
- This temperature sensor terminal is required when using lead-acid batteries.
- Place the sensor between two batteries as shown.
- Secure with tape and place away from the battery terminals to prevent overheating.
- This sensor has no polarity. The temperature sensor helps perform voltage charging adjustments and capacity calculations due to changes in temperature.
- Lithium Batteries DO NOT require our external temperature sensor.

Energy Meter Connection

If you need to use an energy meter for import/export monitoring instead of using CTs, it should be connected to the **485A** and **485B** terminals on the inverter for communication. These terminals support RS485



6.7.5 Emergency Stop and Rapid Shutdown Connection



Emergency Stop (E-Stop) & RSD Connection Guidelines

The Emergency Stop (E-Stop) *button must be* connected to the designated control pins on the inverter.

The Rapid Shutdown (RSD) transmitter should be connected to the corresponding RSD terminal, which supplies 12/24V DC power.

The maximum current output from the RSD terminal is 500mA.

Ensure that the connected RSD transmitter or device does not exceed this limit.

The emergency stop (E-Stop) pins on the KR-L8-15K-SPUS are a normally open (NO) contact. When the circuit is closed, it will trigger the Rapid Shutdown (RSD) function.

The KR-L8-15K-SPUS provides two power terminals for RSD (Rapid Shutdown) transmitter connection: +12V – 12V

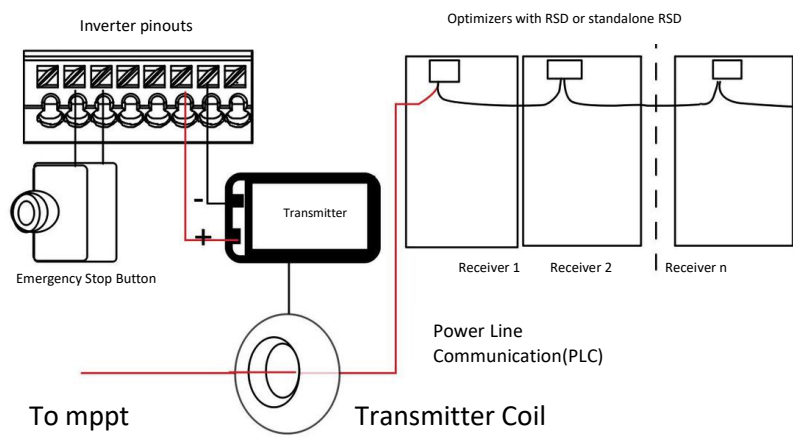
For 12V RSD Transmitters:

Connect the transmitter to: +12V and GND

For 24V RSD Transmitters, connect the transmitter to: +12V and – 12V

⚠ Make sure the RSD transmitter's current consumption does not exceed 500mA, which is the maximum output supported by the inverter's RSD terminal.

There is a way to set the transmitter to achieve fast shutdown, taking care to avoid the transmitter exceeding the current provided by the RSD terminal, the wiring diagram is provided below for user reference.



NOTICE

The Built-in 12Vdc power supply of the KR-L8-15K-SPUS is rated for 500mA . Do not exceed!

If you are unsure of the current (A) rating of the transmitter, contact the manufacturer before connecting.

6.8 Use a Generator

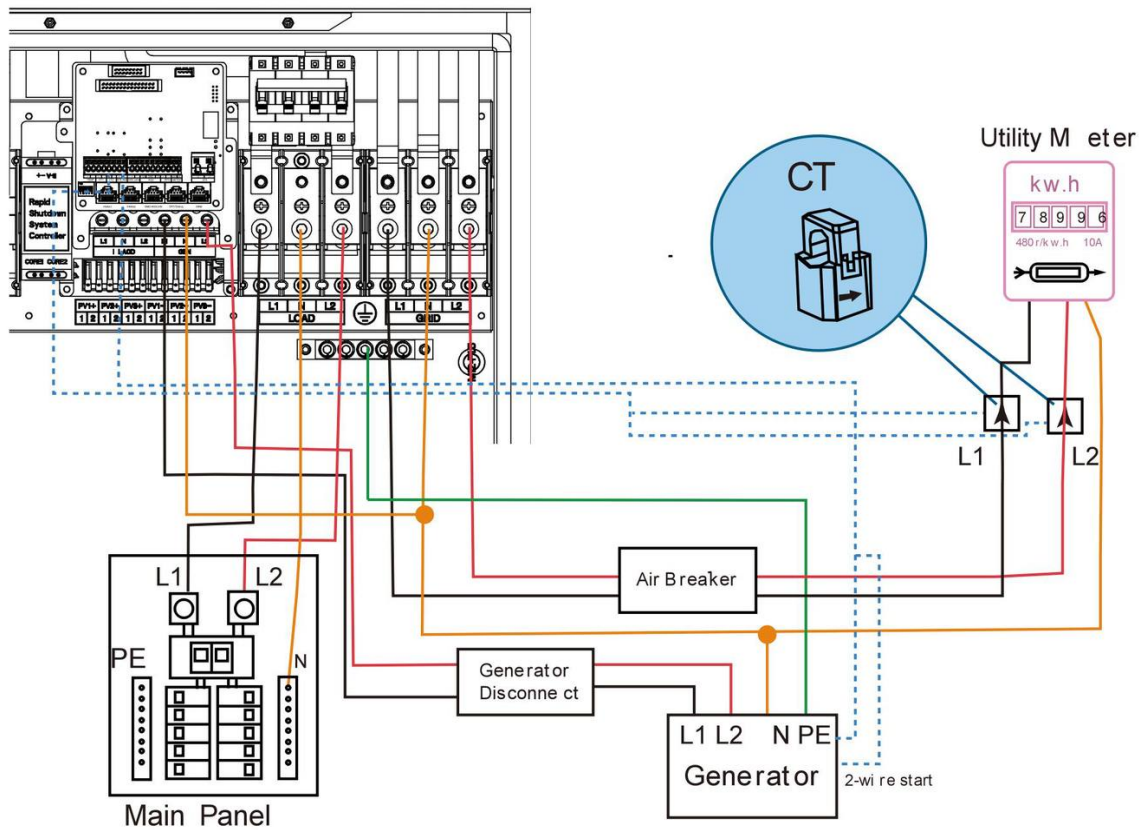
6.8.1 Generator System Connection

This hybrid inverter can be used with a generator. The inverter has a **Gen** port for connecting the generator.

Stator Requirements:

- The generator should use a **neutral-grounding connection** and provide both **240V/120V output**. The generator's capacity may be greater than the current system capacity.
- A THD (Total Harmonic Distortion) of less than 15% is preferred.
- Make sure the generator is correctly connected and meets the voltage and capacity requirements to ensure proper operation with the inverter. Always follow the manufacturer's guidelines for installation and connection.
- DO NOT use "Grid Sell" in Off-Grid systems. Potential to damage the generator.

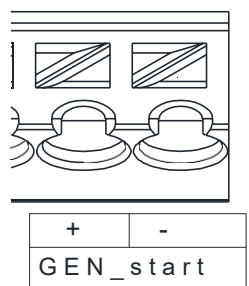
Generator Connection-Whole Home Backup



When the generator starts, all loads connected to the EPS will be powered by the generator, and the battery will be charged simultaneously.

The AC circuit breaker on the **Gen** port is rated for 63A. When the generator is running, ensure that the total load and charging current do not exceed 63A.

If the user requires remote starting of the generator, the generator start signal should be connected to the **GEN_start** port on the I3 board.



6.8.2 Generator Start and Stop Settings

These conditions depend on the **Battery (Bat) Operation Mode** settings. The system uses the **battery's State of Charge (SOC)** or **battery voltage** to determine whether the generator needs to start or stop.

Generator Start Conditions:

The generator will start under the following circumstances:

Power failure: When there is a loss of grid power or other power sources.

Battery discharge cutoff: When the battery discharges to the cutoff voltage or SOC setting.

Forced charging requirement: If there is a need for forced charging of the battery.

Battery voltage or SOC below the generator start threshold: If the battery voltage or SOC falls below the set **generator charging start voltage/SOC**.

Generator Stop Conditions:

The generator will stop under the following conditions:

When the **battery voltage** or **SOC** is higher than the **charging endpoint voltage/SOC** setting, meaning the battery is sufficiently charged and the generator is no longer needed.

By using these settings, the system ensures the generator only operates when necessary, optimizing battery performance and reducing unnecessary generator run time.

Please refer to Chapter 9.9 **Generator Port Settings Menu** for the operation method of the screen.

APP Setting:

<

Generator Control

Save

Generator Setting

Generator

* Gen Warm Time(0-1000S)

10

* Start Gen SOC(10-100%)

25

* Stop Gen SOC(10-100%)

95

* Gen Max Using Power(0.0-800.0kW)

8

* Gen Max Running Time(0-1440min)

600

6.8.3 AC Coupling Installation and Connection

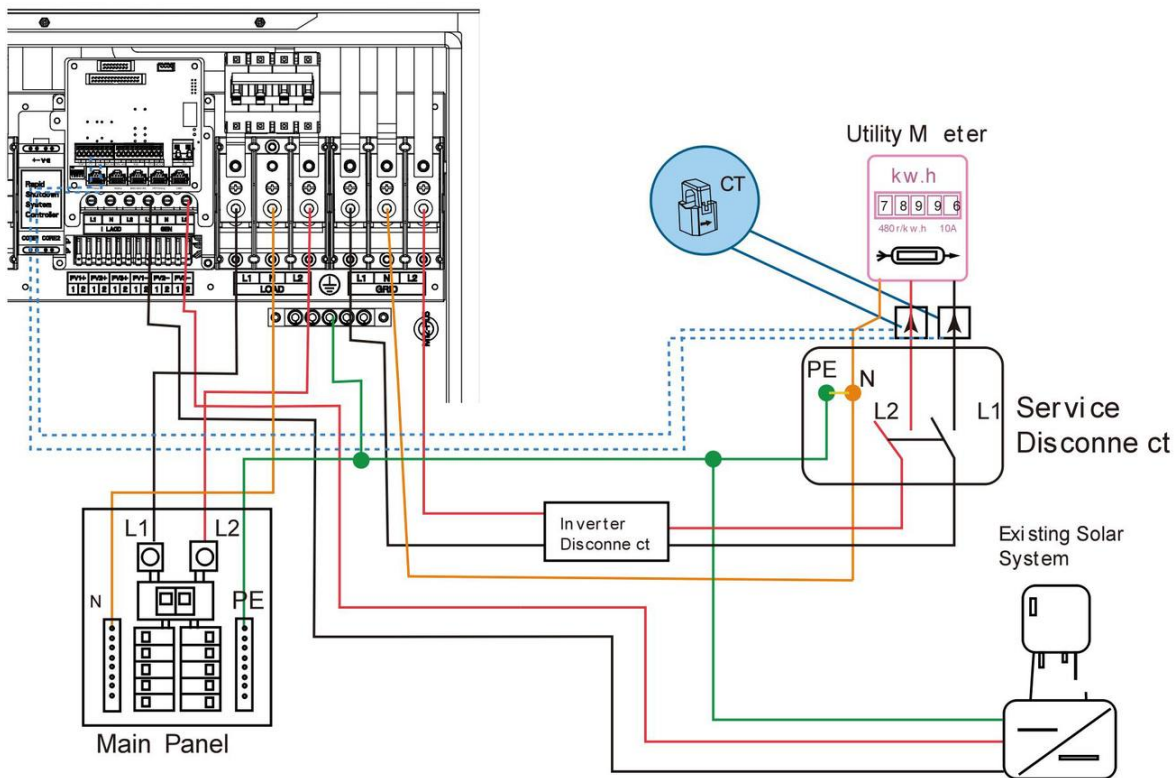
The KR-L8-15K-SPUS is a system that supports the addition of AC coupled solar panels. The max solar input power can be expanded by coupling micro or string inverters into the "GEN" or "LOAD" terminals. A full AC coupled solar system is not recommended as power control and monitoring is limited. Having DC coupled modules or a combination of DC coupled and AC coupled solar panels is always preferred.

Inverters that can be used for AC coupling should meet relevant industry specifications. This certification confirms the inverters' ability to disconnect from the grid based on frequency and ensures that the KR-L8-15K-SPUS will safely be able to frequency shift to control the AC coupled production.

AC coupling to "GEN" or "LOAD"

- Can produce solar power during grid outages.
- Can produce solar power for Off-Grid systems.
- Can monitor solar production.
- The "GEN" input cannot be used when AC is coupled to "LOAD". In addition, the backup power conversion time is extended.

Connection for AC coupled solar system



When the "Grid" is in the **ON** state, the **GEN** terminal connects to the internal "**Grid**" terminal of the inverter. In this case, the hybrid inverter will bypass the interactive inverter grid and EPS (Emergency Power Supply) system.

When the grid is disconnected, the **GEN** terminal connects to the inverter's internal **EPS** terminal. In this situation, the loads will be powered first by solar energy. If the solar power generated exceeds the load consumption, the excess energy will be stored in the battery.

When the solar power exceeds the sum of load power and the battery's maximum charging power (for example, when the battery is close to full), the inverter will signal the interactive inverter to reduce its power output using a **frequency shifting power reduction mechanism**. This helps maintain the balance between power generation and consumption within the microgrid system, ensuring that the system operates efficiently while preventing overproduction.

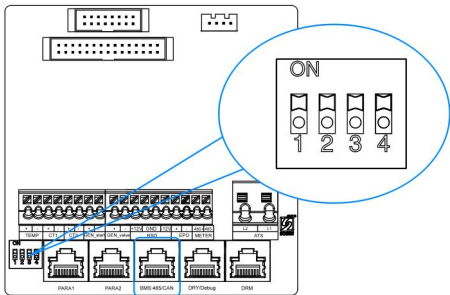
6.9 System Parallel Operation

6.9.1 Parallel System Connection

NOTICE

- The hybrid inverter supports parallel operation, allowing for scalable power and energy capacity to meet various usage scenarios. A total of up to 6 units can be connected in parallel, achieving a capacity of up to 90kW.
- Make sure all units in parallel have the same software version by verifying the “MCU” numbers on System Setup.
- Parallel systems REQUIRE a joint battery bank.
- All INPUTS/OUTPUTS must be shared among ALL parallel inverters, except for DC solar inputs.

DIP Switch Configuration for Parallel Systems



Inverter1(Master)	Inverter2	Inverter3	Inverter4	Inverter5	Inverter6
OFF					
ON	ON				
ON	OFF	ON			
ON	OFF	OFF	ON		
ON	OFF	OFF	OFF	ON	
ON	OFF	OFF	OFF	OFF	ON

Please refer to Chapter 9.10 **Advanced Settings Menu** for the operation method of the screen.

Parallel Systems in APP Setting

Master

Slave 1

Slave 2

<

Parallel Setting

Save

Parallel Type

Parallel Master

Parallel Number

3

Parallel Address

1

Shared Battery

Combined Three Phase

Local Phase

Phase A

<

Parallel Setting

Save

Parallel Type

Parallel Slave

Parallel Number

3

Parallel Address

2

Shared Battery

Combined Three Phase

Local Phase

Phase A

<

Parallel Setting

Save

Parallel Type

Parallel Slave

Parallel Number

3

Parallel Address

3

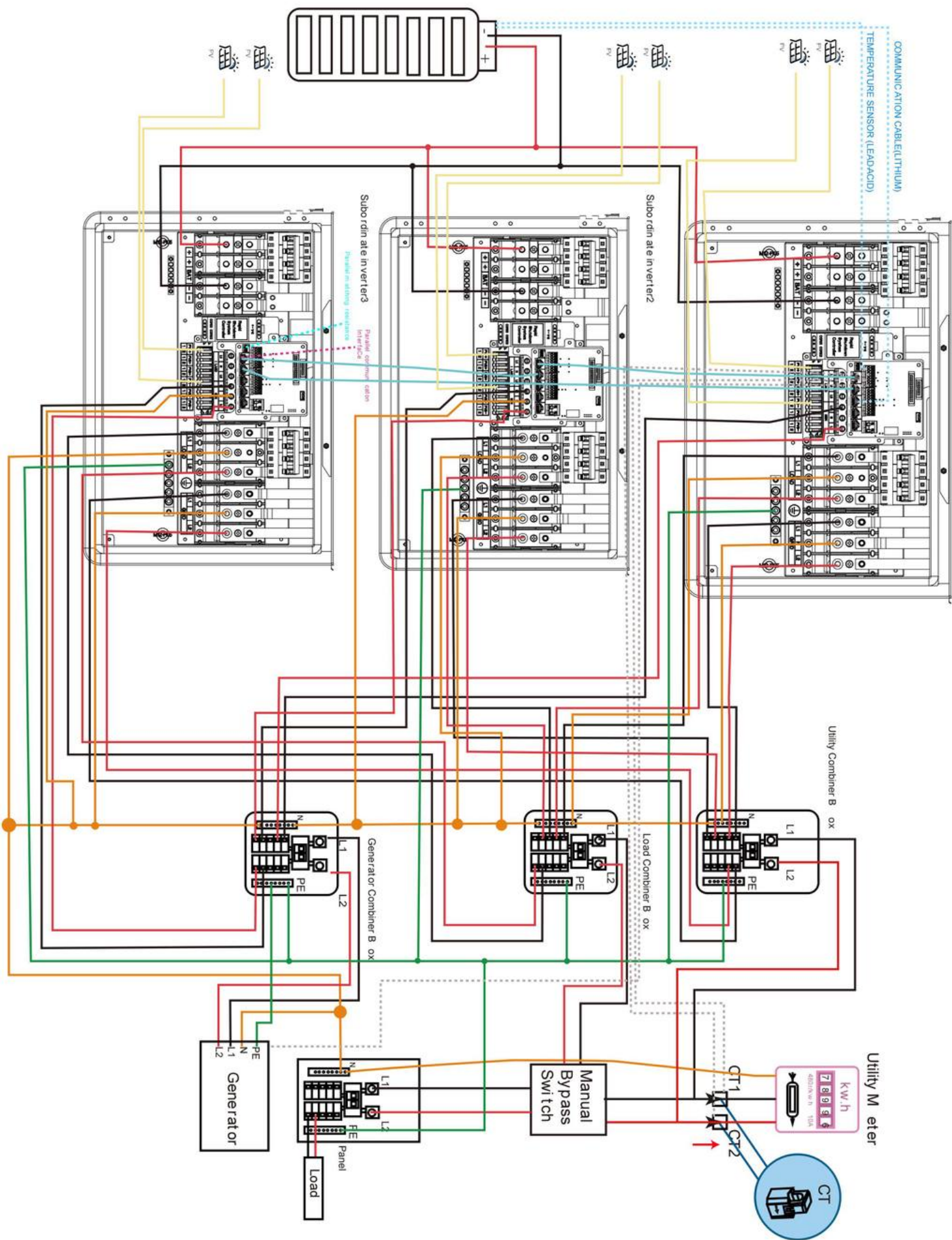
Shared Battery

Combined Three Phase

Local Phase

Phase A

The wiring diagram for parallel connection is provided below.



Prima ry in verter1

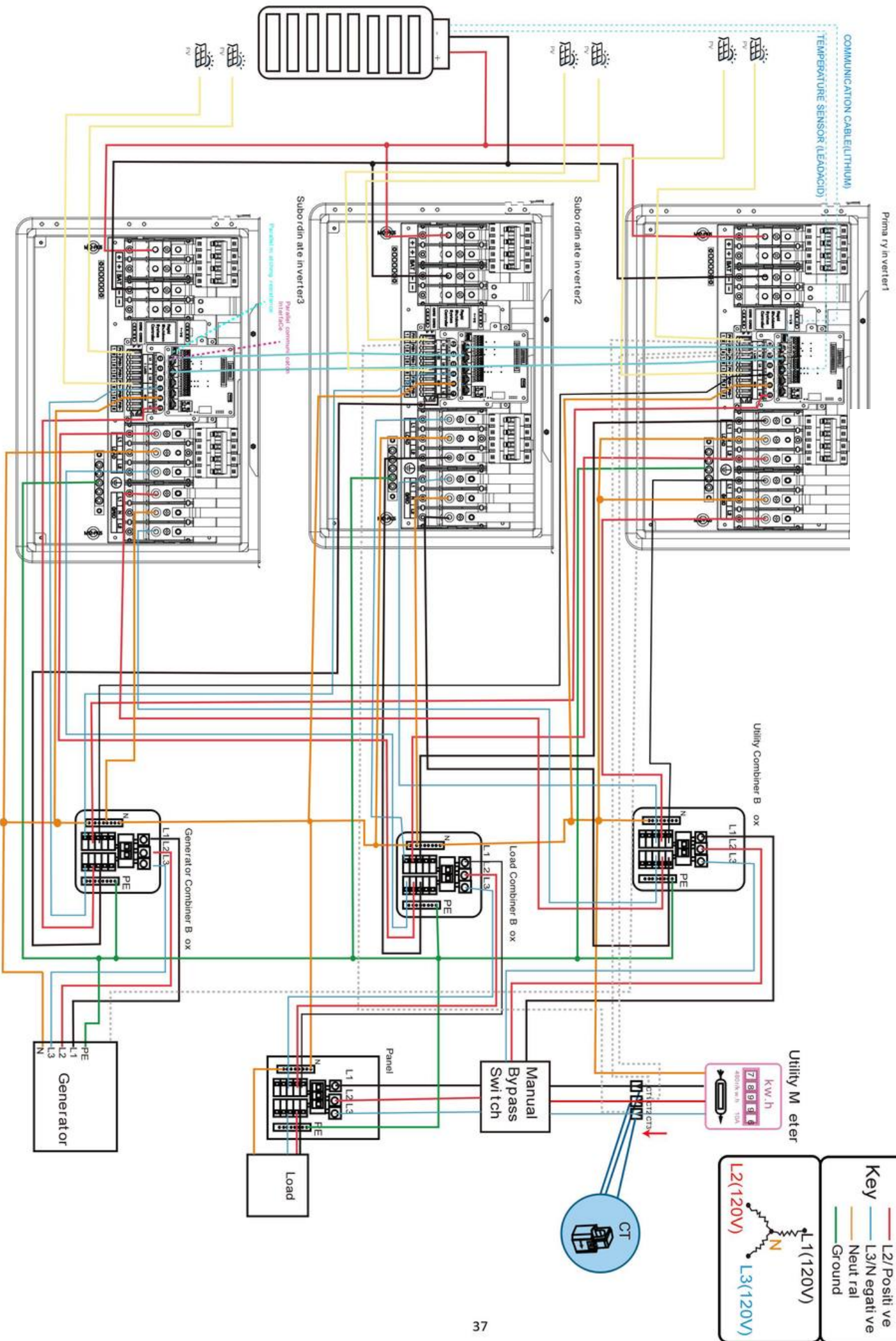
A.15KWx 3 Parallel system installation connection (3 @ 240V/120V)

For the **daisy chain** circuit, ensure that the **2-position CAN termination resistors** on the **first** and **last** inverters in the chain are set to the **ON** position.

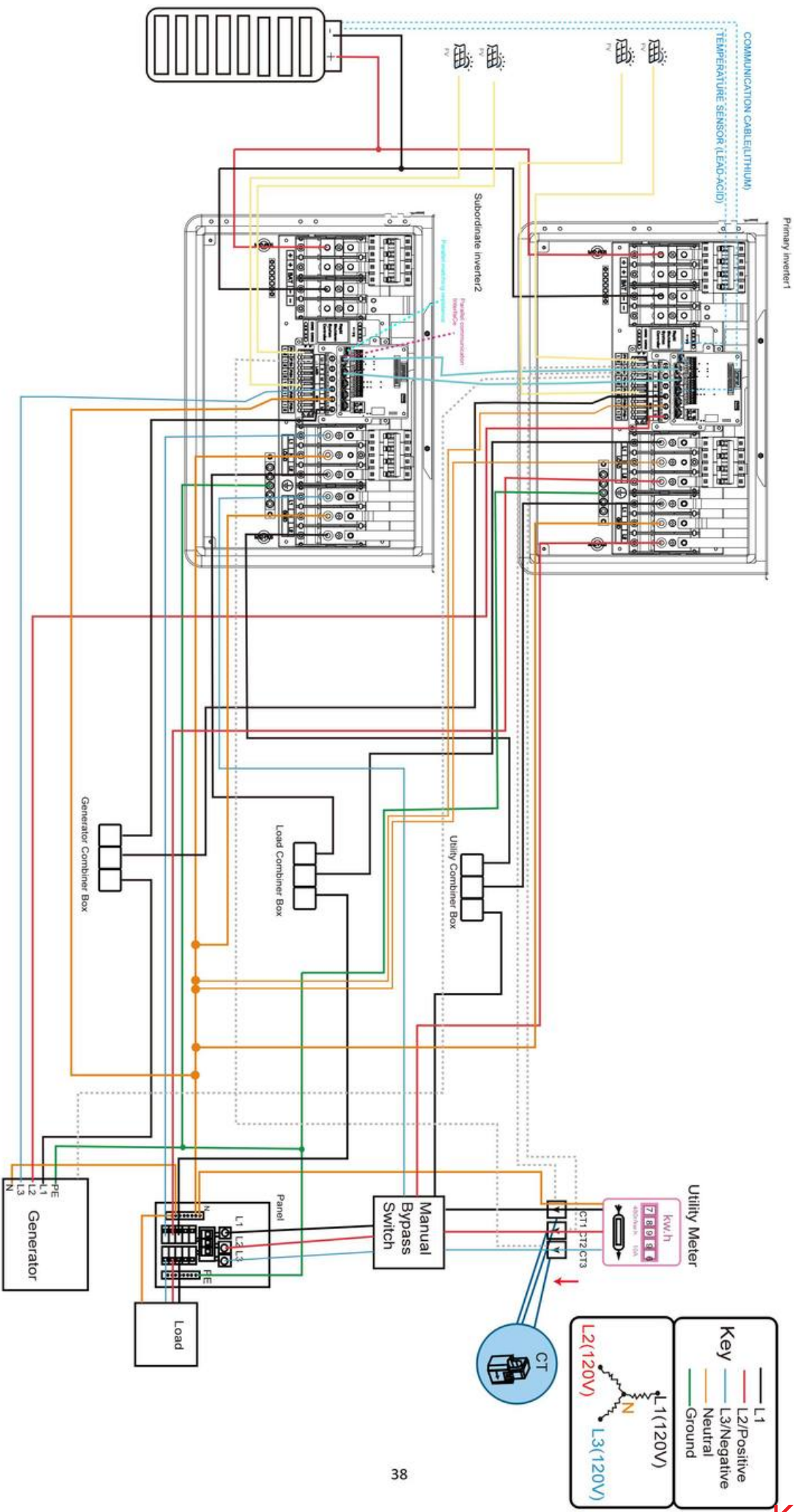
Steps:

- 1.**First Inverter:** Set the CAN termination resistor switch to **ON**.
- 2.**Last Inverter:** Set the CAN termination resistor switch to **ON** as well.
- 3.**Intermediate Inverters:** Leave the termination resistors in the **OFF** position for all inverters between the first and last units in the chain

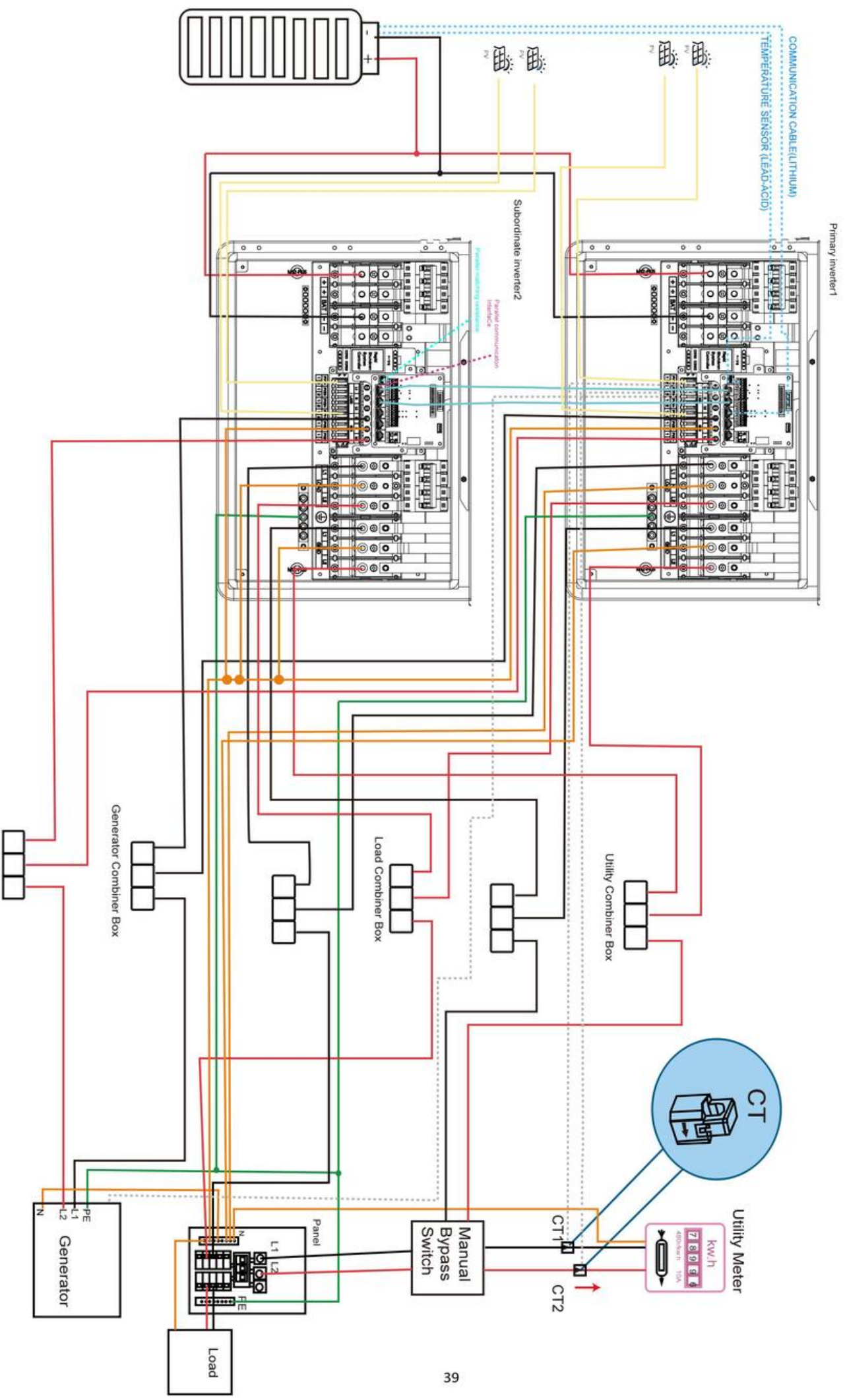
B.15KW x 3 Parallel system installation connection (3 @ 208V/120V)



C.15KW x 2 Parallel system installation connection (2 @ 208V/120V)



D,15KW x2 Parallel system installation connection (2 @ 240V/120V)



To properly configure the CAN communication for a daisy chain setup, follow these steps:

First Inverter:

Set the CAN communication pins to the ON position for the first inverter in the chain. This will enable communication and establish the starting point of the CAN bus.

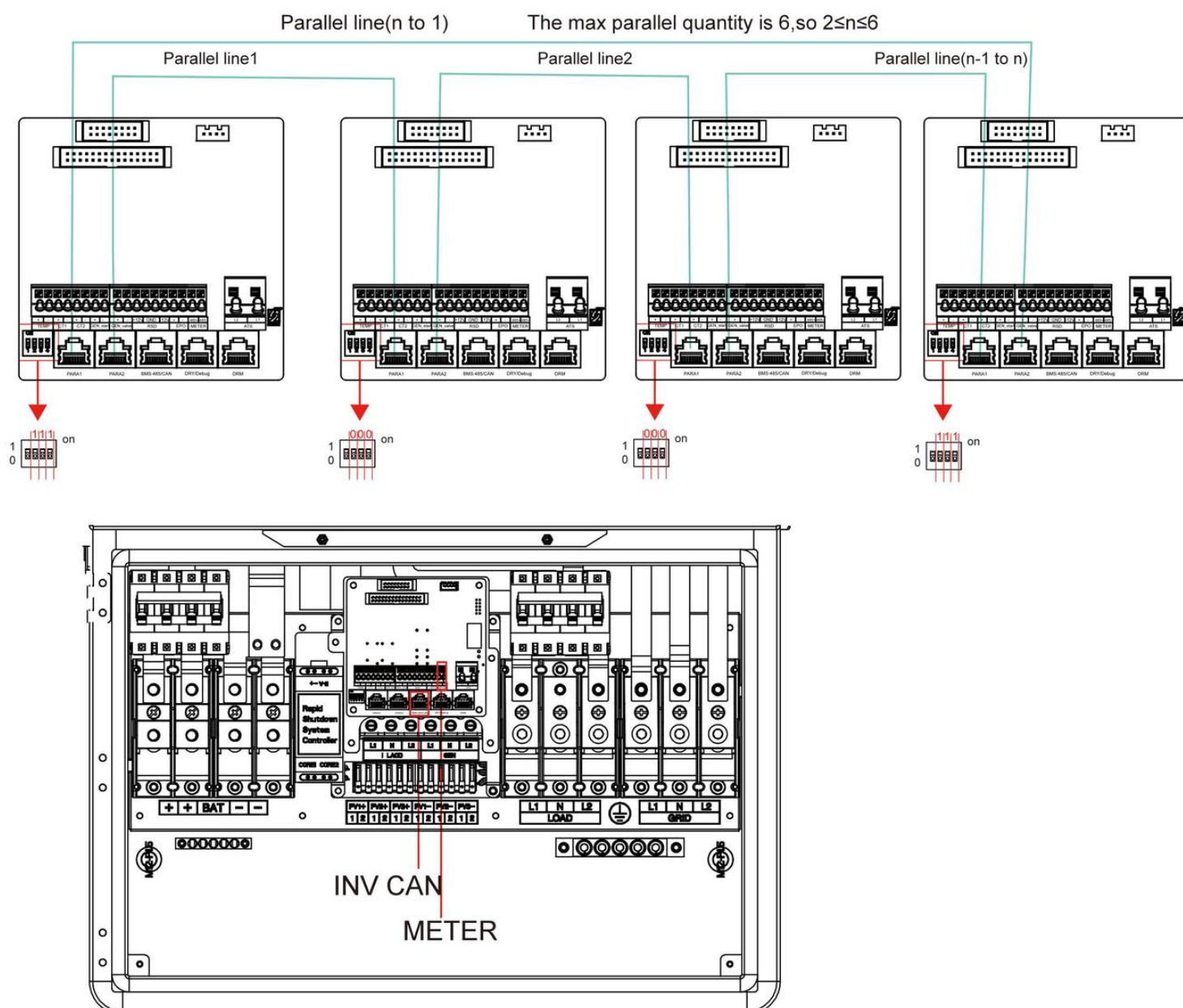
Last Inverter:

Set the CAN communication pins to the ON position for the last inverter in the chain. This ensures the communication line is properly terminated at the end of the bus.

Intermediate Inverters:

For the inverters in between, the CAN communication pins should be left in the OFF position, as they act as repeaters but do not need to have the termination enabled.

If the **parallel cables** are not long enough or unavailable, you can create a **direct pin-to-pin cable** to extend the connections between the inverters.



7. Operating Modes and Functions

7.1 Operate the inverter and conduct a test

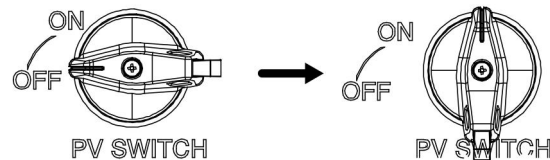
At least one energy source must be available to run this inverter: Battery, PV, or Grid.

7.1.1 Check the voltage of the battery bank

- Voltage of the battery must be between 40VDC - 60VDC.
- If applicable, turn ON internal switches of the batteries. Measure individual voltages.
- Verify that the voltage of the battery bank at the KR-L8-15K-SPUS terminals is adequate.
- DO NOT reverse polarity. DO NOT turn OFF battery disconnect if any current is flowing in or out of the battery.

7.1.2 Check the voltage of each PV input circuit

- Input voltage must not exceed 550VDC.
- Input voltage must be above the startup voltage of 120VDC.
- Do not ground PV+ or PV-.
- Verify polarity in each PV string. Backward polarity will measure 0Vdc by the KR-L8-15K-SPUS and will cause long term damage.
- PV alone turns LCD screen only. Inverter requires grid and/or batteries to operate.
- PV DC disconnect switch on the side of the inverter will turn the PV ON or OFF.

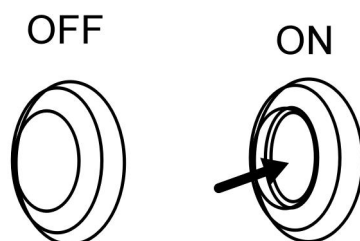


7.1.3 Check GRID input voltage

- Use the "GRID" terminals to measure AC voltage with a multimeter.
- Measure line (L) to neutral (N) voltages on "GRID" terminals. Ensure 120VAC on all phases.
- Measure line (L) to line (L) voltages on "GRID" terminals. Ensure 240VAC. (If voltage reading is close to 220V or 210V, verify if grid is single-phase or three-phase instead).
- Verify that voltage between Neutral and Ground is 0VAC.
- Verify that voltage between "GRID" L1 and "LOAD" L1 is 0V. Do the same for L2

7.1.4 Power ON KR-L8-15K-SPUS

- Turn ON the battery breaker(s).
- PRESS down the power button to the ON position. Wait for the LED indicator to turn on. This may take a few minutes.
- Turn ON the PV DC disconnect switch.
- Turn ON the external "GRID" disconnect.
- Turn ON the internal "LOAD" and external "GEN" breakers.



7.1.5 Power Cycle Sequence

1. TURN OFF all AC breakers / disconnects ("GRID", "GEN" and "LOAD").
2. TURN OFF the built-in PV DC disconnect switch on the side of the inverter.
3. PRESS the power button, making sure it is in the OFF position. An "OFF" message will appear after the LED turns off.
4. TURN OFF the battery breakers.
5. Wait a moment (~1 min) to ensure the inverter is completely de-energized.
6. Make sure that the KR-L8-15K-SPUS is properly connected to the batteries, solar panels, "GRID", "GEN", and "LOAD".
7. Reverse the steps to turn ON the KR-L8-15K-SPUS .

7.2 Operating Modes and Functions

This inverter offers different operating modes and functions to meet a variety of customer needs. The operating modes and their functions are as follows:

7.2.1 Self-Use Mode (Default)

In **Self-Use Mode**, the system prioritizes energy flow as follows:

For powering loads: Solar → Battery → Grid

For using solar energy: Load → Battery → Grid

When solar power is available, it is first used to power local loads. Excess solar energy is stored in the battery, and any remaining surplus is exported to the grid. When solar power is insufficient, the system uses battery energy to support the loads. Only when both solar and battery are insufficient does the system draw power from the grid.

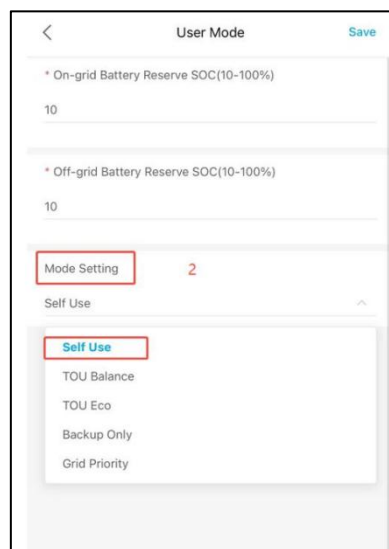
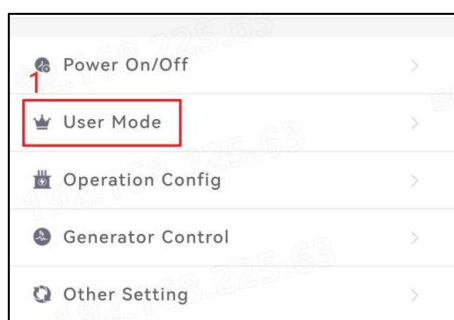
This mode is set as the default operating mode. It is designed to **maximize self-consumption of solar energy**, thereby reducing reliance on the grid and lowering electricity costs.

Application Scenario

Self-Use Mode is ideal for residential or commercial users who want to reduce energy bills and improve energy independence by making full use of on-site solar generation.

Please refer to Chapter 9.6 **Work Mode Settings Menu** for the operation method of the screen.

APP Setting:



*On-grid Battery Reserve SOC (10-100%): When the battery is connected to the grid, it maintains its charge.

100%):When the battery is connected to

When the battery is low on soc, the battery will be charged and discharging is not allowed .

*Off - grid Battery Reserve SOC (10-100%):The battery maintains its charge when off-grid, and will charge when the soc is low, and discharging is not allowed .

7.2.2 TOU ECO Mode

TOU Economy Mode is designed to optimize economic returns by adjusting energy usage and storage behavior according to time-of-use (TOU) electricity pricing.

During peak price periods, the system prioritizes:Load → Grid → Battery
Maximize revenue by selling solar energy to the grid at high prices.

During flat price periods, the system prioritizes:Load → Battery → Grid
Use solar power primarily for self-consumption and store excess in the battery.

During valley price periods, the system prioritizes:Battery → Load → Grid
Charge the battery from the grid when electricity prices are low, and use stored energy during expensive periods.

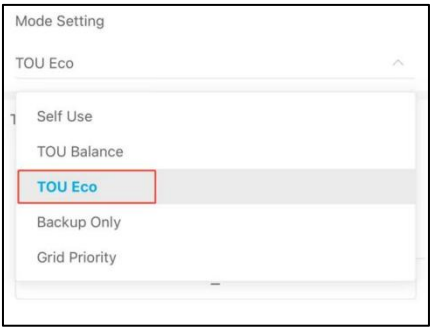
Users can define the peak, flat, and valley time periods according to local electricity rates and their own usage patterns via the system settings or monitoring platform.

Application Scenario

TOU Eco Mode is suitable for users in regions with significant differences in electricity prices across time periods. It helps reduce electricity costs and maximize profits by charging when electricity is cheap and discharging or selling when it is expensive.

Please refer to Chapter 9.6 **Work Mode Settings Menu** for the operation method of the screen.

APP Setting:



Time Setting

Set the time period for peak and valley electricity prices.

“+” is to add time period “-” is to delete time period

Time Setting

00:00

12:30

Peak Price

-

12:30

23:59

Flat Price

+

7.2.3 Backup Only Mode

Backup Only Mode is designed to maximize battery charging and reserve energy specifically for backup use during grid outages or instability.

In this mode, the system prioritizes storing solar energy in the battery rather than supplying it to loads or exporting to the grid during normal operation. The battery will **only discharge when the grid fails or becomes unavailable**.

PV power priority:Battery → Load → Grid

Application Scenario

Backup Only Mode is ideal for users in areas with frequent power outages, unstable grid conditions, or where critical loads (such as medical equipment, servers, or refrigeration) must be kept running at all times. It prioritizes energy availability and system reliability over cost savings or feed-in profits.

Please refer to Chapter 9.6 **Work Mode Settings Menu** for the operation method of the screen.

APP Setting:

Power On/Off

User Mode

Operation Config

Generator Control

Other Setting

Mode Setting

Backup Only

Self Use

TOU Balance

TOU Eco

Backup Only

Grid Priority

7.2.4 Grid priority Mode

Grid Priority Mode is designed to prioritize exporting electricity to the grid when sufficient power is available, in order to maximize feed-in profits.

In this mode, the system supplies power in the following order:

Power Supply Priority: Load → Grid → Battery

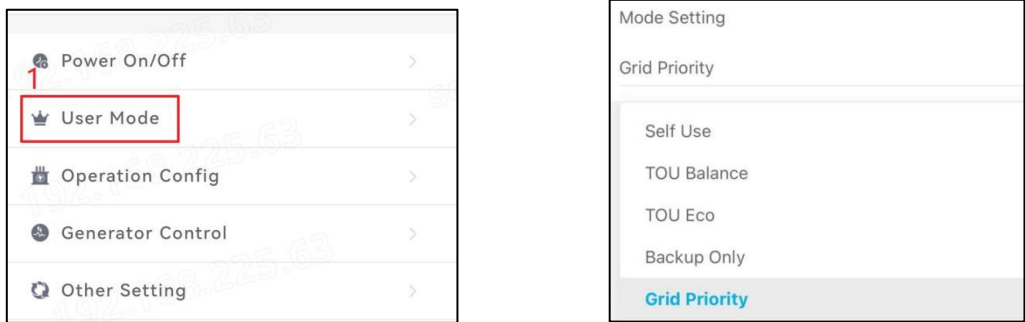
When solar energy is generated, it is first used to meet local load demand. Any excess energy is then exported to the grid for revenue. The battery will only be charged if there is additional surplus after meeting both load and grid feed-in needs.

Application Scenario

Grid Priority Mode is suitable for regions with **favorable feed-in tariffs (FiTs)** or net metering policies. It is ideal for users who aim to **maximize income from solar generation**, rather than prioritize local energy consumption or backup capacity.

Please refer to Chapter 9.6 **Work Mode Settings Menu** for the operation method of the screen.

APP Setting:



7.3 Generator Function

The **Generator&Smart Load** function allows the power input connection point to act as a load connection point. When enabled, the inverter will supply power to the connected load when the **battery state of charge (SOC)** exceed the user-defined thresholds. **Please refer to Chapter 9.9 Generator Port Settings Menu for the operation method of the screen.**

Function Description:

Gen Warm Time:Generator warm-up time. After the inverter sends the start signal, it waits for this duration before accepting generator input.

Start SOC / Stop SOC:Battery SOC thresholds for starting and stopping the generator automatically.

Gen Rated Power:Maximum power output of the generator.

Max Run Time:Maximum continuous run time for the generator. Applies when the inverter is automatically controlling the generator.

Cooling Time:Generator cool down period after stopping. Applies to inverter-controlled generator operation.

Gen Force:Force Start Generator function. Should be enabled during the first-time connection of a generator to ensure recognition and activation.

For example:

- **Generator Start SOC = 90%**
- **Generator Stop SOC = 50%**

Operation:

- When the **battery SOC reaches 90%**, the Generator Port automatically turns on and supplies power to the connected load.
- When the **battery SOC falls below 50%**, the Generator Port will automatically shut off.

Application Scenario:

- This function is useful when you want to prioritize powering non-critical loads with solar energy when both the battery is sufficiently charged and solar power generation is above a certain level, ensuring that only excess power is used for loads.

Important Note:

- **Do not connect a generator** when enabling the Smart Load function, as this can **damage the equipment**.

Please refer to Chapter 9.6 **Work Mode Settings Menu** for the operation method of the screen.

7.4 Smart Load Function

The **Smart Load** function allows the power input connection point to act as a load connection point. When enabled, the inverter will supply power to the connected load when the **battery state of charge (SOC)** exceed the user-defined thresholds. **Please refer to Chapter 9.9 Generator Port Settings Menu for the operation method of the screen.**

Function Description:

Stop Output SOC: When the battery SOC drops to this threshold, the smart load output will automatically disconnect.

Stop Output Volt: If voltage-based control is selected, the smart load output will disconnect when the battery voltage reaches this value.

Control Start Time: Start of the time window during which the SOC or voltage control conditions are valid.

Control End Time: End of the time window during which the SOC or voltage control conditions are valid.

Note:

Outside the defined time window, the smart load output is not restricted by SOC or voltage conditions, and will follow the same discharge behavior as the EPS output.

For example:

- **Smart Load Start SOC = 90%**
- **Smart Load Stop SOC = 50%**

Operation:

- When the **battery SOC reaches 90%**, the Smart Load Port automatically turns on and supplies power to the connected load.
- When the **battery SOC falls below 50%**, the Smart Load Port will automatically shut off.

Application Scenario:

- This function is useful when you want to prioritize powering non-critical loads with solar energy when both the battery is sufficiently charged and solar power generation is above a certain level, ensuring that only excess power is used for loads.

7.5 Off-Grid Installation Tips

- **Limit sensors (CTs)** are **not required** for completely off-grid installations, **unless**:

"Grid Priority" is enabled using a generator connected to the **GRID** terminal.

- It is **recommended** to connect generators to the **GRID** terminal. This simplifies integration with a service panel that is configured as a "GEN" input.
- A **transfer switch is not required** in off-grid systems. The **LOAD** output can be connected directly to the main panel.
- **Do not use "Grid Sell" mode** in off-grid installations.
- When using a **generator in an off-grid scenario**, it is recommended to set the **Grid Mode** to "**General Standard**".

- The **Auto Generator Start (AGS)** function activates when the battery voltage or state of charge drops below the user-defined **“Start Voltage / SOC”** threshold.
- If integrating a **wind turbine**, a **48V wind charge controller with dump load** must be installed to prevent battery overcharging. The wind charge controller must be **connected directly to the battery bank**.
- Ensure that the **battery capacity** is correctly configured and that charge/discharge current limits are set within safe operating ranges.

7.6 Battery Charge Controller

4-Stage Charging

The MPPT has a 4-stage battery charging algorithm for rapid, efficient, and safe battery charging. The next figure shows the stage sequence.

Bulk Charge Stage

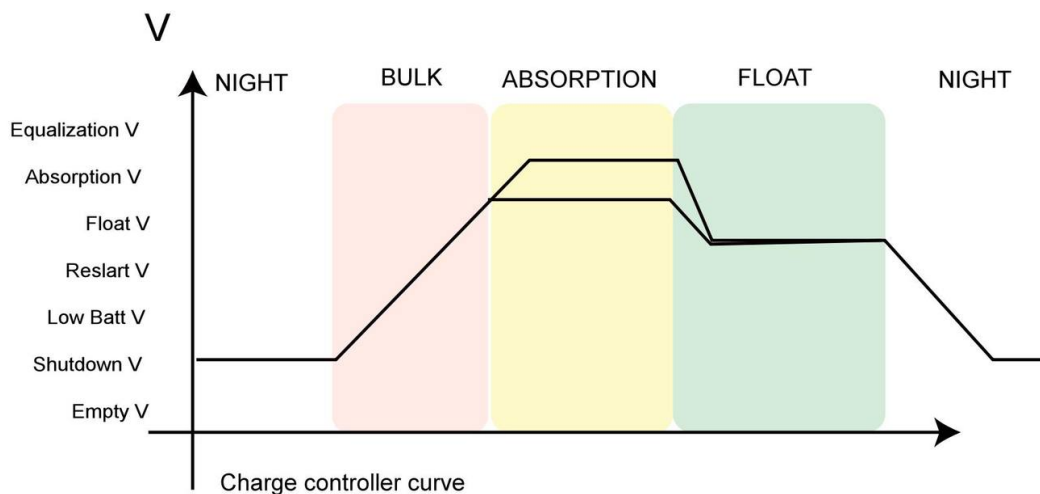
In the Bulk Charge stage, the battery is not at a 100% state of charge and has not yet reached the Absorption voltage setpoint. The controller will deliver 100% of available solar power to recharge the battery.

Absorption Stage

When the battery has reached the absorption voltage setpoint, the inverter uses constant-voltage regulation to maintain battery voltage at the absorption setpoint, preventing overheating and excessive battery gassing. The battery is allowed to come to a full state of charge at the absorption voltage setpoint. Absorption lasts until the battery charge amperage (A) rate reaches 2% of the programmed capacity (Ah).

Float Stage

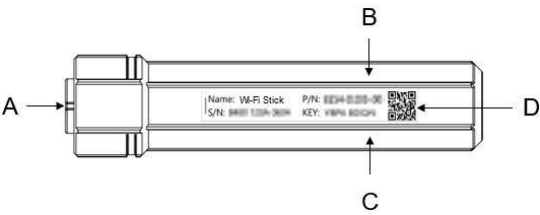
After the Absorption stage charges the battery fully, the MPPT reduces the battery voltage to the float voltage setpoint. If the batteries have 100% charge, there can be no more chemical reactions, and all the charging current turns into heat and gassing. The Float stage provides a minimum rate of maintenance charging while reducing the heating and gassing of a fully charged battery. The purpose of the Float stage is to protect the battery from long-term overcharge



8.Wi-Fi&BLE stick installation

8.1 Indication

- A: Circular Connector Interface: Connect to inverter and communication
B: Red LED: Inverter communication indication
C: Green LED: Network communication indication
D: Product label: Show product information

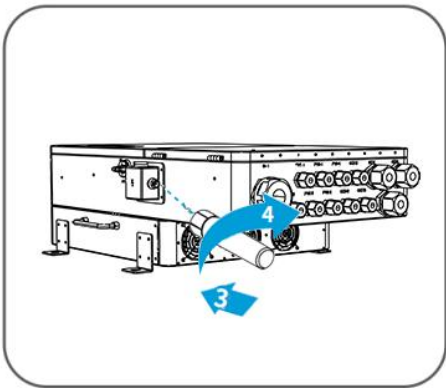
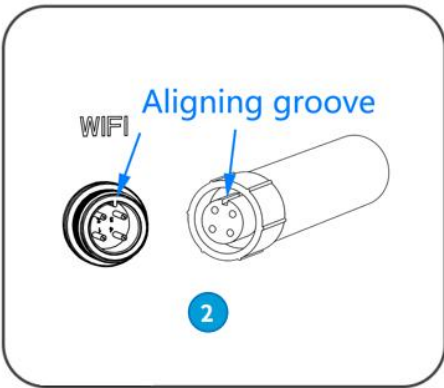
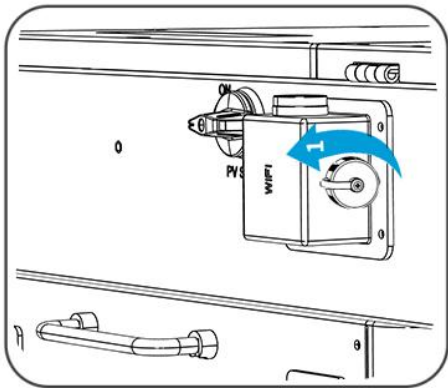


- 1.LED glow only when the Wi-Fi&BLE stick is powered on.
2.When the Wi-Fi&BLE stick is powered on, the green LED glows for 3S as a power on indication.
3.The more detail LED indication please refer chapter 9 “LED indication and trouble shooting”.

8.2 Install the Wi-Fi&BLE stick

Follow the installation steps!

1.Remove the waterproof cover.	2.Aligning groove.
3.Plug in WiFi module.	4.Revolve to lock the WiFi module.

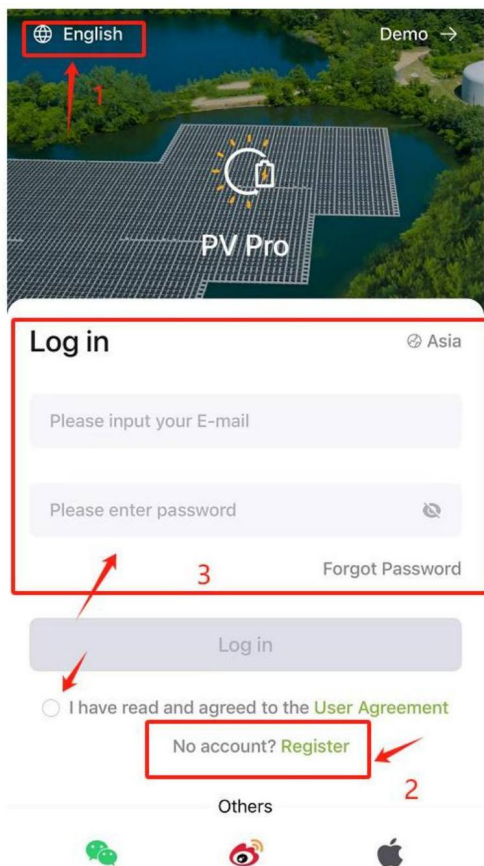


8.3 Web/APP

Item	Web View	APP
QR Code		
Website		iOS: search "PV Pro" in Apple Store Android: search "PV Pro" in Google Play

8.4 Wi-Fi Connection

8.4.1 Log in to your account



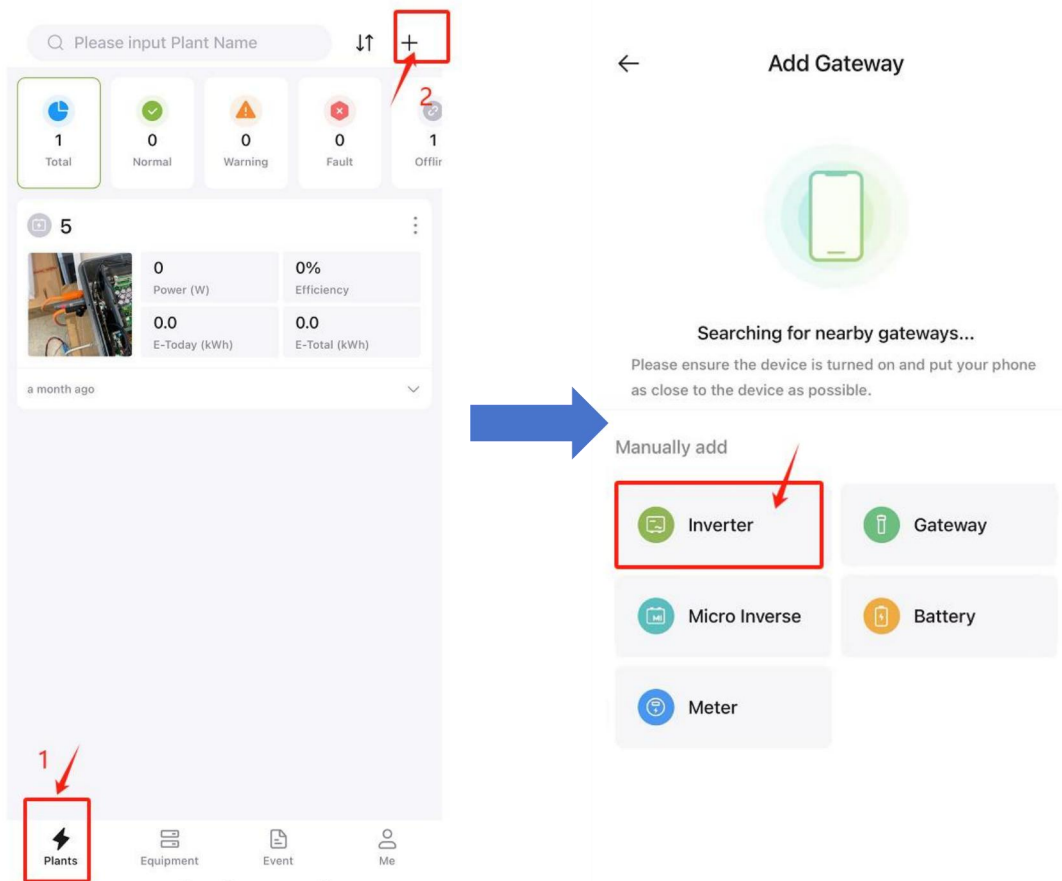
The screenshot shows the PV Pro login interface. At the top left, there is a language selector with a globe icon and the word 'English'. To its right is a 'Demo' link with a right arrow. The main header features a large image of solar panels with the 'PV Pro' logo. Below the header is a 'Log in' section. It includes two input fields: 'Please input your E-mail' and 'Please enter password'. To the right of the password field is a 'Forgot Password' link. Below these fields is a 'Log in' button. Under the button is a checkbox labeled 'I have read and agreed to the User Agreement'. Below the checkbox is a link that says 'No account? Register'. At the bottom, there is a section labeled 'Others' with icons for WeChat, a red eye icon, and the Apple logo.

Step 1: Select language

Step 2: Create an account

Step 3: Enter your account and password
to log in

8.4.2 Create a plant



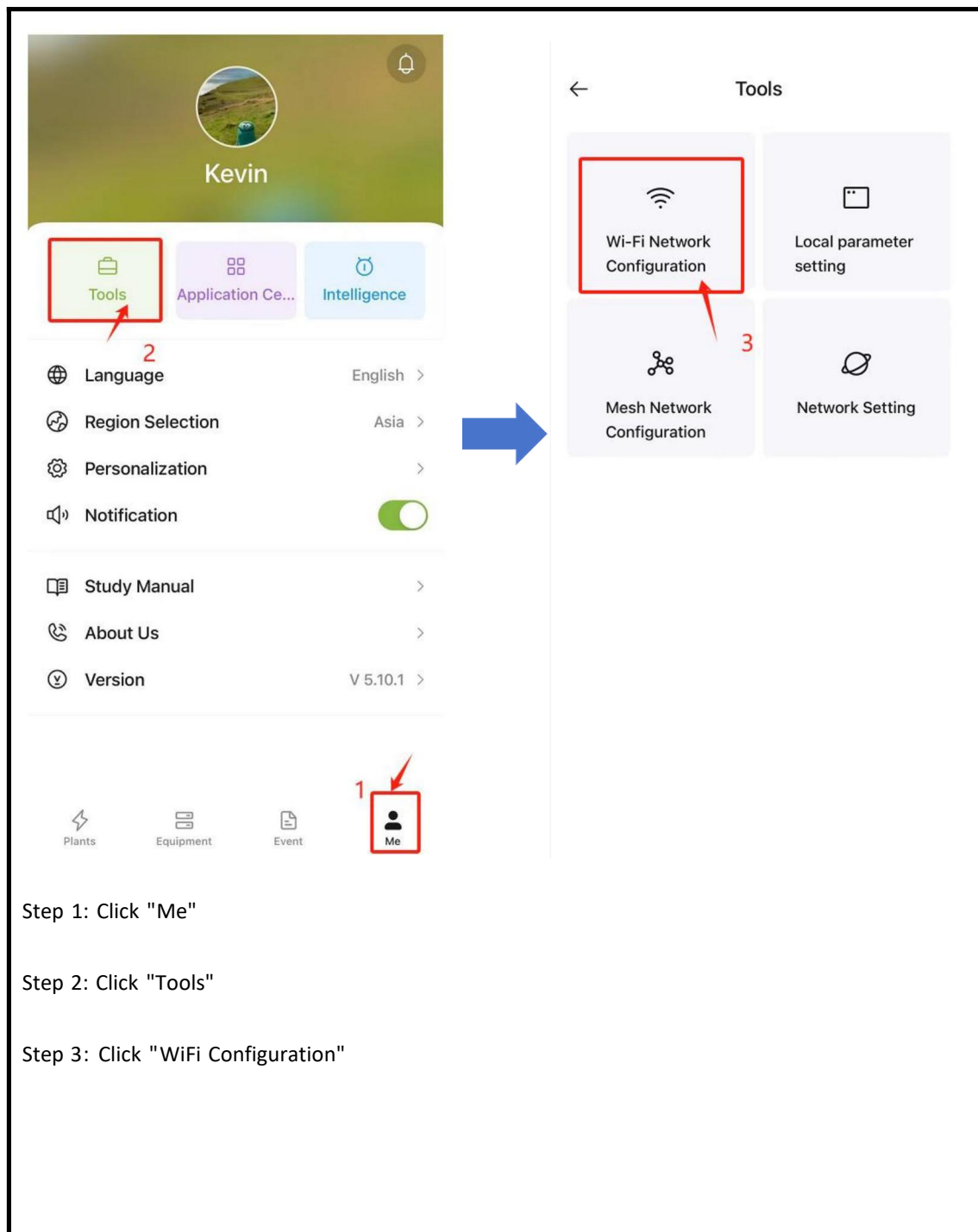
Step 1: Click "Plant"

Step 2: Click the "+" in the upper right corner

Step 3: Click "inverter"

Step 4: Scan the QR code of the data logger or manually enter the serial number to create a power plant

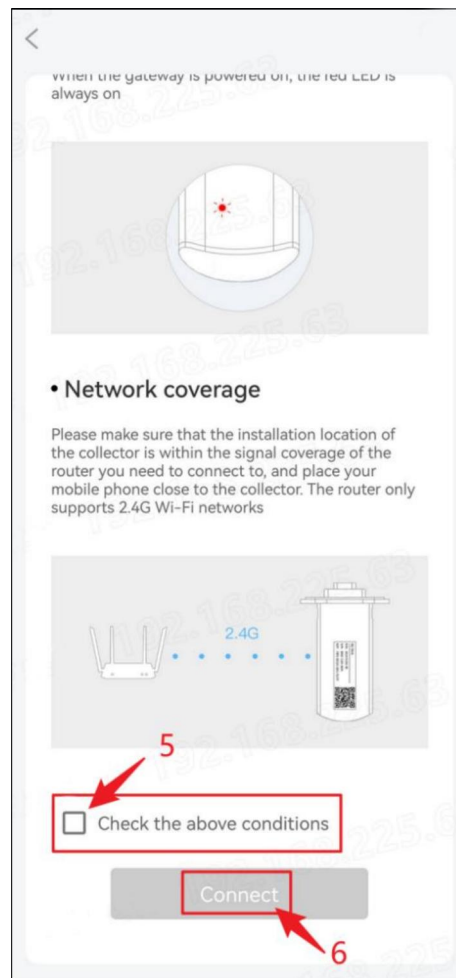
8.4.3 Configure Network



Step 1: Click "Me"

Step 2: Click "Tools"

Step 3: Click "WiFi Configuration"



Step 4: Choose a way to configure the network

Step 5: Check "Check the above conditions"

Step 6: Click "connect"

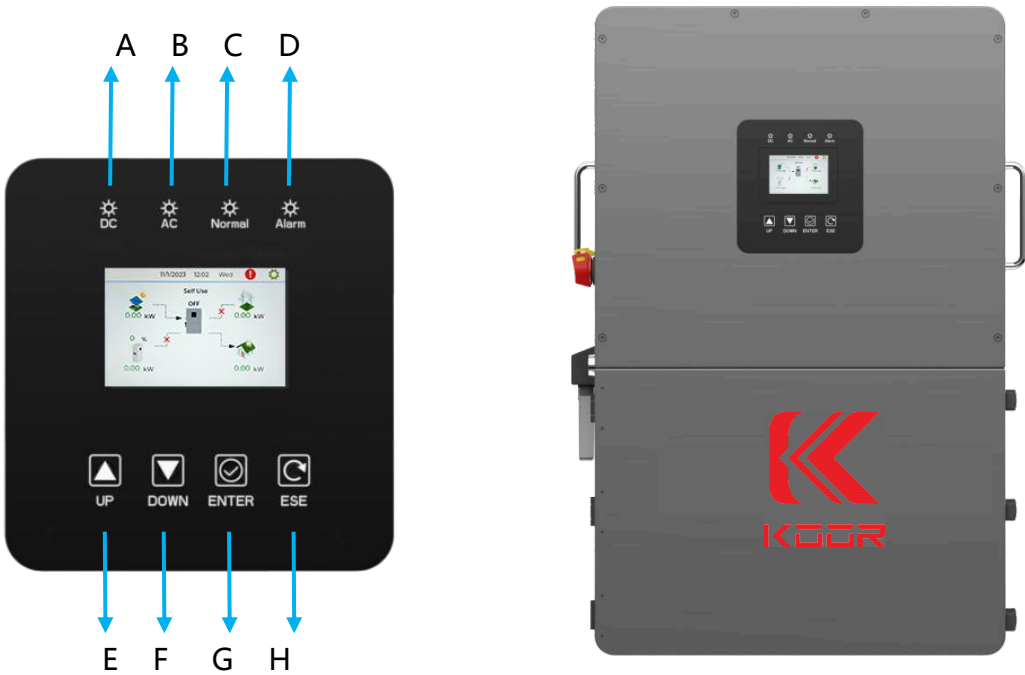
*Bluetooth is used for local configuration without WIFI

8.5 LED indication and trouble shooting

LED	State	Indication	
	Red LED: Inverter communication indication		Green LED: Network communication indication
	Cycle for 2S: flash once quickly, then glowing	Cycle for 2S: flash once quickly, then glowing	
	Not glow more than 20S	The power supply to the Wi-Fi&BLE stick is abnormal or damaged: 1. Check whether the power supply of the Aerial Plug Interface on the inverter is normal 2. Wi-Fi&BLE stick abnormal, contact Fantastech	
	Cycle for 2S: flash once quickly, then off	Communication failure: Check whether the connection between the Wi-Fi&BLE stick and inverter is loose or poor contact	
	When powered on, continuously glows 3S, and then off	Power on indication	
	Glow more than 5S	Communication is normal	
	During the long glowing, flash occasionally	Network transmit data	
	Cycle for 20S: flash once quickly, then off	The route is not connected: 1. Check whether the password is right 2. Check the strength of the router	
	Cycle for 20S: flash 3 times continuously, then off	Connect to the route, but can't connect to the cloud server: 1. Check whether the router has Internet access permission 2. Check the firewall setting	
	Cycle for 20S: flash 4 times continuously, then off	Wi-Fi&BLE stick information error: Please contact Fantastech	

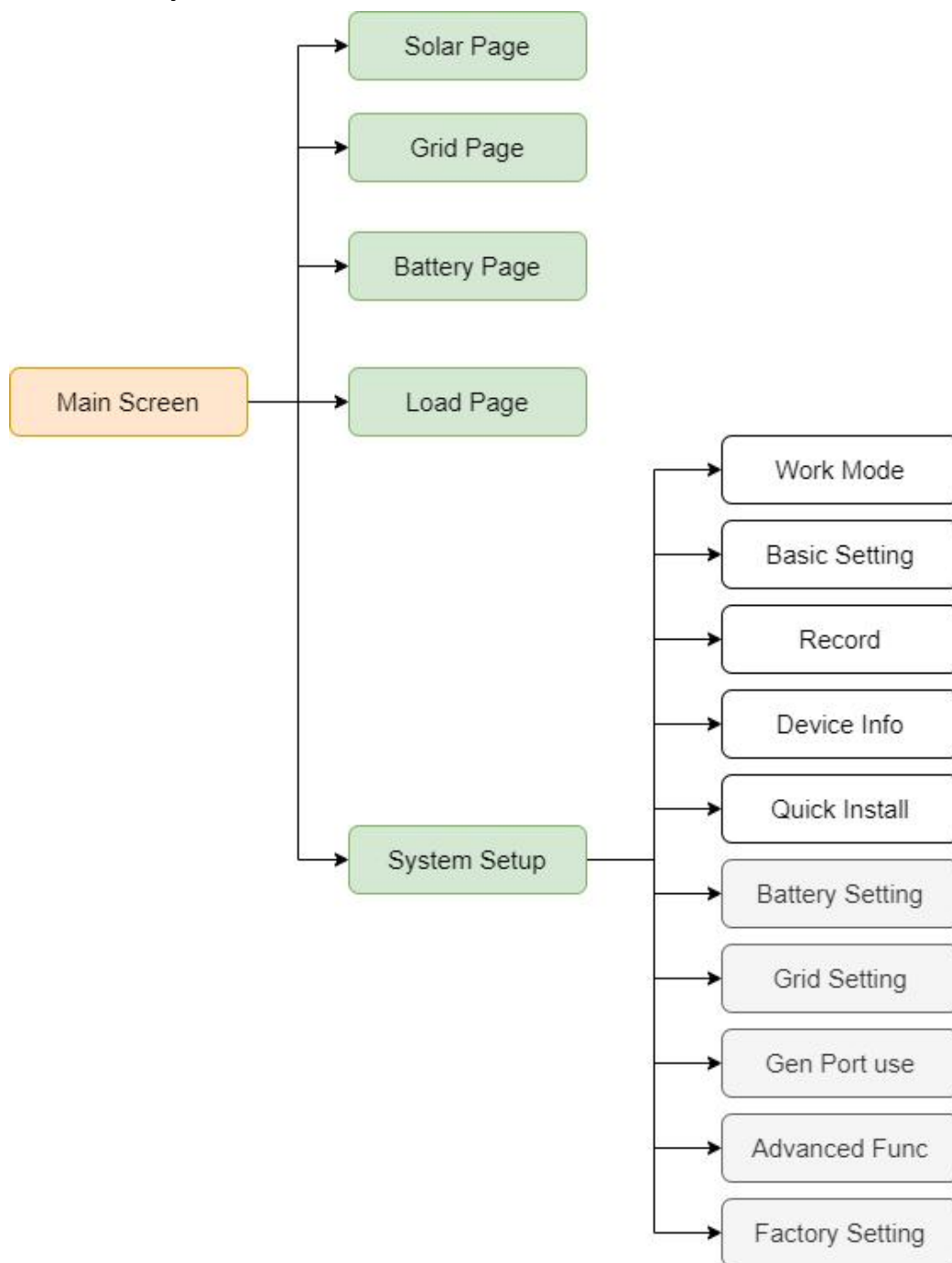
9.Operation

9.1 Indicator panel



Name	Object	Function
Indicator LED	A	Green : ON, The inverter is running ; Flash is Standby.
	B	Blue : ON, Communication with BMS normal.
	C	Yellow : ON, The inverter is in EPS mode.
	D	Red : The inverter is in fault mode.
	E	UP:To go to previous selection
	F	DOWN:To go to next selection
	G	ENTER:To confirm the selection
	H	ESC:To exit setting mode

9.2 LCD operation flow chart



9.3.1 Main Screen

11/1/2023 12:02 Wed ! ⚙

Self Use

0.00 kW

0.00 kW

0.00 kW

0.00 kW

Possible modes include:TOU ECO,Self-Use,Backup,etc.
This helps users understand how the system is currently operating.

-Battery power negative means charge, positive means discharge.

Includes configuration options such as operation mode, battery parameters, grid settings, etc.

Click to access detailed load information.

9.3.2 PV Information Page Overview

Solar		
PV1-V: 0.0 V	PV1-I: 0.0 A	PV1-P: 0 W
PV2-V: 0.0 V	PV2-I: 0.0 A	PV2-P: 0 W
PV3-V: 0.0 V	PV3-I: 0.0 A	PV3-P: 0 W
Power: 0.00 kW		
Today: 0.0 kWh	Total:	0.0 kWh

(Subject to variation depending on project configuration)
When the PV icon on the main interface is clicked, the system navigates to the PV information page. Here, PV1 to PV3 display the voltage, current, and power data for each of the four PV input channels.

Below these readings, users can view the total current PV power, the energy generated today, and the historical cumulative generation. Page navigation arrows on the side of the screen allow users to scroll through additional data pages.

9.3.3 Grid Information Page Overview

Grid		
Status: Off	Power: 0.09 kW	Freq: 49.97 Hz
L-V: 120.3V	L-I: 0.8A	L-P: 101 W
BUY Today: 0.0 kWh	Total: 0.0 kwh	
SELL Today: 0.0 kWh	Total: 0.0 kWh	

(Subject to variation depending on project configuration)
The Power field displays the total grid power. Freq refers to the present grid frequency. L-V represent the single phase voltage, each showing their respective voltage, current, and power values.

At the bottom of the page, the energy purchased from the grid is displayed, including both the daily purchased energy and cumulative purchased energy. (Note: These fields may vary depending on the specific project configuration.)

9.3.4 Battery Information Page Overview

Battery	
Battery 1: Idle	
Bat-V: -0.1 V	
Bat-I: -0.0 A	
Bat-P: -0.0 W	
temp: 0.0C	
SOC: 0.0%	

This page displays the operating information of Battery 1 (subject to variation depending on the project configuration).

Key parameters include the current battery status, voltage, current, power, temperature, and State of Charge (SOC).

9.3.5 Load Information Page Overview

Load

EPS Load: 0 W

Home Load: 0 W

Smart Load: 0 W

Total Load: 0 W

Toady: 0.0 kWh

Total: 11.5 kWh

↑

↓

✕

EPS-RU:0.0 V

EPS-SU:0.0 V

EPS-RI: 0.1A

EPS-SI: 0.1A

Pwr:0 W

Pwr:0 W

(Subject to variation depending on the project configuration)
This page includes readings for EPS load power, household load power, smart load power, and the total load power.

Below these values, the page also displays the voltage, current, and power for the three AC phases: R, and S (corresponding to L1, and L2,).

9.4 System Setup Menu

System Settings Page (Standard User Access)

This is the settings page available under standard user permissions.

System Setup

Work Mode

Basic Setting

Record

Device Info

To enter basic settings mode, click the icon  in the upper-left corner and enter the password **0000**.

0000

7

8

9

4

5

6

1

2

3

0

.

DEL

CANCEL

OK

Advanced User Access



System Setup

Work Mode

Battery Setting

Grid Setting

Gen Port Use

Advanced Func

Factory Setting

Basic Setting

Record

Device Info

Quick Install

To enter advanced settings mode, click the icon  in the upper-left corner and enter the password 7777.

Once verified, the system will open the Advanced Settings Page, where users can access additional configuration options not available under standard permissions.

7777

Range

7

8

9

4

5

6

1

2

3

0

.

DEL

CANCEL

OK

9.5 Basic Settings Menu



Basic setting

Hime setting

2025

7

1

9

31

32

LCD Light

100 %

Beep

Yes

Auto Dim Time

600 s

Language

English



- Time Setting:
Set the system’s date and time, including year, month, day, hour, and minute.
- LCD Light:
Adjust the screen brightness level.
- Beep:
Enable or disable the buzzer.
Enabled by default; the buzzer will sound upon touch or key press.
- Auto Dim Time:
Set the duration before the screen automatically dims or turns off.
- Language:
Select the system display language.

9.6 Work Mode Settings Menu

Work Mode Settings-Page 1

 Work Mode 1/3

0.0kW

Max Solar Power

Zero Export to Grid

No

Max Sell Power

0.0kW

Zero-export Power

Energy pattern

Load First



-Max Solar Power:

Sets the maximum PV input power.

Zero Export to Grid:

When enabled,the inverter limits the amount of power exported to the grid.

The actual export limit is defined by the Zero-export Power setting.

-Zero-export Power:

Specifies the inverter's export power limit to the grid. This setting takes effect only when Zero Export to Grid is enabled.

-Max Sell Power:

Sets the maximum power that can be exported to the grid.

-Energy Pattern:

Defines the system's energy priority strategy.Currently,only **Load First** mode is supported.

-Load First:

PV power is used first to supply the load.

Any surplus PV energy is then used to charge the battery.

If PV is insufficient,the grid will supply the load.

Work Mode Settings-Page 2

Work Mode 2/3

Power On/Off	OFF	
Auto Power On	No	
Work Mode	TOU ECO	
On-Grid Bat Restart Soc	Self Use	
Off-Grid Bat Restart Soc	TOU ECO	
Charge Max soc	Backup Power	
	Grid Priority	

Work Mode 2/3

Power On/Off	OFF	
Auto Power On	No	
Work Mode	TOU ECO	
On-Grid Bat Restart Soc	10 %	
Off-Grid Bat Restart Soc	10 %	
Charge Max soc	100 %	

-Power On/Off:
Manually switches the inverter ON or OFF.

-Auto Power On:
Enables automatic startup and shutdown.
When enabled,the inverter will power on automatically.

-Work Mode:
Sets the system's working mode.Available options include:
Self-USE,TOU ECO,Backup Power,and Grid Priority.

-On-Grid Bat Restart SOC:
When the inverter is connected to the grid:
Battery is allowed to discharge if SOC 10%.
If battery SOC drops below(Restart SOC-3),the inverter will trigger forced charging to raise SOC back to 10%.

-Off-Grid Bat Restart SOC:
When the inverter is off-grid (not connected to the utility):
Battery is allowed to discharge if SOC 10%.

-Charge Max SOC:
Sets the maximum battery SOC for charging.
Once this limit is reached,charging will stop automatically.

Work Mode Settings-Page 3(TOU Mode Settings)



Work Mode--TOU 3/3

	Time		Type	Curr	SOC
-	00:00	12:30	Disch	50 A	20 %
-	12:30	16:30	Self Use	50 A	40 %
+	16:30	23:59	Charge	30 A	90 %



To enable TOU scheduling, the Work Mode on Work Mode Settings – Page 2 must be set to **TOU ECO**.

Only then will the time-based configuration become effective.

-Time:Defines the time slots used for energy management.

The full 24-hour day must be divided into continuous,non-overlapping periods.Each time slot includes a start time and an end time.Up to 6 time slots can be added by clicking the "+"icon.

-Type:Defines the behavior of the system during the time slot:

-Discharge:Battery discharges according to the specified parameters.

Example Schedule:

Time Period	Priority Type	Zero Export Disabled	Zero Export Enabled
00:00 – 12:30	Discharge	- Battery discharges if SOC > 20% (up to 50A limit) - Excess PV is exported to the grid	- Same discharge behavior - Excess PV charges battery to avoid waste
12:30 – 16:30	Self Use	- Battery discharges if SOC > 40% when PV cannot meet load - Battery charges if SOC < 36% using PV or grid - Remaining PV is exported	- Same behavior - No export allowed: Excess PV not used for load or charging is curtailed
16:30 – 23:59	Charge	- Battery charges if SOC < 90% using PV + grid (up to 30A) - Excess PV is exported	- Same charging behavior - No export allowed: Any excess PV after charging is curtailed

9.7 Battery Settings Menu

Battery1 Setting1 – Page 1

 Battery1 Setting1 1/2

Max A Charge	275 A	Batt 1Type	DC Source
Max A Disch	275 A	Activate Batt	No
Max Power Ch	15 kW	Com Type	1-CAN
Max Power Disch	15 kW	Company	PYLON HV
Grid Charge	No		
Parallel bat1&bat2	No		



1.-Max A Charge / Max A Disch:

Set the **maximum battery charging current** and **maximum battery discharging current**, respectively.

-Max Power Ch / Max Power Disch:

Set the **maximum charging power** and **maximum discharging power** for the battery.

-Grid Charge:

Enables or disables battery charging from the utility grid.

2.-Parallel Bat & Bat2:

For inverter models with two battery ports, enabling this function allows a single battery to be connected to both **Battery Port 1** and **Battery Port 2** simultaneously. In this case, the battery communication cable should be connected to the inverter's **BMS1 port**.

-Batt1 Type:

Select the battery type. Available options include: **Lithium**, **Lead-acid**, **DC Source**, and **No Battery**.

-Activate Batt:

Manually triggers battery activation or wake-up.

3.-Com Type:

Select the battery communication protocol: **CAN** or **RS485**

-Company:

Select the battery manufacturer. The system will automatically match the correct communication protocol based on the selected brand.

Battery1 Setting2 – Page 2

Battery1 Setting2 2/2

Float V	56.0 V	Ch Ratio	0.20	   
EqualizationV	57.0 V	TEMPCO(mV/C/Cell)	2	
Shutdown	40.0 V	Parallel Num	1	
High Batt	61.0 V	Batt capacity	100 Ah	
Low Batt	38.0 V			

Float V:
Sets the **battery float charging voltage**.

Ch Ratio:
Sets the **charging C-rate** (charging ratio), applicable **only to lead-acid batteries**.

Equalization V:
Sets the **equalization charging voltage**.

TEMPCO:
Battery **temperature compensation coefficient**, applicable **only to lead-acid batteries**.

Shutdown:
Sets the **battery discharge cutoff voltage** for protection.

Parallel Num:
Sets the **number of parallel-connected battery groups**.
This setting is required **only for lead-acid batteries** used in multiple parallel clusters.

High Batt / Low Batt:
Define the **battery port over-voltage** and **under-voltage protection thresholds**, respectively.

Batt Capacity:
Set the **battery capacity**, applicable **only to lead-acid batteries**.

9.8 Grid Settings Menu

Grid Settings – Page 1



Grid Code selection 1/4

Grid Mode	US 1547			
Grid Volt	240V	Grid Sensor	None	
Grid Freq	60Hz	CT Ratio	2000	
Zero Export	No	Meter Addr	1	
Zero Export Power	0.0W	Meter Select	CHNT	
Max Sell Power	0.0W	Max Buy Power	0.0W	

Grid Mode: Select the national grid standard according to the installation region (e.g., Italy-CEI0-21, Spain UNE217002).

Grid Sensor: Configuration of the grid-side current sampling method. Options include:

None: Inverter samples current internally at the grid port.

CT: External current transformer (CT) is used for sampling.

Meter: External smart meter connected via RS485 for grid data acquisition.

Grid Volt: Set the nominal grid voltage according to the country standard.

Grid Freq: Set the nominal grid frequency based on the local grid specification.

CT Ratio: Set the CT transformation ratio.

Zero Export: When enabled, the inverter will limit the power exported to the grid.

The limit is defined by the Zero-export Power setting.

Zero-export Power: Sets the maximum inverter export power. This is only effective when Zero Export is enabled.

Meter Addr: Set the RS485 address of the external energy meter.

Meter Select: Select the meter manufacturer. Please ensure the selected model matches the actual hardware.

Max Sell Power: Sets the maximum allowed output power exported to the grid.

Max Buy Power: Sets the maximum allowed input power drawn from the grid to the inverter.

Grid Settings – Page 2

Grid connect 2/4

Out Active Power	15kW	L/HVRT Mode	<input type="checkbox" value="No"/>	
Normal Ramp rate	60s	L/HPRP Mode	<input type="checkbox" value="No"/>	
Normal Connection Time	60s	P-V Mode	<input type="checkbox" value="No"/>	
Reconnect Ramp rate	10s	P-P Mode	<input type="checkbox" value="No"/>	
Reconnection Time	60s	Q-V Mode	<input type="checkbox" value="No"/>	
PF	1.0	APF Mode	<input type="checkbox" value="No"/>	

Reserved:This function is reserved.It is not recommended.Please inform the manufacturer/installer before setting up

- Out Active Power:
- Sets the maximum apparent power output to the grid.
- Normal Ramp Rate:
- Sets the power ramp-up time after inverter startup.
- Normal Connection Time:
- Sets the time delay before the inverter injects active power to the grid after startup.
- Reconnect Ramp Rate:
- Sets the ramp-up time for active power after fault recovery.
- Reconnection Time:
- Sets the reconnection delay time after a fault is cleared.

Grid Protection1 3/4

High volt 1	110.0 %	Low volt 1	88.0 %	
High volt 2	120.0 %	Low volt 2	88.0 %	
		Low volt 3	50.0 %	
High volt time 1	13.00	Low volt time 1	21.00	
High volt time 2	0.16 s	Low volt time 2	21.00	
		Low volt time 3	2.00 s	
High volt recv	105.0%	Low volt recv	91.7 %	

Reserved:This function is reserved.It is not recommended. Please inform the manufacturer/installer before setting up

High Volt 1/High Volt 2:Level 1 and Level 2 overvoltage protection thresholds (as a percentage of nominal voltage).

High Volt Time 1/High Volt Time 2:Time delays for triggering Level 1 and Level 2 overvoltage protection.

High Volt Recv:Grid overvoltage recovery threshold.

Low Volt 1 Low Volt 2:Level1 and Level 2 undervoltage protection thresholds,also defined as percentages of the nominal voltage.

Example

To adjust the acceptable voltage range of the inverter for grid connection:

Simply configure High Volt 1 and Low Volt 1 values.

If the nominal voltage is 230V:

High Volt 1 at 120%-Overvoltage point 276V

Low Volt 1 at 80%-Undervoltage point 184V

If the nominal voltage is 220V:

High Volt 1 at 120%-Overvoltage point 264V

Low Volt 1 at 80%-Undervoltage point 176V

Grid Settings-Page 4



Grid Protection2 4/4

High freq 1	61.0Hz	Low freq 1	58.5Hz	
High freq 2	62.0Hz	Low freq 2	56.5Hz	
High freq time 1	56.5	Low freq time 1	56.5	
High freq time 2	56.5	Low freq time 2	56.5	
High freq recv	60.5Hz	Low freq recv	59.5Hz	

Reserved: This function is reserved. It is not recommended. Please inform the manufacturer/installer before setting up

High Freq 1 and High Freq 2 refer to the first- and second-level over-frequency protection thresholds (in Hz). These values are typically based on the system's nominal frequency (e.g., 50Hz or 60Hz).

High Freq Time 1 and High Freq Time 2 define the delay time

before triggering protection actions when the grid frequency exceeds the corresponding threshold.

High freq recv: The recovery point frequency value of the power grid overfrequency protection. When the frequency recovers from too high to below this value, the system will allow reconnection to the grid or resume operation.

Low Freq 1 and Low Freq 2 represent the first- and second-level under-frequency protection thresholds. If the frequency falls below these values, protection will be triggered accordingly.

Low Freq Time 1 and Low Freq Time 2 set the delay durations before activating under-frequency protection once the frequency drops below the set limits.

Low Freq Recv: The recovery threshold for under-frequency protection. The system will reconnect or resume operation when the grid frequency rises above this value.

Example:

If the system nominal frequency is 50Hz, and the settings are as follows:

High Freq 1 = 52Hz, Low Freq 1 = 47Hz

The system will trigger first-level protection when the frequency goes above 52Hz or below 47Hz, based on the defined time delays.

High Freq 2 = 53Hz, Low Freq 2 = 46Hz

More severe deviations will activate second-level protection mechanisms. High Freq 1 and High Freq 2 refer to the first- and second-level over-frequency protection thresholds (in Hz). These values are typically based on the system's nominal frequency (e.g., 50Hz or 60Hz).

High Freq Time 1 and High Freq Time 2 define the delay time before triggering protection actions when the grid frequency exceeds the corresponding threshold.

High Freq Recv is the recovery point. The inverter will reconnect to the grid or resume operation only after the frequency drops below this value.

Low Freq 1 and Low Freq 2 represent the first- and second-level under-frequency protection thresholds. If the frequency falls below these values, protection will be triggered accordingly.

9.9 Generator Port Settings Menu

Gen Port Function

Select the functional mode of the generator port. Available options:



Generator Setting 1/2

Gen Port Function

No

Gen Warm Time

10 s

Start SOC

25 %

Gen Rated Power

80 W

Stop SOC

95 %

Max Run Time

600 min

Cooling Time

10 min

Gen Force

No



Generator setting:The port is used as an input for generator power.Once the generator starts,its power is used to charge the battery.

When Gen Port Function is set to Generator, the following parameters become effective:

Gen Warm Time:
Generator warm-up time. After the inverter sends the start signal, it waits for this duration before accepting generator input.

Start SOC / Stop SOC:
Battery SOC thresholds for starting and stopping the generator automatically.

Gen Rated Power:
Maximum power output of the generator.

Max Run Time:
Maximum continuous run time for the generator. Applies when the inverter is automatically controlling the generator.

Cooling Time:
Generator cooldown period after stopping. Applies to inverter-controlled generator operation.

Gen Force:
Force Start Generator function. Should be enabled during the first-time connection of a generator to ensure recognition and activation.

Smart Load Configuration

This configuration page becomes effective only when Gen Port Function is set to Smart Load.

Smart Load 2/2

Stop Output Soc

0 %

Stop Output Volt

0 V

ControlStart Time

00:00

Control End Time

00:00

Smart Load:The port is used as a dedicated load output,with user-defined working conditions for smart load control.

Stop Output SOC:
When the battery SOC drops to this threshold, the smart load output will automatically disconnect.

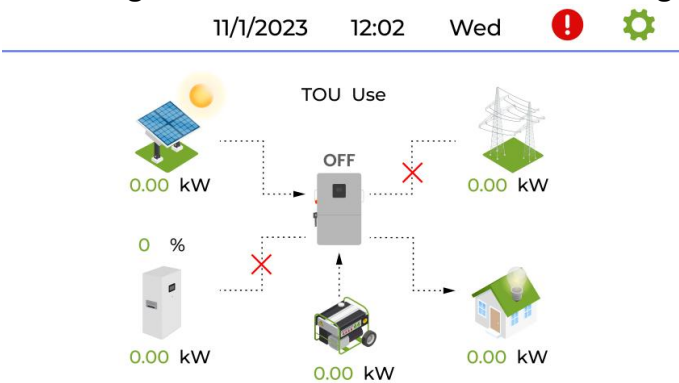
Stop Output Volt:
If voltage-based control is selected, the smart load output will disconnect when the battery voltage reaches this value.

Control Start Time:
Start of the time window during which the SOC or voltage control conditions are valid.

Control End Time:
End of the time window during which the SOC or voltage control conditions are valid.

Note:
Outside the defined time window, the smart load output is not restricted by SOC or voltage conditions, and will follow the same discharge behavior as the EPS output.

Accessing Generator Data from the Home Page



Once the generator function is enabled, a generator icon will appear on the home screen. Clicking this icon will navigate to the Generator Details Page.

Generator Details Page

Generator

L1-V: 0.0 V

L2-U: 0.0 V

L3-U: 0.0 V

L1-I: 0.0 A

L2-I: 0.0A

L3-U: 0.0 A

L1-P: 0.0 kW

L2-P: 0.0 kW

L3-U: 0.0 kW

Today: 0.0 kWh

Total: 0.0 kWh

Total Power: 0.0 kW

(Subject to variation depending on the project configuration)

L1 / L2 / L3:
Displays the voltage, current, and power of each phase.

Additional Information:

Today's Generated Energy

Total Historical Generation

Total Generator Power Output

9.10 Advanced Settings Menu

This page allows configuration of parallel settings, Wi-Fi baud rate, and EMS communication parameters.

Advanced Function

Parallel Type

Single

Parallel Num

1

Parallel Addr

1

WiFi Baud

19200

EMS Port

AFCI_485

EMS Baud

19200

2.WiFi Baud:

Baud rate for communication between inverter and data logger.

EMS Port:

Designates the spare RS485 port on the inverter for EMS (Energy Management System) communication.

Default setting: AFCI_485 (can be changed based on actual use).

EMS Baud:

Sets the communication baud rate for the EMS Port.

Default value: 19200.

1.Parallel Settings

Parallel Type:

Sets the parallel operation mode of the system.

Parallel Num:

Specifies the total number of inverters in the system (Master + Slaves).

Parallel Addr:

Sets the inverter address in the system.

The master inverter is always set to address 1.

Slave inverters follow in sequential order (2, 3, ...).

3. Parallel Configuration Screens – Example of 3-Inverter System

The following screenshots demonstrate how to configure a parallel system with one master inverter and two slave inverters. All settings must be consistent across devices, except for Parallel Addr, which should be unique per unit.

(1)Master Inverter Configuration

Advanced Function

Parallel Type	Master
Parallel Num	3
Parallel Addr	1
WiFi Baud	19200
EMS Port	AFCI_485
EMS Baud	19200



Parallel Type: Master

Parallel Num: 3 (total number of inverters)

Parallel Addr: 1

WiFi Baud: 19200

EMS Port: AFCI_485

EMS Baud: 19200

This inverter is designated as the master unit, responsible for synchronization and control across the entire system.

(2) Slave Inverter 1 Configuration

Advanced Function

Parallel Type	Slave
Parallel Num	3
Parallel Addr	2
WiFi Baud	19200
EMS Port	AFCI_485
EMS Baud	19200



Parallel Type: Slave

Parallel Num: 3

Parallel Addr: 2

WiFi Baud: 19200

EMS Port: AFCI_485

EMS Baud: 19200

This inverter operates as Slave 1, assigned address 2.

(3)Slave Inverter 2 Configuration

Advanced Function

Parallel Type	Slave
Parallel Num	3
Parallel Addr	3
WiFi Baud	19200
EMS Port	AFCI_485
EMS Baud	19200



Parallel Type: Slave

Parallel Num: 3

Parallel Addr: 3

WiFi Baud: 19200

EMS Port: AFCI_485

EMS Baud: 19200

This inverter operates as Slave 2, assigned address 3.

-Important Notes:

1.

All inverters must have the same WiFi Baud rate and EMS Baud rate.

2.

Ensure proper wiring and grounding between all units.

3.

Make sure firmware versions are consistent across all inverters.

4.

Communication cables (e.g. RS485 or CAN) must be securely connected.

9.11 Record Menu

Records

Current Faults
History Faults
Operate Logs



Current Fault	
Occurred	Alarms code
23/11/25 19:58:25	F057-dc bat1 under volt
23/11/25 19:59:20	F068 wifi cornm lost
25/07/01 09:25:22	W004-grid under freq
23/11/25 19:58:25	W010-dc bat under volt
23/11/25 19:58:23	W011-bat none-connected
1/1	



History Fault	
Occurred	Alarms code
19:25:22 00/00/00	W004-grid under freq
19:59:20 00/00/00	F068 wifi cornm lost
09:58:28 23/11/25	W006-grid under freq
09:58:28 23/11/25	W004-grid under freq
19:58:25 00/00/00	W010-dc bat under volt
19:58:23 00/00/00	F057-dc bat under volt
1/107	



Operate Logs	
Occurred	Description
25/07/01 09:26:49	INV75 05010 06050
25/07/01 09:26:48	INV74 04950 05950
25/07/01 09:26:03	INV2A 01500 00000
25/07/01 09:26:02	INV04 00000 00001
25/07/01 09:26:02	DC23 01500 00000
25/07/01 09:26:02	DC02 00000 00001
1/24	



Fault and Operation Logs

Current Fault:

Displays the inverter's active faults and alarms. It includes the timestamp, fault code, and fault description of the current issue.

History Fault:

Shows the inverter's fault and alarm history. It records the start time, recovery time, and details of each fault event.

Operate Logs:

Contains logs of user operations such as changing inverter modes, powering the system on/off, or adjusting system settings. These logs are useful for tracking system changes and user actions.

9.12 Device Info Setup Menu



Device Info

Device type:

SN:

Model:

ARM: V2.01

INV: V3.69

DCDC: V3.69

Run Time: 00:00:39:39



This page displays basic information about the inverter, including:

Device Type: The category or class of the inverter.

Model Number: The specific model identifier.

Serial Number: The unique identifier assigned to each unit.

Software Version: The currently installed firmware/software version. (ARM, INV, DCDC)

Run Time: Total runtime of the inverter since initial operation.

This page is useful for system diagnostics, firmware tracking, and technical support.

9.13 Factory Setting Menu

Reserved: This function is reserved. It is not recommended.
Please inform the manufacturer/installer before setting up



Factory Setting 1/3

DRM	No	Preventive PID	No	
Factory Reset	No	Active island	No	
AFCI	No	BatTempCompen	No	
Bat Warn	No	Bat Wake-up	No	
GRID Warn	No	ISO	No	
Ext-Control	No	Leakage Curr	No	



Factory Setting 2/3

Device Reset	No	MPPT Ctrl Mode	MPPT	
Record Clear	No	MPPT Para Mode	Two-Para	
AFCI Alarm Clear	No	MPPT Num	2	
Rolay self-test	No	MPPT Power	25 %	
Device lock	No	MPPT Disturbed	8	
Exec quick	No	CVT Volt	500V	



Factory Setting 2/3

Device Reset	No	MPPT Ctrl Mode	MPPT	
Record Clear	No	MPPT Para Mode	Two-Para	
AFCI Alarm Clear	No	MPPT Num	2	
Rolay self-test	No	MPPT Power	30kW	
Device lock	No	MPPT Disturbed	8	
Exec quick	No	CVT Volt	500V	

10.Trouble Shooting

This section contains information and procedures for solving possible problems with the KR-L8-15K-SPUS series inverters, and provides you with trouble shooting tips to identify and solve most problems that could occur with the KR-L8-15K-SPUS series inverters.

This section will help you narrow down the source of any problems you may encounter. Please read the following trouble shooting steps.

Check the warning or fault messages on the System Control Panel or Fault codes on the inverter information panel. If a message is displayed, record it before doing anything further. Attempt the solution indicated in below table.

Error code	Description	Solutions
F01	inv over volt	The inverter is faulty. Turn off the PV, grid, and battery, and wait 5 minutes before turning on the inverter. Check whether the problem is resolved. Or seek help from us, if not go back to normal state.
F02	inv under volt	
F03	short circuit	
F04	inv over curr	
F05	over load	If the load power is too high or the device is downgraded, please reduce the power consumption. Or seek our help if you are unable to return to a normal state.
F07	inv bus over volt	The inverter is faulty. Turn off the PV, grid, and battery, and wait 5 minutes before turning on the inverter. Check whether the problem is resolved. Or seek help from us, if not go back to normal state.
F08	inv bus under volt	
F09	bus unbalance	
F10	leakage self-test fail	
F11	leakage over limit	
F12	relay self-test fail	The inverter is faulty. Turn off the PV, grid, and battery, and wait 5 minutes before turning on the inverter. Check whether the problem is resolved. Or seek help from us, if not go back to normal state.
F13	inv locked	
F14	busbar buffer fail	
F15	inv over temp	
F17	parallel total count err	
F18	parallel comm fail	Check whether the parallel communication cable is properly connected. Or seek help from us, if not go back to normal state.
F19	parallel SW mismatch	
F20	parallel HW mismatch	
F21	parallel model mismatch	
F22	parallel para sync	
F23	master conflict	
F24	slave id conflict	
F25	parallel signal err	
F26	master no salve	
F33	inv flash fault	The internal communication and storage are abnormal. Turn off the PV, grid, and battery, and wait 5 minutes before turning on the inverter. Check whether the problem is resolved. Or seek help from us, if not go back to normal state.
F34	inv-dc comm fault	
F35	inv phase-locked	
F36	inner para match fail	

F38	inv master fault	The internal communication and storage are abnormal. Turn off the PV, grid, and battery, and wait 5 minutes before turning on the inverter. Check whether the problem is resolved.Or seek help from us, if not go back to normal state.
F39	inv parallel comm fail	
F40	inv parallel signal err	
F41	inv curr DC component high	
F42	inv parallel overload	
F43	inv load curr unbalance	
F44	inv parallel system fault	
F49	dc bus over volt	The inverter is faulty. Turn off the PV, grid, and battery, and wait 5 minutes before turning on the inverter. Check whether the problem is resolved.Or seek help from us, if not go back to normal state.
F50	dc bus under volt	
F51	midbus over volt	
F52	midbus under volt	
F53	dc bat disch OC	
F54	dc bat ch OC	
F55	LLC HW over curr	
F56	dc bat over volt	
F57	dc bat under volt	
F60	bat reverse connect	Check whether the positive and negative terminals of the battery power line are connected in reverse mode.
F63	dc over temp	Ensure that the inverter is installed in a place without direct sunlight. Make sure the inverter is installed in a cool/well-ventilated area. Ensure that the inverter is installed vertically and the ambient temperature is lower than the upper limit of the inverter temperature.
F64	Fan fault	Please first power off the inverter,check whether the software version is the latest one with us.If not,please upgrade firmware.
F65	meter comm lost	Check whether the meter communication line is normal.
F66	arm-inv comm lost	The internal communication and storage are abnormal. Turn off the PV, grid, and battery, and wait 5 minutes before turning on the inverter. Check whether the problem is resolved.Or seek help from us, if not go back to normal state.
F67	arm-dcdc comm lost	
F68	wifi comm lost	Please check wifi LED light status.Or seek help from us, if not go back to normal state.
F69	bms1-can comm lost	Make sure the battery you use is compatible with the inverter. Check whether the communication cables or ports between the battery and the inverter are properly connected
F70	dsp para match err	The internal communication and storage are abnormal. Turn off the PV, grid, and battery, and wait 5 minutes before turning on the inverter. Check whether the problem is resolved.Or seek help from us, if not go back to normal state.

F71	EPO	Seek help from us, if not go back to normal state.
F72	ambient temp high	Ensure that the inverter is installed in a place without direct sunlight. Make sure the inverter is installed in a cool/well-ventilated area. Ensure that the inverter is installed vertically and the ambient temperature is lower than the upper limit of the inverter temperature.
F73	bms1-485 comm lost	Make sure the battery you use is compatible with the inverter. Check whether the communication cables or ports between the battery and the inverter are properly connected
F74	arm flash error	The internal communication and storage are abnormal. Turn off the PV, grid, and battery, and wait 5 minutes before turning on the inverter. Check whether the problem is resolved.Or seek help from us, if not go back to normal state.
F75	NTC disconnect	
F76	dsp para set fail	
F81	dc-inv comm lost	
F82	dc flash fault	
F83	dc fault locked	
F86	PV1 over volt	The inverter is faulty. Turn off the PV, grid, and battery, and wait 5 minutes before turning on the inverter. Check whether the problem is resolved.Or seek help from us, if not go back to normal state.
F87	PV2 over volt	
F88	PV1 over curr	
F89	PV2 over curr	
F90	PV1 reverse connect	Check whether the PV cables are correctly connected.
F91	PV2 reverse connect	
F93	PV3 overvoltage protection	
F94	PV3 overcurrent protection	
F95	PV3 reverse connect protection	
F96	DC insulation fault	Seek help from us, if not go back to normal state.
Warning code	Description	Solutions
W01	Overload alarm	If the load power is too high or the device is downgraded, please reduce the power consumption. Or seek our help if you are unable to return to a normal state.
W02	INV overtemperature alarm	Ensure that the inverter is installed in a place without direct sunlight. Make sure the inverter is installed in a cool/well-ventilated area. Ensure that the inverter is installed vertically and the ambient temperature is lower than the upper limit of the inverter temperature.

W03	Grid over frequency alarm	
W04	Grid under frequency alarm	
W05	Grid overvoltage alarm	
W06	Grid undervoltage alarm	If the power grid is abnormal, the inverter automatically returns to the normal working state after the power grid recovers. Or seek help from us, if not go back to normal state. *GARDA(Power grid amplitude quick check alarm)
W07	Gen freq abnormal alarm	Measure the generator output voltage using a multimeter. Ensure the generator is operating within its rated voltage (typically 220/230V or 380/400V depending on the system). Check for loose or faulty wiring between the generator and inverter input.
W08	DC BAT overvoltage alarm	If the internal fault of the lithium battery occurs, Turn off the PV, grid, and battery, and wait 5 minutes to turn on the inverter and lithium battery. Check whether the problem is resolved. Or seek help from us, if not go back to normal state.
W09	DC BAT undervoltage alarm	
W10	BAT none-connected	
W11	DC over-temp alarm	
W12	grid phase rapid inspection	If the power grid is abnormal, the inverter automatically returns to the normal working state after the power grid recovers. Or seek help from us, if not go back to normal state.
W13	Grid fast inspection alarm	
W14	BAT Ch&Disch prohibited alarm	If the internal fault of the lithium battery occurs, Turn off the PV, grid, and battery, and wait 5 minutes to turn on the inverter and lithium battery. Check whether the problem is resolved. Or seek help from us, if not go back to normal state.
W19	Grid phase deficiency alarm	If the power grid is abnormal, the inverter automatically returns to the normal working state after the power grid recovers. Or seek help from us, if not go back to normal state. *GARDA(Power grid amplitude quick check alarm)
W20	Grid long time OV alarm	
W21	Grid phase abnormal alarm	
W22	Grid amplitude fast inspection alarm	

W23	Grid DC component fast detection	If the internal fault of the lithium battery occurs, Turn off the PV, grid, and battery, and wait 5 minutes to turn on the inverter and lithium battery. Check whether the problem is resolved. Or seek help from us, if not go back to normal state.
W24	Grid phase fast inspection alarm	Check the generator frequency output with a frequency meter. Make sure the generator engine speed is stable (RPM too high or low will affect frequency). Adjust the engine governor if needed. Avoid overloading the generator, which may cause frequency fluctuations.
W25	PV1 undervoltage alarm	If the internal fault of the lithium battery occurs, Turn off the PV, grid, and battery, and wait 5 minutes to turn on the inverter and lithium battery. Check whether the problem is resolved. Or seek help from us, if not go back to normal state.
W26	PV2 undervoltage alarm	
W27	PV3 undervoltage alarm	



NOTICE

If your inverters information panel is not displaying a Fault light, check the following list to make sure that the present state of the installation allows proper operation of the unit.

Is the inverter located in a clean, dry, and adequately ventilated place?

Have the DC input breakers been opened?

Are the cables adequately sized and short enough?

Are the input and output connections and wiring in good condition?

Are the configurations settings correct for your particular installation?

Are the display panel and the communications cable properly connected and undamaged?

Contact Fantastech Customer Service for further assistance. Please be prepared to describe details of your system installation and provide the model and serial number of the unit.

11.Maintenance

11.1 Power ON the Inverter for first time



Important: Please follow these steps to turn on the inverter.

Step 1: make the PV SWITCH to the ON position.

Step 2: Turn on the battery. Turn on the DC switch between battery and inverter.

Step 3: Turn on the AC circuit breaker between the inverter port and the power grid.

Step 4: Open the AC circuit breaker between the inverter load port and the emergency load.

Step 5: Manually send the startup command through the APP (for safety, it can be set to automatic startup after the initial power-on).

Step 6: The inverter should start running now.

11.2 Power Off the Inverter



- Power off the inverter before operations and maintenance. Otherwise, the inverter may shocks or occur.
- Delayed discharge. Wait until the components are discharged after power off.

Step 1: Turn off the AC breaker on the ON-GRID side of the inverter.

Step 2: Turn off the AC breaker on the BACK-UP side of the inverter.

Step 3: Turn off the battery breaker between the inverter and the battery.

Step 4: Turn off the PV switch of the inverter.

11.3 Removing the Inverter



- Make sure that the inverter is powered off.
- Wear proper PPE before any operations.

Step 1: Disconnect all the cables, including DC cables, AC cables, communication cables, the communication module, and PE cables.

Step 2: Remove the inverter from the mounting plate.

Step 3: Remove the mounting plate.

Step 4: Store the inverter properly. If the inverter needs to be used later, ensure that the storage conditions meet the requirements.

11.4 Disposing of the Inverter

If the inverter cannot work anymore, dispose of it according to the local disposal requirements, The inverter cannot be disposed of together with household waste.



- Make sure that the inverter is powered off.
- Wear proper PPE before any operations.

12 Appendix

12.1 Routine Maintenance

Maintaining Item	Maintaining Method	Maintaining Period
System Clean	Check the heat sink, air intake, and air outlet for foreign matter or dust.	Once 6-12 months
PV Switch	Turn the DC switch on and off ten consecutive times to make sure that it is working properly.	Once a year
Electrical Connection	Check whether the cables are securely connected. Check whether the cables are broken or whether there is any exposed copper core.	Once 6-12 months
Sealing	Check whether all the terminals and ports are properly sealed. Reseal the cable hole if it is not sealed or too big.	Once a year

12.2 Limited Warranty

The warranty does not apply to any Product or Product part that has been modified or damaged by the following:

- Installation or Removal (examples: wrong voltage batteries, connecting batteries backward, damage due to water/rain to electronics, preventable damage to solar wires.)
- Alteration or Disassembly.
- Normal Wear and Tear.
- Accident or Abuse.
- Unauthorized Firmware updates/software updates or alterations to the software code.
- Corrosion.
- Repair or service provided by an unauthorized repair facility.
- Operation or installation contrary to manufacturer product instructions.
- Fire, Floods, or Acts of Nature.
- Shipping or Transportation
- Shipping or Transportation.
- Incidental or consequential damage caused by other components of the power system.
- Any product whose serial number has been altered, defaced, or removed.



KOOR expressly disclaims any responsibility for performance issues arising from improper installation. Installers and users are solely responsible for following proper installation procedures outlined in provided documentation. KOOR disclaims any liability for changes in the installation that might result in electrical malfunctions or any other issues related to the KOOR product.



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